To Stage or Not to Stage?
Optimal Contracts and Project Management

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Abstract

We study how a principal optimally manages a project when the agent's effort is unobservable and completion is uncertain. The principal can allow the agent to pursue the final product directly or require a verifiable intermediate milestone first (staging). A milestone is evidence of effort that can be rewarded with more time, but producing it may slow the project. We show that the optimal contract switches between the two routes at most twice. Initially, the principal may allow the agent to pursue the final product directly. If completion fails to materialize, she may require an intermediate milestone. If this too fails to arrive, she drops the requirement and gives the agent a final opportunity to complete the project before an endogenously chosen deadline. The demand for milestones is thus nonmonotonic over a project's life. (JEL: D82, D86, O32)

Keywords: Dynamic Contracting, Milestones, Deadlines, Monitoring.

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1. Introduction

Many economic activities require sustained effort that is difficult or impossible for the party financing the activity to observe. A firm hires a software developer to produce an application, a government agency contracts with a defense manufacturer to develop a new system, or an investor finances an entrepreneur pursuing a commercial innovation. In each case, the principal ultimately cares about a completed product, but success may take considerable time and is uncertain even when the agent works diligently. How should the principal respond when the project remains unfinished and there is little observable evidence of progress? One possibility is to give the agent more time to pursue the final objective. Another is to reorganize the project into stages, requiring a verifiable intermediate accomplishment before committing further resources. The latter approach provides an earlier opportunity to observe productive progress, but it may also slow the project by diverting effort toward an intermediate deliverable that is not itself the principal's ultimate objective.

Such choices are common in practice. A software contractor initially given discretion to develop a complete application may subsequently be required to deliver a working prototype or independently testable module. A defense contractor that has not delivered a functioning system may be directed to demonstrate a critical subsystem or pass an intermediate technical test. Similarly, an investor financing a research-intensive startup may begin by allowing the entrepreneur to pursue commercialization directly, but eventually insist on proof of concept or another verifiable technical milestone as a condition for continued support. In each setting, the principal can make progress more observable by changing how the project is organized. Importantly, requiring an intermediate deliverable is not necessarily the fastest way

to produce the final product. Its attraction may instead lie in the information and incentives generated along the way.

This paper studies the optimal organization of such projects when effort is unobservable and the principal can commit to a contingent production plan. Our central insight is that the principal may optimally respond to prolonged noncompletion by progressively changing the conditions under which the relationship continues. Initially, she may allow the agent to pursue the final product directly. If no success materializes, she may require an observable intermediate breakthrough that earns the agent additional time to complete the project. Notably, if this breakthrough also fails to arrive, the principal may abandon the intermediate requirement and give the agent one final opportunity to deliver the completed product. Thus, the optimal production plan can exhibit a nonmonotonic pattern: an initial period of autonomy, followed by a demand for verifiable progress, and finally a return to direct production under a terminal relatively tight deadline.

The final switch deserves particular emphasis. It does not occur because an externally imposed deadline is approaching. Rather, the principal optimally commits in advance to limiting the time during which intermediate accomplishments can earn further support. A milestone is useful because it provides evidence of productive effort and can justify extending the relationship. But obtaining a milestone is not the same as completing the project. Eventually, the principal may prefer to make continued compensation contingent exclusively on final delivery rather than allow the agent additional opportunities to prolong the relationship through intermediate achievements. In this sense, the principal turns up the heat as the project remains incomplete, first demanding evidence of progress and ultimately insisting on

completion itself. The terminal deadline is an endogenous incentive instrument, not an exogenous constraint.

We formalize these ideas in a continuous-time principal-agent model in which a project can be completed along one of two routes. The direct route requires a single productive breakthrough, arriving according to a Poisson process when the agent exerts effort. The staged route requires two successive breakthroughs: a verifiable intermediate milestone followed by final completion. Each stage has a higher arrival rate than direct completion, so staging offers an earlier opportunity to observe a productive accomplishment. However, staging requires two successes rather than one and may take longer in expectation. The agent's effort is unobservable, and shirking provides a private flow benefit. The principal chooses the production route over time, an initial deadline, completion-contingent payments, and the terms of continuation following a milestone. These contractual provisions must induce the agent to exert effort even though the principal cannot directly monitor his actions.

The model isolates three forces governing the optimal choice of production route. The first is the *milestone effect*. An intermediate milestone provides verifiable evidence of productive activity, allowing the principal to reward progress by extending the deadline. This extension supplies incentives without requiring an immediate monetary payment. The second is the *deadline effect*. Because direct production requires only one breakthrough whereas staging requires two, the principal may optimally commit to a terminal phase in which the agent must complete the project directly rather than earn additional time through an intermediate milestone. Importantly, the principal chooses the deadline and the production schedule jointly, anticipating that direct production will become preferable as the contractual horizon shrinks. The third is the *efficiency effect*. When staging takes longer in expectation, the principal may initially

favor direct production, allowing the agent to pursue the final objective without requiring intermediate evidence of progress. The interaction of these forces determines not only whether staging is desirable, but also when it should be introduced, when it should be abandoned, and when the relationship should terminate.

We begin with a benchmark in which the two routes have the same expected completion time. Thus, staging is technologically innocuous: it changes the distribution and observability of progress without increasing expected production time. In this case, we show that the optimal contract switches routes at most once, from staged to direct production (Theorem 1). For projects with relatively low returns, direct production is optimal throughout. For projects with intermediate returns, the principal initially requires a milestone but switches to direct production if it has not arrived by a specified date. For sufficiently valuable projects, staging remains optimal throughout the relationship. The principal therefore uses intermediate milestones more extensively when the potential return to successful completion is greater. Even when staging is technologically innocuous, however, she may eventually stop rewarding milestones with extensions and insist on final completion.

The contracting problem admits a particularly simple characterization. Along the optimal contract, the agent's promised continuation utility declines deterministically at the rate of his private benefit from shirking until a productive breakthrough occurs. Successful completion generates a final payment, while an intermediate milestone generates an upward jump in continuation utility by extending the deadline exactly one expected stage duration. Because the incentive constraints bind, these jumps exactly compensate, in expectation, for the downward drift. Thus, the agent's continuation utility is a martingale under the recommended effort strategy, regardless of which production route is assigned. The principal can change the timing and form of

incentive rewards without changing their expected rate of accrual. This observation allows us to reduce the contracting problem to a simpler problem of choosing the production route and deadline under observable effort.

Our most distinctive results arise when staging is technologically inefficient. In many applications, producing a prototype, demonstrating a subsystem, or establishing proof of concept requires activities that would not be necessary along the fastest path to the final product. In such cases, staging can be worthwhile even though it delays expected completion, because its intermediate deliverable creates an earlier opportunity to verify progress and reward effort. We show that the optimal contract switches routes at most twice, regardless of the magnitude of the efficiency loss (Theorem 2). When this loss is sufficiently large and the project return sufficiently high, the optimal contract follows the direct–staged–direct sequence described above (Proposition 5). The principal initially tolerates the absence of intermediate evidence while the agent pursues the efficient direct route. If the project remains unfinished, she introduces staging to obtain a verifiable intermediate accomplishment. If the milestone also fails to materialize, she ultimately returns to direct production, terminating the possibility of further extensions for intermediate progress.

This result has a natural interpretation in the applications motivating our analysis. A software developer may initially be allowed to pursue a complete application, subsequently be required to demonstrate a working module, and finally be given an ultimatum consisting of a fixed period to deliver the application without further intermediate requirements. A defense contractor may similarly move from unrestricted development to a period of technical demonstrations before facing a final delivery deadline. And an entrepreneur may initially have considerable discretion, subsequently be required to establish proof of concept to earn additional

financing, and ultimately face a final opportunity to achieve commercialization. These examples illustrate that the optimal demand for observable progress need not increase monotonically with the duration of an unsuccessful relationship. Intermediate milestones can be most valuable in the middle of a project, when the principal is no longer content to wait without evidence of progress but is still willing to reward a verifiable breakthrough with additional time.

The analysis also identifies less familiar contractual patterns. When the efficiency loss from staging is small, the optimal arrangements resemble those in the technologically innocuous benchmark (Proposition 4). When the loss is large, staging is used only during an intermediate window, if at all. At intermediate levels of efficiency loss, a reversed one-switch contract can arise in which the principal begins with direct production and subsequently introduces staging without returning to the direct route. Although this arrangement lacks a final direct-production phase, it reflects the same underlying tradeoff: the value of obtaining an observable milestone changes as the contractual horizon declines. Our characterization shows that these possibilities exhaust the optimal switching patterns within the model.

Several additional findings clarify the scope and interpretation of our results. First, the optimal contract rewards intermediate progress through additional time rather than an immediate cash payment. This connects the analysis to milestone-contingent financing, probationary arrangements, and contracts in which continued support depends on demonstrable progress. Second, the results remain implementable when the agent privately observes the timing of breakthroughs or privately chooses the production route, provided he cannot fabricate a milestone. The inability to fabricate progress is important, since false milestones would otherwise allow the agent to obtain extensions without exerting effort. Third, the deadline effect reflects differences

in the distributions of completion times rather than the exponential assumption alone. Finally, numerical analysis shows that discounting can generate contractual patterns resembling those arising from technologically inefficient staging. Impatience can therefore create an incentive to begin with direct production even when the two routes have identical expected completion times.

The broader implication is that project organization and incentive provision cannot generally be separated. When effort is unobservable, the principal can influence incentives not only through compensation and deadlines, but also by choosing which productive accomplishments are made verifiable. As a project remains unfinished, she may optimally move from granting the agent autonomy, to demanding evidence of progress, to insisting on final completion. Thus, the principal chooses not only how long to support a project, but also what the agent must accomplish as time passes without success.

Related Literature. Our closest connection is to the literature on dynamic contracting for projects with deadlines and intermediate breakthroughs (Hu 2014; Green and Taylor 2016; Wolf 2018; Moroni 2022). In particular, the staged route builds on the two-stage project with observable progress in Green and Taylor (2015). Whereas the sequence of stages is technologically fixed in that framework, here the principal can choose between staged and direct production and change routes following noncompletion. Endogenizing the stage structure reveals when the principal optimally creates, and subsequently forgoes, opportunities to reward intermediate progress. Related models allow agents to observe or misreport progress privately (Green and Taylor 2016; Varas 2018; Mayer 2022; Feng et al. 2024; Feng and Zhang 2024); we focus on verifiable milestones and discuss fabrication in Remark 3.

The milestone's role as evidence of productive activity connects our analysis to dynamic monitoring (Orlov 2022; Piskorski and Westerfield 2016; Dilmé and Garrett 2019; Marinovic and Szydlowski 2022; Varas et al. 2020; Marinovic and Szydlowski 2023; Chen et al. 2020; Wong 2023). Rather than purchasing a signal about effort while holding the technology fixed, the principal reorganizes production to generate a contractible intermediate outcome. This distinction matters because staging changes not only observability but also the distribution of completion times. Indeed, the deadline effect arises even with observable effort; the additional incentive value of the milestone comes from the continuation it can earn.

Staging also features prominently in venture financing (Gompers 1995; Kaplan and Strömberg 2003). Existing accounts emphasize the option to withdraw financing after unfavorable news (Admati and Pfleiderer 1994; Bergemann and Hege 1998) or the mitigation of entrepreneurial hold-up (Neher 1999). Here the principal can already terminate the project. The distinctive benefit of an intermediate breakthrough is instead that it can justify continued support while preserving effort incentives. The possibility of fabricated progress, emphasized by Cornelli and Yosha (2003), marks a limit of this mechanism, as discussed in Remark 3.

Finally, our route-choice problem is related to Carnehl and Schneider (2023), who study a choice between one-success and multiple-success approaches under an exogenous horizon and obtain a related bound on reversals, though their mechanism involves experimentation and learning rather than moral hazard. The complementarity between successive breakthroughs also links our setting to work on the sequencing of innovation (Green and Scotchmer 1995; Gilbert and Katz 2011; Bryan and Lemus 2017; Poggi 2021). In a patent race with the same two-route technology, Kim and Poggi (2026) study incentives to conceal an intermediate

discovery from a rival. By contrast, our milestone is reported to a principal committed to rewarding it with more time, making concealment unprofitable (Proposition 6).

2. Setup

Environment. A principal (she) hires an agent (he) to complete a project. Work takes place in continuous time over an unbounded horizon, $t \in [0, \infty)$. Completing the project yields the principal a gross payoff $\Pi > 0$ and ends the relationship; until the project is completed or abandoned the principal incurs an operating flow cost $c > 0$. Both players are risk neutral, do not discount the future (Section 6.3 relaxes this), and have outside options normalized to zero. The principal has unlimited funds; the agent is protected by limited liability, so every transfer to him must be nonnegative.

Effort and Routes. The agent's action at time t is his effort $\beta_t \in [0, 1]$, where $\beta_t = 1$ is full effort and $\beta_t = 0$ is full shirking.[1] Effort is his private information. Withholding the fraction $1 - \beta_t$ leaves him a private flow benefit $(1 - \beta_t)\varphi$, and we assume $c > \varphi > 0$, so that the cost of keeping the project alive exceeds the agent's gain from shirking. The friction is cash diversion in the sense of DeMarzo and Sannikov (2006): the principal outlays c per unit of time, and during an instant of full shirking the agent privately captures φ of that outlay. In that reading φ/c is the diversion rate and $1 - \varphi/c$ the resulting deadweight wedge.

At each instant the principal directs the agent along one of two routes to completion. Under the **direct route**, the agent works on the project as a whole

1. Both the arrival rate and the private benefit are linear in β_t, so a corner is optimal and restricting to $\beta_t \in \{0, 1\}$ is without loss (Lemma OA.2).

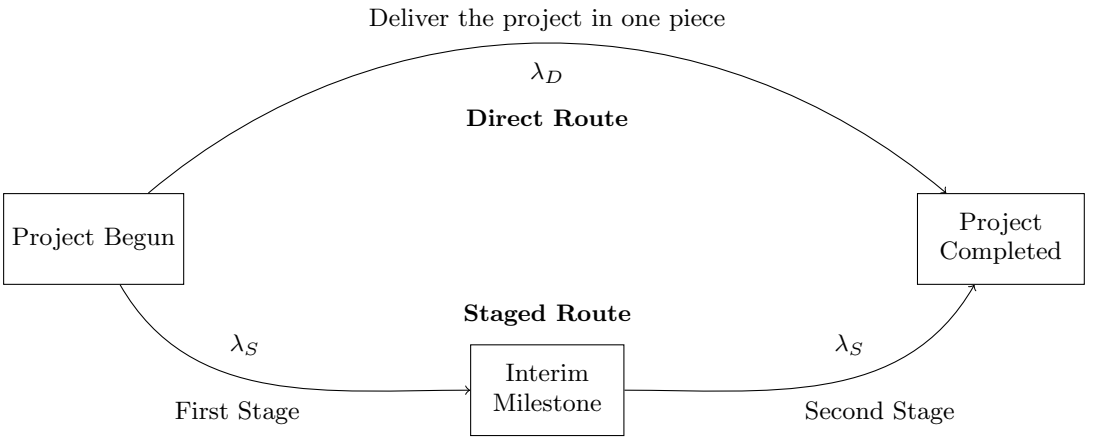


FIGURE 1. Routes to Project Completion

and delivers it in finished form; completion arrives at Poisson rate $\lambda_D \beta_t$. Under the **staged route**, he first produces an *interim milestone* (a prototype, a demonstration, a report on work in progress) and only then completes the project; each stage arriving at rate $\lambda_S \beta_t$. Figure 1 illustrates the two paths. We assume $\lambda_S > \lambda_D$: otherwise, the staged route would be slower stage for stage and in total, and never used.

The milestone and project completion are publicly observable; the agent's effort is not. Payments may thus depend on which deliveries have occurred and when; absent a delivery, calendar time is the only payoff-relevant public information.

Efficiency. Absent any agency friction, and with no deadline in force, the choice between routes would turn on expected completion time alone. The direct route completes the project in expected time $1/\lambda_D$, at expected cost c/λ_D; the staged route requires two successes and so takes $2/\lambda_S$, at expected cost $2c/\lambda_S$. The first best is therefore $\Pi - \min\{c/\lambda_D, 2c/\lambda_S\}$.

Sections 3 and 4 assume the routes are equally efficient, $\lambda_S = 2\lambda_D$. Staging is then *technologically innocuous*: a principal who could dictate effort would be indifferent

between the routes, so any staging that appears in the optimal contract is there for incentive reasons alone. Section 5 allows staging to be costly, $\lambda_S < 2\lambda_D$.

We also assume the project is worth undertaking:

$$\Pi > \frac{c}{\lambda_D} \qquad \Longleftrightarrow \qquad \pi \equiv \frac{\lambda_D \Pi}{c} > 1, \tag{1}$$

where π denotes the *project return*.

Contracts. Limited liability makes deadlines indispensable. If the contract were never to terminate, the agent could shirk at every date and still be retained, collecting φ per unit of time in perpetuity; because payments cannot be negative, no finite reward for completion would deter him.[2] Every incentive-compatible contract must therefore abandon the project in finite time, trading the surplus lost when a viable project is abandoned against the rents the principal must otherwise concede.

At $t = 0$ the principal commits to a contract; if the agent rejects it, both receive zero. We restrict attention to contracts that recommend full effort at every date.[3]

A contract specifies when to abandon the project, which route is assigned at each date, how much the agent is paid on project completion, and how the deadline and payments change once the milestone arrives. Formally, a (deterministic) contract is an object $\Gamma \equiv \left\{T, \{a_t, R_t, \hat{\Gamma}^t\}_{0 \le t \le T}\right\}$, with the following components.[4]

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2. Absent limited liability, the principal could attain the first best by selling the project to the agent upfront at its first-best value; limited liability means he cannot pay that price.

3. It is without loss of generality: because $c > \varphi$, any interval on which the agent is meant to shirk can be excised, advancing all subsequent events and saving $c - \varphi$ per unit of time.

4. Randomization cannot improve upon this class, though the argument requires care; see Remark 2 and Online Appendix OA.2.

1. $T \in \mathbb{R}_+$: the deadline at which the project is abandoned absent a delivery.
2. $a_t \in \{0, 1\}$: the route assigned at t: *direct* if $a_t = 1$, *staged* if $a_t = 0$.[5]
3. $R_t \geq 0$: the payment if he completes the project at t with no milestone delivered.[6]
4. $\hat{\Gamma}^t \equiv \left\{T^t, \{R_s^t\}_{t \leq s \leq T^t}\right\}$: the continuation contract if the milestone arrives at t: a revised deadline $T^t \geq t$ and payments $R_s^t \geq 0$ for completion at s.

Any delivery inconsistent with the assigned route (a milestone under the direct route, or completion with no milestone on record under the staged route) terminates the contract without payment, so the agent works the assigned route. T^t is unrestricted: whether and by how much the deadline is extended is a result, not an assumption.

3. Observable Effort with Exogenous Deadlines

Before turning to the contracting problem under agency, we ask how a principal who can *observe* and *contract* on effort deploys the two routes. Since effort can be mandated, no obedience constraint arises and route choice turns on technology alone.

An interesting trade-off appears in the presence of deadlines. Fix an exogenous deadline T, extended by $\Delta \geq 0$ if the interim milestone is delivered before T.[7] The principal faces two effective deadlines: T if no milestone has been delivered, and $T + \Delta$

5. We assume the principal observes which route the agent works, so his only choice is whether to work or shirk. Section 6.2 relaxes this.

6. Since both parties are risk neutral and do not discount, all payments may be backloaded to the moment of a delivery without loss of generality (see, e.g., Ray 2002).

7. The case $\Delta = 0$ is permitted, so that staging confers no additional runway. One could also allow $\Delta < 0$; we show in Section 4 that under agency it is optimal to extend, not curtail, the deadline upon delivery of the milestone.

if one has. Both deadlines are inefficient; whenever (1) holds, efficiency demands running the project to completion without truncation, and it is this truncation that makes the routes differ. Given (T, Δ), the principal chooses a route policy $a : [0, T] \to \{0, 1\}$ to maximize her expected payoff. Throughout this section staging is technologically innocuous, $\lambda_S = 2\lambda_D$.

3.1. Deadline and Milestone Effects

To build intuition, we first identify the two forces behind route choice: the *deadline effect* and the *milestone effect*. To isolate them, compare the two policies that never switch: the *direct-only* policy ($a_t = 1$ for all t), under which the agent works on the project as a whole throughout, and the *staged-only* policy ($a_t = 0$ for all t), under which he works toward the milestone and then, if delivered, toward completion. The principal's payoff also reflects operating costs, but the probability of completing in time is what shapes her preference.

Under the direct-only policy, a single success is needed by date T, yielding the completion probability

$$\mathcal{P}^D(T) \equiv \int_0^T \lambda_D e^{-\lambda_D \tau}\, d\tau = 1 - e^{-\lambda_D T}. \tag{2}$$

Under the staged-only policy, the milestone must arrive at some date $\tau_m \leq T$, and the project must then be completed by the extended deadline $T + \Delta$, yielding the

completion probability

$$\mathcal{P}^S(T,\Delta) \equiv \int_0^T \left[\int_{\tau_m}^{T+\Delta} \lambda_S e^{-\lambda_S(\tau-\tau_m)}\, d\tau \right] \lambda_S e^{-\lambda_S \tau_m}\, d\tau_m$$

$$= 1 - \left(1 + \lambda_S T e^{-\lambda_S \Delta}\right) e^{-\lambda_S T}. \tag{3}$$

Setting $\Delta = 0$ strips the staged route of any deadline extension and isolates the technological comparison.

LEMMA 1 (Deadline effect). *Suppose $\Delta = 0$. There exists $\check{T} > 0$ such that $\mathcal{P}^D(T) > \mathcal{P}^S(T,0)$ for all $T < \check{T}$, and $\mathcal{P}^S(T,0) > \mathcal{P}^D(T)$ for all $T > \check{T}$.*

The direct route's advantage is that it requires only one arrival. Over a short horizon the probability of one arrival is first-order in remaining time while the probability of two is second-order, so needing one slow success beats needing two fast ones. The staged route's advantage is that each of its stages is faster, and over a long horizon two fast successes are likelier than one slow one. Figure 2 shows the two completion-time densities. Under equal efficiency ($\lambda_S = 2\lambda_D$) they share a mean (dotted line), but the direct route places more mass both near zero and in the far tail (shaded), whereas the staged route concentrates around the mean: direct completion time is a mean-preserving spread of staged completion time. A tight deadline rewards dispersion and a generous one rewards concentration. We call this the *deadline effect*.

The extension Δ confers a second advantage on the staged route, and it is not technological. Since $\mathcal{P}^S(T,\Delta)$ is strictly increasing in Δ while $\mathcal{P}^D(T)$ does not depend on it, any $\Delta > 0$ tilts the comparison toward staging. The source is information: the

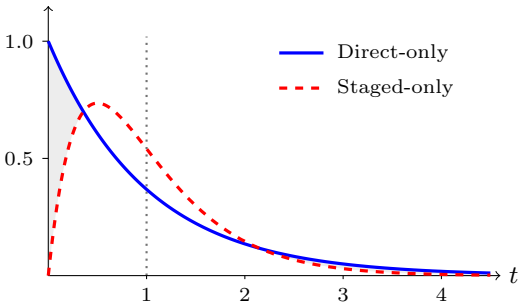


FIGURE 2. Completion-time densities under equal efficiency ($\lambda_D = 1$, $\lambda_S = 2$)

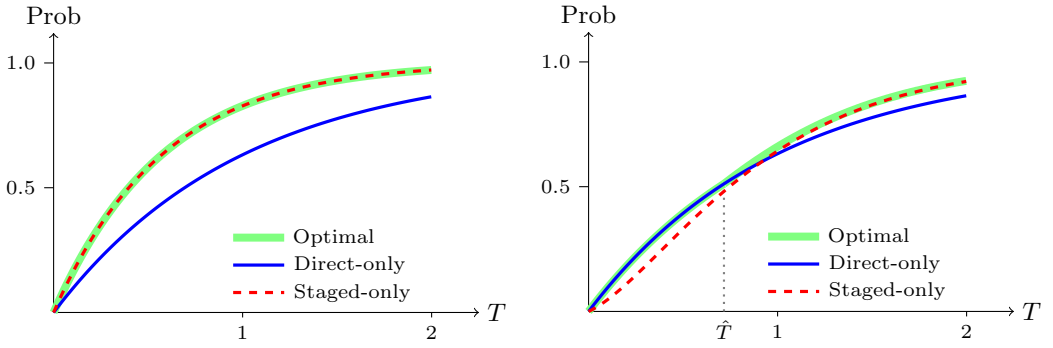
milestone reveals to the principal that concrete progress has occurred, and Δ is the value she places on that news. We call this the *milestone effect*.

3.2. Optimality of One-Switch Policy

The two effects pull in opposite directions along the horizon. Staging is effective when the deadline is distant, where the milestone effect dominates; the direct route is preferable when the deadline is near, where the deadline effect dominates. This motivates staging first and switching to the direct route as time runs short.

DEFINITION 1. A policy is called a *one-switch policy* if there exists an intermediate deadline $S \in [0, T]$ such that: (i) the principal chooses the staged route up to S ($a_t = 0$ for $t \leq S$); (ii) if the interim milestone is delivered before S, she proceeds with the second stage until the extended deadline $T + \Delta$; and (iii) if the milestone is not delivered by S, she switches to the direct route until the original deadline T.

The two benchmark policies are special cases: the direct-only policy sets $S = 0$ and the staged-only policy sets $S = T$. The following proposition establishes that the optimal policy invariably takes the form of a one-switch policy. We prove this result with a guess-and-verify approach to the principal's dynamic optimization problem.



(A) Long extension ($\Delta = 1 > \bar{\Delta}$) (B) Short extension ($\Delta = 0.1 < \bar{\Delta}$)

FIGURE 3. Completion probabilities ($\lambda_S = 2,\ \lambda_D = 1,\ \Pi = 3,\ c = 1$; $\bar{\Delta} \approx 0.46$, $\hat{T} \approx 0.72$)

PROPOSITION 1. *Suppose that the two routes are equally efficient ($\lambda_S = 2\lambda_D$). When the principal faces a deadline T and a deadline extension Δ contingent on milestone delivery, the optimal policy is characterized as follows:*

(a) ***(Long extension)*** *If $\Delta \geq \bar{\Delta} \equiv (1/\lambda_S)\log[(2\pi - 1)/(\pi - 1)]$, the staged-only policy is optimal.*

(b) ***(Short extension)*** *If $\Delta < \bar{\Delta}$, set $\hat{T} \equiv 2(\bar{\Delta} - \Delta) > 0$. Then:*

  (i) *when $T < \hat{T}$, the direct-only policy is optimal;*

  (ii) *when $T > \hat{T}$, the one-switch policy with the intermediate deadline $S = T - \hat{T}$ is optimal.*

Figure 3 plots completion probabilities against the deadline for the two benchmark policies and for the optimum. When the extension is long (Panel 3a), the milestone effect outweighs the deadline effect at every horizon: the staged-only policy completes with higher probability for all T, and the optimum coincides with it. When the extension is short (Panel 3b) the two benchmarks cross, and the optimum follows whichever is better.

The cutoff $\hat{T}$ is an amount of time remaining, not a fraction of the horizon: the principal stages until $\hat{T}$ units are left, whatever the deadline. It also has a simple form, $\hat{T} = 2(\bar{\Delta} - \Delta)$, so it shrinks to zero as the extension approaches $\bar{\Delta}$ and the two cases of Proposition 1 meet continuously. Note that the optimum is chosen to maximize surplus rather than completion probability; when the project return is low the principal accepts a lower chance of completion in exchange for a shorter expected run, and the optimal policy then completes less often than staging throughout.

Finally, the extension cutoff $\bar{\Delta}$ falls as the project becomes more valuable.

LEMMA 2. *Suppose that $\lambda_S = 2\lambda_D$ and $\pi > 1$. The extension cutoff $\bar{\Delta}$ is strictly decreasing in the project return π.*

Proof. Differentiating, $\partial\bar{\Delta}/\partial\pi = -1/[\lambda_S(2\pi - 1)(\pi - 1)] < 0$, since $\pi > 1$ by (1). □

The comparative static reflects how the two terms in the principal's objective scale with the project return. Expected surplus is Π times the completion probability less c times the expected time in operation, and the staged route trades one against the other: over a long horizon it completes more often, by the milestone effect, but it also runs longer, since a first success that is never followed by a second consumes time and delivers nothing. Raising π raises what the completion probability is worth without changing what the time costs, so the balance tilts toward the route that completes more often. A smaller extension then suffices to make staging worthwhile at every horizon, and $\bar{\Delta}$ falls.

4. Agency

We now make effort unobservable, keeping $\lambda_S = 2\lambda_D$: the principal can no longer compel the agent to work and must incentivize him with rewards and deadlines.

As in the tangible-progress case of Green and Taylor (2015), the optimal contract rests on three pillars: (i) the contract carries a finite deadline; (ii) the reward for completing the project declines linearly in calendar time; and (iii) the deadline is extended by exactly $1/\lambda_S$ when the interim milestone is delivered.

What is new relative to Green and Taylor (2015) is the choice of route: at each date, the principal assigns the route the agent works. The central finding is that the optimal contract inherits the *one-switch* structure of Section 3, staging first and switching to the direct route as the deadline nears. We define this class formally.

Definition 2. A contract is a *one-switch contract* with final deadline T and intermediate deadline $S \in [0, T]$ if

(i) the agent is assigned to work on the staged route up to S;

(ii) if the interim milestone is delivered before S, the deadline is extended by $1/\lambda_S$ and the reward for completing the project at t is $R_t^S \equiv \varphi\left(T - t + 2/\lambda_S\right)$;

(iii) if the milestone is not delivered by S, the direct route is assigned until T, and the reward for completing the project at t is $R_t^D \equiv \varphi\left(T - t + 1/\lambda_D\right)$; and

(iv) the contract terminates at $T + 1/\lambda_S$ in case (ii) and at T in case (iii).

When $S = T$ it is a *staged-only contract*, and when $S = 0$ a *direct-only contract*.

The optimal contract is a one-switch contract, and the project return $\pi = \lambda_D \Pi / c$ decides whether it stages throughout, switches once, or never stages.

THEOREM 1 (Optimal contract). *Suppose staging is technologically innocuous ($\lambda_S = 2\lambda_D$), and define*

$$\pi_F \equiv 1 + \frac{\varphi}{c}, \qquad \pi_D \equiv \frac{1}{e}\left[e + \frac{\varphi}{c} + \sqrt{\frac{\varphi}{c}\left(e + \frac{\varphi}{c}\right)}\right], \qquad \pi_S \equiv \frac{e-1}{e-2} \approx 2.392. \tag{4}$$

Then $\pi_F < \pi_D < \pi_S$, and the optimal contract can be implemented as follows:

(a) ***(Stage throughout)*** *when $\pi > \pi_S$, a staged-only contract is optimal;*

(b) ***(Stage, then switch)*** *when $\pi_S > \pi > \pi_D$, a one-switch contract is optimal;*

(c) ***(Never stage)*** *when $\pi_D > \pi > \pi_F$, a direct-only contract is optimal; and*

(d) ***(Abandon)*** *when $\pi < \pi_F$, the project is infeasible.*

Two features of (4) stand out. First, the thresholds depend on the primitives λ_D, λ_S, c, Π and φ only through the project return π and the diversion rate φ/c, and the upper threshold π_S is a constant, independent of all primitives. The same constant appears in Green and Taylor (2015). Their condition (C.2) for the incentive condition to bind everywhere is $\Pi \geq (2c/\lambda_S)(e-1)/(e-2)$, which reduces to $\pi \geq \pi_S$ when staging is technologically innocuous. Above that threshold our contract and theirs coincide (staging throughout, extension by one expected stage); below it they diverge, for reasons taken up in Remark 1.

Second, the ordering $\pi_F < \pi_D < \pi_S$ follows from $\varphi < c$ (Lemma B.1). The rest of the section explains the theorem; the proof is in Appendix B.

Why the extension is one expected stage. We track the agent's *promised utility*, the expected payoff he is owed from a given date onward, and use it as the state variable (Spear and Srivastava 1987). Write u_t for promised utility at t before the milestone, and u_M^t for the promise that takes effect if it arrives at t. Suppose the direct

route is assigned. Working over $[t, t+dt)$ completes the project with probability $\lambda_D\,dt$ and pays R_t, but completion ends the relationship and forfeits u_t, for an expected gain of $\lambda_D(R_t - u_t)\,dt$; shirking pays $\varphi\,dt$ and keeps u_t. Working is optimal if and only if

$$R_t \geq u_t + \frac{\varphi}{\lambda_D}. \tag{IC$_D$}$$

On the staged route the agent is paid not in cash but by a jump in promised utility, from u_t to u_M^t, and the milestone arrives at rate λ_S, so the same comparison gives

$$u_M^t \geq u_t + \frac{\varphi}{\lambda_S}. \tag{IC$_S$}$$

Both say: what the agent gets for succeeding must exceed what he already holds by the premium φ/λ, larger on the slower route, where success is rarer.

We *guess* that the constraints bind, and *verify* below that it is indeed optimal for them to do so. The agent is then indifferent between working and shirking at every date, so his promise must run down at exactly the rate he could earn by shirking:

$$\dot{u}_t = -\varphi, \qquad \text{that is,} \qquad u_t = u_0 - \varphi t. \tag{5}$$

Therefore, promised utility is a clock: it runs down at the constant rate φ, and the contract ends when it hits zero (the agent's outside option), at $T = u_0/\varphi$. In effect, the principal chooses how long to run the project, and u_0 is that length denominated in utils. One unit of promised utility buys $1/\varphi$ units of the agent's time.

All three pillars follow. The deadline is finite because a clock started at a finite value runs out; it is not an instrument the principal adds but what promise-keeping leaves behind. Rewards decline linearly because substituting $u_t = \varphi(T - t)$ into (IC$_D$)

at equality gives $R_t^D = \varphi(T - t + 1/\lambda_D)$: the agent's outside option is to shirk out the remaining clock, and that shrinks as the deadline nears. And the extension is one expected stage because (IC_S) requires the promise to jump by φ/λ_S on delivery, which at the exchange rate $1/\varphi$ is

$$t + \frac{u_M^t}{\varphi} = t + \frac{u_t}{\varphi} + \frac{1}{\lambda_S} = T + \frac{1}{\lambda_S}.$$

The extension is independent of Π and c because incentive compatibility prices the milestone against the agent's alternative, and the only rates in that comparison are λ_S and φ; since φ is also the exchange rate between utility and time, it cancels. What Π and c determine is not the price of a milestone but how much time the principal buys. A more valuable project is run longer, not rewarded more generously per milestone.

Why the project return decides the form. Suppose, as guessed, that the incentive conditions bind. For a given promised utility u, whatever the routes, the contract runs for u/φ and extends the deadline by $1/\lambda_S$ on milestone delivery, and gives the agent exactly u. The principal's payoff is therefore the surplus the contract generates net of u, and route choice only affects the surplus. The next proposition confirms the guess: the principal cannot gain by leaving an incentive condition slack.

PROPOSITION 2 (Value function). *Let $\mathcal{W}^*(T, \Delta)$ denote the optimal expected surplus in the benchmark of Section 3 with deadline T and extension Δ, and let $\mathcal{V}(u)$ be the most the principal can earn while delivering the agent exactly u. Then $\mathcal{V}$ is concave, incentive conditions bind at every u, and*

$$\mathcal{V}(u) = \mathcal{W}^*\left(\frac{u}{\varphi}, \frac{1}{\lambda_S}\right) - u. \tag{6}$$

Moral hazard does not change what the routes are good for. It changes the terms on which the principal buys time, and fixes the extension at a level she would not otherwise have chosen. This is what makes the project return decisive. Proposition 1 said that the form of the policy turns on how the available extension compares with the cutoff $\bar{\Delta}$. Here the available extension is not a free parameter (it is $1/\lambda_S$) so the comparison is between $1/\lambda_S$ and $\bar{\Delta}$, and by Lemma 2 the cutoff $\bar{\Delta}$ falls as the project return rises. A more valuable project therefore needs less runway to justify staging, while the runway on offer stays put. Setting $\bar{\Delta} = 1/\lambda_S$ gives

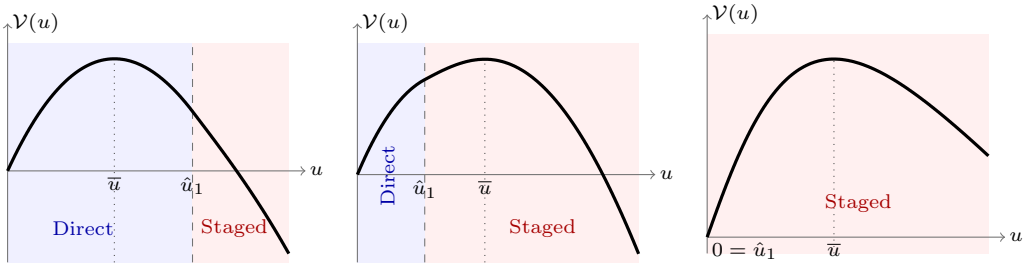
$$\frac{1}{\lambda_S}\log\left[\frac{2\pi-1}{\pi-1}\right] = \frac{1}{\lambda_S} \qquad \Longleftrightarrow \qquad \pi = \frac{e-1}{e-2} = \pi_S.$$

The arrival rate appears on both sides and cancels, which is why π_S is a constant rather than a function of the primitives. Above it the extension always suffices, the staged region is everything, and a staged-only contract is optimal (Theorem 1(a)).

How long a deadline. Below π_S the principal stages early and switches late, and the remaining forms differ in how much time she buys. Since $\mathcal{V}$ is concave, the problem

$$\overline{u} \equiv \arg\max_{u \geq 0} \mathcal{V}(u) \tag{MP}$$

has a unique solution and the deadline is $\overline{u}/\varphi$. Extending by one instant costs the rent needed to keep the agent working over it, which by (5) is φ per unit of time whatever the state of the project: the cost of time is linear. The benefit is the added chance of completing in that instant, and it falls as the deadline lengthens, since a project still unfinished at a distant date is one on which arrivals have repeatedly failed to materialize. A linear cost against a diminishing benefit gives an interior optimum.



(A) Low π: never stage (B) Intermediate π: one switch (C) High π: stage throughout

FIGURE 4. The principal's value function under technologically innocuous staging

Two numbers then fix the form: the optimum $\overline{u}$, and the switching point $\hat{u}_1 \equiv \varphi\hat{T}$ inherited from the benchmark, below which the direct route is used. The contract starts at $\overline{u}$ and runs down to zero, so it stages if and only if $\overline{u} > \hat{u}_1$. Figure 4 plots $\mathcal{V}$ for three project returns; a contract traverses its panel from right to left.

In panel (c) the project return exceeds π_S, so $\hat{u}_1 = 0$ and the staged region is the whole line. Panels (a) and (b) have a kink at $\hat{u}_1 > 0$ and differ only in which side of it $\overline{u}$ falls on: in (b) the contract begins in the staged region and crosses into the direct region as the deadline approaches, while in (a) the milestone is never sought. Because $\overline{u}$ rises with the project return and $\hat{u}_1$ falls with it, the two cross exactly once, at the threshold π_D of Appendix B.2. That gives parts (b) and (c) of Theorem 1.

The project is worth undertaking at all only if $\overline{u} > 0$, and that condition reads transparently at the deadline. As u approaches zero the direct route is in use and the reward approaches φ/λ_D, so the principal's flow profit over the final instant is positive if and only if $\pi > \pi_F \equiv 1 + \varphi/c$: she must cover the operating cost c and, because incentives bind, the shirking benefit φ for every instant the contract runs. Below π_F the project is abandoned, which is part (d).

REMARK 1 (*Incentive conditions always bind*). When $\pi < \pi_S$, the first-stage incentive condition in Green and Taylor (2015) goes slack near zero promised utility. With little time left, a first-stage delivery is worth little without its counterpart, so the value function is convex under binding incentives, and slack supplies the missing concavity. Here the direct route does that work: as the horizon shrinks the principal switches to the route needing a single success (the deadline effect), and the two strictly concave branches join smoothly at $\hat{u}_1$. Concavity comes from a second technology rather than slack, and incentive conditions bind everywhere.

REMARK 2 (*Randomization*). The optimal contract is deterministic, even against contracts that randomize over terms and conceal the realization from the agent. Concealment could in principle make a mixture of individually non-incentive-compatible contracts incentive compatible jointly. Here it cannot: the agent's deviations do not affect his beliefs about the realization, so incentive compatibility constrains only the conditional means of the terms, in the linear form of (IC_D) and (IC_S). Randomization is then a mean-preserving spread of promised utility, which concavity of $\mathcal{V}$ makes costly (Online Appendix OA.2).

5. Efficiency Loss from Staging

Sections 3 and 4 assumed staging to be technologically innocuous. We now relax that assumption and ask what survives. Write $\eta \equiv \lambda_S/\lambda_D - 1$, so that the rate restriction $\lambda_D < \lambda_S \leq 2\lambda_D$ maintained in Section 2 is exactly $\eta \in (0, 1]$. Staging is technologically innocuous at $\eta = 1$; when $\eta < 1$ the two stages take longer in expectation than a single direct attempt, and η measures the loss.

The loss is invisible near the deadline and decisive far from it. With no deadline at all the two routes deliver $\Pi - c/\lambda_D$ and $\Pi - 2c/\lambda_S$, and the second is smaller when $\eta < 1$. Efficiency therefore enters as a third force, and the three sort themselves along the horizon: close to the deadline the deadline effect favors the direct route, further out the milestone effect makes staging worth its cost, and far enough out the deadline effect has washed out while the efficiency loss has not, so the direct route returns. Route choice can therefore change twice, and in that order.

5.1. Route Choice under an Exogenous Deadline

DEFINITION 3. A policy is called a *two-switch policy* if there exist intermediate deadlines $0 \le S_1 \le S_2 \le T$ such that (i) the principal chooses the direct route up to S_1 ($a_t = 1$ for $t \le S_1$), (ii) she chooses the staged route on $(S_1, S_2]$ ($a_t = 0$), (iii) if the interim milestone is delivered before S_2, she works toward completion until the extended deadline $T + \Delta$, and (iv) if the milestone is not delivered by S_2, she switches back to the direct route until T.

This class nests every policy in Proposition 1 ($S_1 = 0$ gives a one-switch policy, $S_1 = 0$ and $S_2 = T$ the staged-only policy, $S_1 = S_2$ the direct-only policy). What is new is the case $0 < S_1 < S_2 < T$, in which staging occupies an interior window and the direct route is used at both ends.

The extension cutoff of Proposition 1 generalizes to

$$\overline{\Delta}(\eta, \pi) \equiv \begin{cases} \dfrac{1}{\lambda_S} \log\left[\dfrac{(1+\eta)\pi - 1}{\eta\pi - 1}\right], & \text{if } \eta\pi > 1, \\ \infty, & \text{otherwise,} \end{cases} \tag{7}$$

and we define a second cutoff,

$$\Delta^*(\eta,\pi) \equiv \begin{cases} \dfrac{1}{\lambda_S}\left\{\log\left[\dfrac{(1+\eta)\left((1+\eta)\pi-1\right)}{\eta(\pi-1)}\right]+\eta\log\left[\dfrac{1-\eta^2}{\eta^2(\pi-1)}\right]\right\}, & \text{if } \eta^2\pi>1, \\ \overline{\Delta}(\eta,\pi), & \text{otherwise,} \end{cases} \tag{8}$$

so that $\Delta^*(\eta,\pi) \leq \overline{\Delta}(\eta,\pi)$, with equality if and only if $\eta^2\pi \leq 1$.

Proposition 3 (Optimal policy under efficiency loss). *Suppose that staging is less efficient ($0 < \eta < 1$). There exist $\hat{T}$ and $\check{T}$ with $0 \leq \hat{T} \leq \check{T} \leq \infty$, both independent of the deadline T, such that for every deadline T the two-switch policy with $S_1 = \max\{T-\check{T},0\}$ and $S_2 = \max\{T-\hat{T},0\}$ is optimal, with $S_i = 0$ when the corresponding threshold is infinite. The extension determines the thresholds:*

(a) ***(Short extension)*** *if $\Delta \leq \Delta^*(\eta,\pi)$, then $\hat{T} = \check{T} = \infty$;*

(b) ***(Intermediate extension)*** *if $\Delta^*(\eta,\pi) < \Delta \leq \overline{\Delta}(\eta,\pi)$, then $0 \leq \hat{T} < \check{T} < \infty$;*

(c) ***(Long extension)*** *if $\Delta > \overline{\Delta}(\eta,\pi)$, then $\hat{T} = 0 < \check{T} < \infty$.*

Because $\hat{T}$ and $\check{T}$ are amounts of time remaining, the policy can be read off the clock: the direct route while fewer than $\hat{T}$ units remain, staging between $\hat{T}$ and $\check{T}$, and the direct route again beyond $\check{T}$. The two cutoffs order the regimes by how much staging is used: Δ^* is the extension staging must earn to be used at all, and $\overline{\Delta}$ the extension it must earn to be used right up to the deadline.[8] The thresholds depend on

8. Δ^* may be negative, in which case the project return is high enough that staging is used even with no extension and case (a) is empty. The two cutoffs coincide when $\eta \leq 1/\sqrt{\pi}$: the efficiency loss is then large relative to the return, and case (b) disappears.

Δ as well as on η and π; we suppress that dependence here and restore it in Section 5.2, where binding incentive conditions pin Δ at $1/\lambda_S$.

Proposition 3 departs from Proposition 1 in two ways. The first is the outer direct phase beyond $\check{T}$. It is absent under technologically innocuous staging because the two no-deadline surpluses then coincide; formally, $\check{T} \to \infty$ as $\eta \uparrow 1$, and Proposition 3 converges to Proposition 1. The second is case (c). When the extension is generous, staging is attractive near the deadline, since delivering the milestone buys Δ additional units of time exactly when time is scarce. The direct route is then used early rather than late, reversing the ordering of Proposition 1(b)—an ordering that cannot arise when $\eta = 1$.

5.2. The Optimal Contract under Efficiency Loss

The argument of Section 4 goes through unchanged, since none of it uses $\lambda_S = 2\lambda_D$: binding incentive conditions still deliver $u_t = \varphi(T - t)$ and a jump of φ/λ_S upon delivery, and the assigned route leaves promised utility untouched. Proposition 2 therefore holds verbatim, with $\mathcal{W}^*$ now supplied by Proposition 3. The contract again inherits the structure of the benchmark, which is now a two-switch policy. Measuring its thresholds in units of promised utility, at the extension that binding incentives purchase,

$$\hat{u}_1 \equiv \varphi \hat{T}\left(\frac{1}{\lambda_S}\right) \qquad \text{and} \qquad \hat{u}_2 \equiv \varphi \check{T}\left(\frac{1}{\lambda_S}\right), \tag{9}$$

the direct route is assigned when $u < \hat{u}_1$, the staged route when $\hat{u}_1 < u < \hat{u}_2$, and the direct route again when $u > \hat{u}_2$. The contract starts at $\overline{u}$ and runs down to zero, so its form is fixed by where $\overline{u}$ falls relative to the two thresholds.

TABLE 1. Forms of the optimal contract under efficiency loss

Condition	Route sequence	Contract
$\overline{u} = 0$	—	infeasible
$0 < \overline{u} \le \hat{u}_1$	direct	direct-only
$\hat{u}_1 = 0,\ \overline{u} \le \hat{u}_2$	staged	staged-only
$\hat{u}_1 > 0,\ \hat{u}_1 < \overline{u} \le \hat{u}_2$	staged → direct	one-switch
$\hat{u}_1 = 0,\ \overline{u} > \hat{u}_2$	direct → staged	reversed one-switch
$\hat{u}_1 > 0,\ \overline{u} > \hat{u}_2$	direct → staged → direct	two-switch

THEOREM 2 (Optimal contract under efficiency loss). *Suppose staging is less efficient ($0 < \eta < 1$), and let $\hat{u}_1$ and $\hat{u}_2$ be as in (9). Then:*

(a) ***(At most two switches)*** *the optimal contract changes the assigned route at most twice, at every efficiency level $\eta \in (0, 1)$;*

(b) ***(Taxonomy)*** *it takes one of the six forms in Table 1, and which one obtains is determined by the position of $\overline{u}$ relative to $\hat{u}_1$ and $\hat{u}_2$.*

In every case the reward schedules are those of Definition 2 and the deadline is extended by $1/\lambda_S$ upon delivery of the interim milestone.

Part (a) is inherited from Proposition 3, and the argument behind it is short. At any point where the principal is indifferent between two routes, the direction in which the staged advantage crosses zero depends only on time remaining, not on how the project is run afterward, and that direction can change only once, from toward staging to away from it (Lemma C.1 and Corollary C.2 in Appendix C, which proves the theorem). The staged advantage can therefore turn positive at most once and negative at most once, which caps the number of switches at two for every $\eta \in (0, 1)$; at $\eta = 1$ the direction never changes, which is why Theorem 1 has at most one switch.

Locating the thresholds. Theorem 2 classifies the possible contracts but leaves $\hat{u}_1$, $\hat{u}_2$ and $\overline{u}$ implicit. At the two extremes of the efficiency loss they can be located

in terms of the primitives, and each mechanism appears on its own. Define

$$\overline{\eta} \equiv \max\left\{\sqrt{\frac{c}{c+\varphi}},\ \frac{1}{e-1}\right\} \qquad \text{and} \qquad \underline{\eta} \equiv \min\left\{\frac{c}{c+\varphi},\ \frac{1}{e-1}\right\}.$$

We characterize the optimal contract at the two extremes, small loss ($\eta > \overline{\eta}$) and large loss ($\eta < \underline{\eta}$). In the intermediate band, Theorem 2 still applies, but a full characterization requires many subcases; we instead give a numerical example of a form that neither extreme produces.

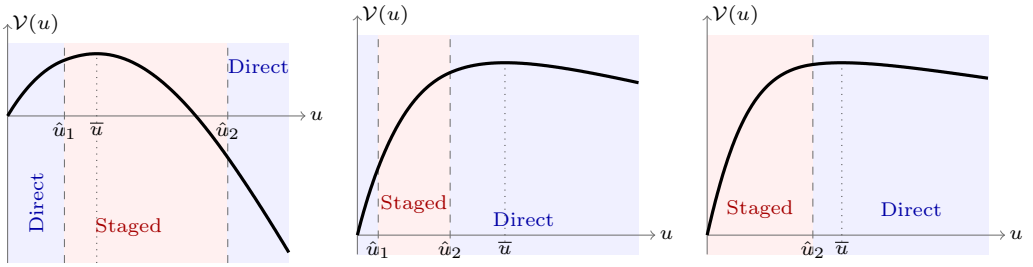
Small efficiency loss. When the loss is small the outer threshold is never reached: $\hat{u}_2$ exceeds $\overline{u}$ at every project return (Lemma OA.11), so the principal never buys enough time for the efficiency effect to bite, and Section 4 applies unchanged.

PROPOSITION 4 (Small efficiency loss). *Suppose $\eta \in (\overline{\eta}, 1)$. Then Theorem 1 holds with π_S and π_D replaced by thresholds $\tilde{\pi}_S(\eta) > \tilde{\pi}_D(\eta) > \pi_F$.*

Large efficiency loss. When the loss is large the direct route is preferred not only near the deadline but also at the beginning of a long project, so for most project returns the contract never stages at all. Only when the return is high enough to sustain a long deadline does staging appear, and then only in an interior window.

PROPOSITION 5 (Large efficiency loss). *Suppose $\eta < \underline{\eta}$. There exists a threshold $\tilde{\pi}_M(\eta) > \pi_F$ such that a two-switch contract is optimal when $\pi > \tilde{\pi}_M(\eta)$, and a direct-only contract is optimal when $\tilde{\pi}_M(\eta) > \pi > \pi_F$.*

Intermediate efficiency loss. The reversed one-switch contract arises only in the intermediate band. It needs $\hat{u}_1 = 0$ and $\overline{u} > \hat{u}_2$ at once, and neither extreme permits



(A) Small loss: one switch ($\lambda_S = 3.8$, $\varphi = 0.5$)

(B) Large loss: two switches ($\lambda_S = 3.1$, $\varphi = 0.5$)

(C) Intermediate loss: reversed ($\lambda_S = 3.4$, $\varphi = 0.3$, $\pi = 10$)

FIGURE 5. Value functions under efficiency loss ($\lambda_D = 2$, $c = 1$)

the combination: a small loss keeps $\overline{u}$ below $\hat{u}_2$, and a large one keeps $\hat{u}_1$ strictly positive.

Figure 5 shows one instance from each regime. In panel (a) the outer direct region is visible beyond $\hat{u}_2$, but $\overline{u}$ lies short of it and the contract is the one-switch contract of Theorem 1. In panel (b) the principal buys enough time that $\overline{u}$ lies beyond $\hat{u}_2$: the contract begins in the outer direct region, crosses into staging, and crosses back. In panel (c) the inner direct region is absent and $\overline{u}$ lies beyond $\hat{u}_2$, so the contract begins on the direct route and turns to staging as the deadline approaches.

6. Discussion

6.1. Contractual Forms

The optimal contract can be read in three ways closer to observed contracts. Each applies to every form in Table 1, since all share the reward schedules of Definition 2.

A buyout of the right to divert. Splitting (IC_D) at equality,

$$R_t^D = \underbrace{\varphi(T-t)}_{\text{remaining diversion}} + \underbrace{\frac{\varphi}{\lambda_D}}_{\text{completion bonus}},$$

the agent holds a right to divert at rate φ until the deadline, and completing the project surrenders it. The principal must therefore buy that right out at face value and add a fixed bonus worth one expected attempt's diversion. The post-milestone schedule is the same two components measured against the extended deadline, $R_t^S = \varphi(T + 1/\lambda_S - t) + \varphi/\lambda_S$, with the bonus at the current route's rate.

The buyout is profitable because completion also ends the operating cost. Against a counterfactual in which the agent diverts until T, completion at t delivers the principal $\Pi - \varphi/\lambda_D + (c - \varphi)(T - t)$: she purchases the remaining time at φ and saves c on it. This is why she prefers early completion even though neither party discounts, and it is where $\varphi \leq c$ does its work: were diversion efficient, the buyout would cost more than the burn it saves.

The milestone is rewarded in time rather than in cash: the extension $1/\lambda_S$ raises the promise by the φ/λ_S that (IC_S) requires. The completion payment changes only in composition: $R_t^S - R_t^D = \varphi(1 - \eta)/\lambda_S$, zero under technologically innocuous staging.

Probation and up-or-out. Read as an employment arrangement, each form is a review window: verifiable progress is asked for only while the time remaining lies between $\hat{u}_1/\varphi$ and $\hat{u}_2/\varphi$. The one-switch contract is a probation period ending in an up-or-out decision at S, much like a tenure clock. An agent who shows progress has his clock extended; one who does not is converted rather than terminated, moved to the direct route and given until T. The two-switch contract adds an initial phase without review, resembling the discretionary period of Feng and Zhang (2024); there it arises from the cost of deterring false progress reports, here from the efficiency loss. In the reversed one-switch contract review comes last, with no conversion stage.

Runway and milestone-contingent funding. Read as a financing arrangement, the buyout is a runway: the principal advances an account $M \equiv \varphi T + \varphi/\lambda_D$ that depletes at the diversion rate, $R_t^D = M - \varphi t$, and abandons the project when it falls to the bonus φ/λ_D. At $\eta = 1$ the milestone adds no cash; it lowers the exhaustion threshold to φ/λ_S and so lengthens the runway. The route rule becomes a rule on the balance: fund unconditionally while the runway is long, tie its extension to a milestone once it enters the band, and fund a final attempt near exhaustion.

This is how staged financing is written. In venture capital, the release of committed funds is routinely conditioned on interim performance, and withholding funds shortens the time until the venture must raise again (Kaplan and Strömberg 2003); a milestone-contingent release of capital is thus a milestone-contingent extension of the horizon. Both forms are used: in most first rounds the full commitment is released at closing, while in the rest roughly half is held back against milestones. Our results say the choice should turn on the remaining runway.

6.2. What the Principal Cannot Observe

Throughout the analysis the principal observes the route she assigns and each output (the milestone or the finished project) as soon as it arrives. We first remove the milestone's observability altogether, to isolate what it contributes, and then show that the principal loses nothing when the agent privately observes his outputs or chooses his route.

(i) The milestone is observed by neither party. Neither party learns when the first stage is complete. Nothing can be conditioned on the milestone, so the contract is a single deadline with a declining reward, what Green and Taylor (2016)

call a *simple contract*, and route choice is the case $\Delta = 0$ of Section 3. By Lemma 1, under technologically innocuous staging the direct route is preferred at short horizons and the staged route at long ones: the ordering is a property of the completion-time distributions (Figure 2) and owes nothing to monitoring. What monitoring adds is the extension $1/\lambda_S$, and with it the range of returns over which staging is used; without it, less efficient staging may be used at no horizon (Proposition 3(a) at $\Delta = 0$).

(ii) Outputs are observed only by the agent. Suppose next that only the agent observes his outputs and chooses when, if ever, to present them, though he cannot fabricate one (Remark 3); the route remains observable. Two deviations are new: delaying a delivery and never revealing a milestone.

PROPOSITION 6 (Concealed outputs). *Suppose the agent privately observes his outputs and chooses when, if ever, to present them, but cannot fabricate one. Then the contracts of Theorem 2, with the extension granted upon presentation of the milestone, remain incentive compatible, and the principal's value function is unchanged.*

The proof is in Appendix D.1. Delay is free because binding incentive conditions make the promise run down at exactly the rate φ that shirking earns while the agent waits, so the value of an output does not depend on when it is presented. Never revealing a milestone is worse because an unrevealed milestone is useful only as a step toward completion, on a schedule whose premium is calibrated to a single direct arrival; two stages take at least as long on average and may miss the deadline.

(iii) The route is also unobservable. Suppose in addition that the principal cannot verify the route. He could then leave the recommended route and hold its

output until the contract asks for it: finish the project now and present it when the contract turns to the direct route, or bank a milestone until the staged phase opens.

Neither pays, and the reason is the property that organizes the whole analysis. Binding incentive conditions price a delivery on route j at $u_t + \varphi/\lambda_j$, so the flow value of working that route is $\lambda_j \cdot \varphi/\lambda_j = \varphi$, whatever j is; and shirking pays φ by construction. The premium and arrival rate offset, leaving the agent indifferent.

PROPOSITION 7 (Route choice is incentive-irrelevant). *Fix any contract of Theorem 2, and suppose that, in addition to the assumptions of Proposition 6, the agent privately chooses which route to work at each date. Consider a date t at which route j is recommended, and let $k \neq j$.*

(a) *If the contract calls for route k's output at some later date, the agent is indifferent among working route j, working route k and withholding its output until it is called for, and shirking.*

(b) *If it never does, working route k is strictly worse than working route j.*

In neither case is a deviation profitable.

The proof is in Appendix D.2. Part (a) says that the switching structure is not vulnerable to being anticipated: producing the wrong output early and banking it is exactly break-even. Part (b) rests on the termination rule of Section 2, which makes output the contract never calls for worthless on its own; the agent can then profit from staged output only by completing the project himself, and for the reason given after Proposition 6 that detour earns less than the assigned route.

REMARK 3 (*Fabrication*). Propositions 6 and 7 rely on the agent's inability to present a milestone he does not have. If he can claim one falsely, the contracts above

are no longer incentive compatible: an agent without a milestone could report one and then shirk, and since promised utility declines at exactly the rate φ that shirking earns, the false report would collect the full jump φ/λ_S at no cost. Deterring it requires a no-false-progress constraint of the kind studied by Green and Taylor (2016), which limits the extension the principal can grant and breaks the reduction (6). As Feng and Zhang (2024) emphasize, whether the agent can fabricate news is also what determines whether the optimal contract must randomize: in their model, as in Green and Taylor (2016), deterring false reports calls for random termination, whereas ours, without fabrication, remain deterministic. That problem is left to future work.

6.3. Scope of the Assumptions

Three assumptions shape how the results are stated more than what drives them: Poisson arrivals, the rate restriction $\lambda_S \leq 2\lambda_D$, and patience. We take them in turn.

The arrival technology. Memorylessness makes promised utility a sufficient state; it does not drive the choice of route. The deadline effect rests on the staged completion time being less dispersed than the direct one with the same mean, and that property is general. If each stage draws from the direct distribution scaled by one half, the staged time is the average of two independent draws, a mean-preserving contraction of one. What is special to the exponential case is that the two completion probabilities cross exactly once (Lemma 1), which orders the routes cleanly along the horizon; other distributions may cross more than once, a case we do not pursue. The milestone effect requires only that the interim delivery be verifiable. Splitting the project into more than two stages strengthens the deadline effect further (Online Appendix OA.5).

Efficiency gains from staging. The restriction $\lambda_S \leq 2\lambda_D$ keeps the two routes in competition; it is not where the analysis stops. If $\lambda_S > 2\lambda_D$, staging completes the project sooner in expectation and is preferred far from the deadline, but the direct route need not disappear. The comparison at the deadline is the one in Lemma OA.4, which carries over: with the extension at $1/\lambda_S$, the staged route wins the last instant only when $\pi \geq \tilde{\pi}_S(\eta) = (e-1)/(\eta(e-1)-1)$. Since every feasible project has $\pi > 1$, the direct route loses the last instant for every project return only once $\tilde{\pi}_S(\eta) \leq 1$, that is, once $\eta \geq e/(e-1) \approx 1.58$, or $\lambda_S \geq \lambda_D(2e-1)/(e-1) \approx 2.58\,\lambda_D$. Below that bound, efficient staging coexists with the deadline effect, and the direct route is still used near the deadline whenever $\pi < \tilde{\pi}_S(\eta)$. Above it, the problem approaches the single-route analysis of Green and Taylor (2015).

Discounting. With a common discount rate $r > 0$ the incentive conditions are unchanged, but promised utility no longer runs down at a constant rate, so the extension that a milestone buys depends on the state and the reduction to the observable-effort benchmark fails. We solve the problem numerically in Online Appendix OA.4. Discounting favors the direct route far from the deadline: the discount factor is convex in the completion date and so, at equal expected completion times, rewards the more dispersed direct route. Impatience therefore acts on the margin on which an efficiency loss acts, and it produces an outer direct region even at $\lambda_S = 2\lambda_D$. With enough impatience the two-switch and reversed one-switch contracts appear; in every case we compute, the contract takes one of the six forms in Table 1.

7. Conclusion

Project organization is an instrument of dynamic incentive provision. When a project remains unfinished, the principal must decide how long to continue supporting it and what accomplishments to require along the way. Our analysis shows how these decisions interact: the principal jointly chooses the production route, the rewards for success, and the opportunities to earn additional time. The deadline is an endogenous part of this arrangement, determining how long the agent can continue without delivering verifiable progress.

The optimal contract can move through three distinct phases. Initially, the principal may give the agent discretion to pursue the completed product directly, avoiding the efficiency cost of producing an intermediate deliverable. If success does not materialize, she may require a verifiable milestone that earns additional time for completion. Staging is then valuable because it creates an opportunity to reward productive effort before the final product is delivered. If the milestone also fails to arrive, the principal may close this opportunity and give the agent one final chance to complete the project directly. This last phase concentrates effort on a route requiring only one breakthrough while ruling out any further extension.

These changes in contractual demands sustain effort throughout the relationship. The agent does not shirk in equilibrium, and prolonged noncompletion can occur despite diligent work. Nevertheless, the contract must ensure that shirking would not be profitable along precisely these unsuccessful histories. In the absence of a breakthrough, the agent's continuation payoff declines; a verifiable milestone replenishes it by earning an extension. The principal thus responds to missing progress in a way that disciplines the temptation to shirk, even when that temptation is never

acted upon. The eventual insistence on final delivery is part of the incentive scheme that makes continued effort optimal from the outset.

The resulting demand for intermediate evidence can be strongest in the middle of a project. Early discretion preserves productive efficiency, milestones subsequently make continued support contingent on demonstrable progress, and a final delivery requirement brings the relationship to a close.

This perspective also suggests a broader agenda for milestone design: Beyond choosing between alternative production routes, a principal may determine how many intermediate accomplishments to require, how demanding they should be, and how much additional time each should earn. These choices jointly shape the technology of production and the incentives to exert effort. The central lesson is that deadlines, milestones, and production methods are not independent features of a project: they are complementary instruments of dynamic project management.

Appendix A: Proofs for Section 3

A.1. Expected Surpluses of Benchmark Policies

In this section, we provide formal representations of the expected social surpluses under the direct-only and staged-only policies.

Suppose that the principal employs the direct-only policy. Recall that the completion probability $\mathcal{P}^D(T)$ is derived in (2). Next, the expected duration of project completion is

$$\mathcal{D}^D(T) \equiv \int_0^T \tau \cdot \lambda_D \cdot e^{-\lambda_D \tau} d\tau + T \cdot e^{-\lambda_D T} = \frac{1}{\lambda_D}(1 - e^{-\lambda_D T}). \tag{A.1}$$

Then, the expected social surplus of the direct-only policy is

$$\mathcal{W}^D(T) \equiv \Pi \cdot \mathcal{P}^D(T) - c \cdot \mathcal{D}^D(T) = \left(\Pi - \frac{c}{\lambda_D}\right) \cdot (1 - e^{-\lambda_D T}). \tag{A.2}$$

Now suppose that the principal employs the staged-only policy. Recall that the completion probability $\mathcal{P}^S(T, \Delta)$ is derived in (3). Conditional on milestone delivery at τ_m, the expected duration is

$$\mathcal{D}^1(T, \Delta, \tau_m) \equiv \int_{\tau_m}^{T+\Delta} \tau \cdot \lambda_S e^{-\lambda_S(\tau - \tau_m)} d\tau + (T + \Delta) \cdot e^{-\lambda_S(T+\Delta-\tau_m)}.$$

Then, the total expected duration of the project can be derived as follows:

$$\begin{aligned} \mathcal{D}^S(T, \Delta) &\equiv \int_0^T \mathcal{D}^1(T, \Delta, \tau_m) \cdot \lambda_S e^{-\lambda_S \tau_m} d\tau_m + T \cdot e^{-\lambda_S T} \\ &= \frac{2}{\lambda_S}(1 - e^{-\lambda_S T}) - T \cdot e^{-\lambda_S(T+\Delta)}. \end{aligned} \tag{A.3}$$

The expected social surplus of the staged-only policy is thus

$$\begin{aligned} \mathcal{W}^S(T, \Delta) &\equiv \Pi \cdot \mathcal{P}^S(T, \Delta) - c \cdot \mathcal{D}^S(T, \Delta) \\ &= \left(\Pi - \frac{2c}{\lambda_S}\right) \cdot (1 - e^{-\lambda_S T}) - \lambda_S \left(\Pi - \frac{c}{\lambda_S}\right) \cdot T \cdot e^{-\lambda_S(T+\Delta)}. \end{aligned} \tag{A.4}$$

A.2. Proof of Lemma 1

Proof of Lemma 1. Observe that $\mathcal{P}^D(T, 0) \geq \mathcal{P}^S(T, 0)$ is equivalent to $(1 + \lambda_S T) \geq e^{(\lambda_S - \lambda_D)T}$. Equality holds at $T = 0$. The left-hand side linearly increases with slope λ_S, while the right-hand side exponentially increases with an initial slope at $T = 0$ of

$\lambda_S - \lambda_D \in (0, \lambda_S)$. Therefore, for sufficiently small T, $\mathcal{P}^D(T,0) > \mathcal{P}^S(T,0)$, but there exists a unique cutoff $\check{T} > 0$ equating the two sides. Thus, $\mathcal{P}^D(T,0) > \mathcal{P}^S(T,0)$ for all $T < \check{T}$, and $\mathcal{P}^D(T,0) < \mathcal{P}^S(T,0)$ for all $T > \check{T}$. □

A.3. Proof of Proposition 1

Value functions and the HJB equation. Let W_x^1 denote the expected surplus once the interim milestone has been delivered and x units of time remain. By the same steps that deliver (A.2),

$$W_x^1 \equiv \left(\Pi - \frac{c}{\lambda_S}\right) \cdot (1 - e^{-\lambda_S x}). \tag{A.5}$$

If the milestone arrives with z units of time remaining, the deadline extends by Δ, so the continuation surplus is $W_{z+\Delta}^1$.

Before the milestone arrives, the optimal expected surplus $W_z^{0,\Delta}$ with z units of time remaining satisfies

$$\dot{W}_z^{0,\Delta} = \max_{a_z \in \{0,1\}} \left[\lambda_D a_z \cdot (\Pi - W_z^{0,\Delta}) + \lambda_S (1 - a_z) \cdot (W_{z+\Delta}^1 - W_z^{0,\Delta}) - c\right], \quad (\text{HJB}_W)$$

with $W_0^{0,\Delta} = 0$, since the project is abandoned at the deadline. By standard verification arguments (e.g., Proposition 3.2.1 in Bertsekas 1995), if $w : [0,T] \to \mathbb{R}$ is differentiable, satisfies (HJB_W), and has $w(0) = 0$, then $w(z) = W_z^{0,\Delta}$.

The staged advantage. For a continuation value W, define the net flow advantage of the staged route over the direct route,

$$\Lambda(z; W) \equiv \lambda_S \left(W_{z+\Delta}^1 - W\right) - \lambda_D \left(\Pi - W\right) = \lambda_S W_{z+\Delta}^1 - \lambda_D \Pi - (\lambda_S - \lambda_D) W. \tag{A.6}$$

The staged route attains the maximum in (HJB$_W$) at z if and only if $\Lambda(z; W_z^{0,\Delta}) \geq 0$. Two observations organize the proof. First, because $\lambda_S > \lambda_D$, the map $W \mapsto \Lambda(z; W)$ is strictly decreasing: a higher continuation value makes the direct route relatively more attractive, since the direct route delivers Π outright whereas the staged route only advances the project by one stage. Second, and consequently, Λ must be evaluated at the value function that actually prevails at z. Substituting the direct-only value $\mathcal{W}^D(z)$ is legitimate only where the direct route is in fact used, and it *overstates* the staged advantage elsewhere. Lemma A.1 locates the switch point using $\mathcal{W}^D$, and Lemma A.2 then propagates the comparison beyond it using the correct value function.

The one-switch value function. Fix a switching threshold $Z \in [0, T]$ in remaining time: the direct route is used when $z < Z$ and the staged route when $z \geq Z$, which is the one-switch policy with intermediate deadline $S = T - Z$. Let $\mathcal{W}_Z^{SD}(z, \Delta)$ denote its expected surplus with z units of time remaining. It satisfies

$$\dot{\mathcal{W}}_Z^{SD}(z, \Delta) = \lambda_D \cdot \big(\Pi - \mathcal{W}_Z^{SD}(z, \Delta)\big) - c \qquad \text{for } z < Z, \tag{A.7}$$

$$\dot{\mathcal{W}}_Z^{SD}(z, \Delta) = \lambda_S \cdot \big(W_{z+\Delta}^1 - \mathcal{W}_Z^{SD}(z, \Delta)\big) - c \qquad \text{for } z \geq Z, \tag{A.8}$$

and solving with $\mathcal{W}_Z^{SD}(0, \Delta) = 0$ gives

$$\mathcal{W}_Z^{SD}(z, \Delta) = \begin{cases} \mathcal{W}^D(z), & \text{if } z \leq Z, \\ \left(\Pi - \dfrac{2c}{\lambda_S}\right) \cdot \left(1 - e^{-\lambda_S (z - Z)}\right) + \mathcal{W}^D(Z) \cdot e^{-\lambda_S (z - Z)} & \\ \qquad - \lambda_S \left(\Pi - \dfrac{c}{\lambda_S}\right) \cdot (z - Z) \cdot e^{-\lambda_S (z + \Delta)}, & \text{if } z > Z. \end{cases} \tag{A.9}$$

The two benchmark policies are the endpoints of this family: $\mathcal{W}_0^{SD}(z,\Delta) = \mathcal{W}^S(z,\Delta)$ is the staged-only surplus and $\mathcal{W}_T^{SD}(z,\Delta) = \mathcal{W}^D(z)$ is the direct-only surplus. We prove the proposition by exhibiting a $Z \in [0,T]$ for which $\mathcal{W}_Z^{SD}(\cdot,\Delta)$ solves (HJB$_W$).

Optimal route at the deadline. Since $W_0^{0,\Delta} = 0$, the staged route is preferred at the deadline if and only if $\Lambda(0;0) \geq 0$, that is,

$$\lambda_D \Pi \leq (\lambda_S \Pi - c) \cdot \left(1 - e^{-\lambda_S \Delta}\right). \tag{A.10}$$

Under $\lambda_S = 2\lambda_D$, dividing by c and rearranging shows that (A.10) holds if and only if $\Delta \geq \bar{\Delta}$.

Lemma A.1 (Locating the switch point). *Suppose $\lambda_S = 2\lambda_D$ and $\Delta < \bar{\Delta}$, and set $\hat{T} \equiv 2(\bar{\Delta} - \Delta)$. Then $\hat{T} > 0$ and (i) $\Lambda\left(z; \mathcal{W}^D(z)\right) < 0$ for all $z < \hat{T}$; and (ii) $\Lambda\left(\hat{T}; \mathcal{W}^D(\hat{T})\right) = 0$.*

Proof of Lemma A.1. Substituting (A.5) and $\mathcal{W}^D(z) = (\Pi - c/\lambda_D)(1 - e^{-\lambda_D z})$ into (A.6) and writing $x \equiv e^{-\lambda_D z} \in (0,1]$,

$$\begin{aligned}\Lambda_D(x) \equiv \Lambda\left(z; \mathcal{W}^D(z)\right) = (\lambda_S \Pi - c)\left(1 - e^{-\lambda_S \Delta} x^{\lambda_S/\lambda_D}\right) - \lambda_D \Pi \\ - (\lambda_S - \lambda_D)\left(\Pi - \frac{c}{\lambda_D}\right)(1 - x).\end{aligned} \tag{A.11}$$

Under $\lambda_S = 2\lambda_D$ the exponent λ_S/λ_D equals 2 and the constant terms cancel, so (A.11) reduces to the quadratic

$$\tilde{\Lambda}(x) \equiv (\lambda_D \Pi - c) \cdot x - (\lambda_S \Pi - c) \cdot e^{-\lambda_S \Delta} \cdot x^2 = x\left[(\lambda_D \Pi - c) - (\lambda_S \Pi - c) e^{-\lambda_S \Delta} x\right]. \tag{A.12}$$

Since $\pi > 1$ by (1), both $\lambda_D \Pi - c > 0$ and $\lambda_S \Pi - c > 0$, so $\tilde{\Lambda}$ is strictly concave with roots 0 and

$$\hat{x} \equiv \frac{\lambda_D \Pi - c}{\lambda_S \Pi - c} \cdot e^{\lambda_S \Delta}. \tag{A.13}$$

Equality in (A.10) defines $\bar{\Delta}$, and under $\lambda_S = 2\lambda_D$ it reads $e^{-\lambda_S \bar{\Delta}} = (\lambda_D \Pi - c)/(\lambda_S \Pi - c)$, which is precisely the constant multiplying $e^{\lambda_S \Delta}$ in (A.13). Hence $\hat{x} = e^{-\lambda_S (\bar{\Delta} - \Delta)}$, so $\hat{x} < 1$ precisely when $\Delta < \bar{\Delta}$. Thus $\tilde{\Lambda}(\hat{x}) = 0$ and $\tilde{\Lambda}(x) < 0$ for all $\hat{x} < x \leq 1$.

Taking logarithms in $\hat{x} = e^{-\lambda_S (\bar{\Delta} - \Delta)}$ and using $\lambda_S = 2\lambda_D$,

$$\hat{T} \equiv -\frac{\log(\hat{x})}{\lambda_D} = \frac{\lambda_S}{\lambda_D} \left(\bar{\Delta} - \Delta\right) = 2 \left(\bar{\Delta} - \Delta\right) > 0.$$

Because $x = e^{-\lambda_D z}$ is strictly decreasing in z, $z < \hat{T}$ is equivalent to $x > \hat{x}$, which yields (i); and $\tilde{\Lambda}(\hat{x}) = 0$ yields (ii). □

Condition (ii) is precisely the indifference condition delivered by smooth pasting: at $\hat{T}$ the two branches of (HJB_W) have equal flow value, so the candidate value function is continuously differentiable there.

Lemma A.2 (Propagating past the switch point). *Suppose* $\lambda_S = 2\lambda_D$ *and* $\Lambda\left(Z; \mathcal{W}^D(Z)\right) \geq 0$ *for some* $Z \geq 0$. *Then* $\Lambda\left(z; \mathcal{W}_Z^{SD}(z, \Delta)\right) \geq 0$ *for all* $z > Z$.

Proof of Lemma A.2. For $z > Z$ write $y \equiv e^{-\lambda_S (z - Z)} \in (0, 1)$ and $\kappa \equiv (\Pi - c/\lambda_S)\, e^{-\lambda_S (Z + \Delta)} > 0$. Rearranging (A.9) and (A.5) gives

$$\mathcal{W}_Z^{SD}(z, \Delta) = \mathcal{W}^D(Z) + \left(\Pi - \frac{2c}{\lambda_S} - \mathcal{W}^D(Z)\right)(1 - y) + \kappa \log(y)\, y,$$
$$W_{z+\Delta}^1 = W_{Z+\Delta}^1 + \left(\Pi - \frac{c}{\lambda_S} - W_{Z+\Delta}^1\right)(1 - y).$$

Substituting these into (A.6) and collecting terms, $\Lambda\left(z;\mathcal{W}_Z^{SD}(z,\Delta)\right)=\Lambda_{SD}(y)$, where

$$\Lambda_{SD}(y)\equiv\Lambda\left(Z;\mathcal{W}^D(Z)\right)+h\cdot(1-y)-(\lambda_S-\lambda_D)\,\kappa\log(y)\,y, \tag{A.14}$$
$$h\equiv\lambda_S\left(\Pi-\frac{c}{\lambda_S}-W^1_{Z+\Delta}\right)-(\lambda_S-\lambda_D)\left(\Pi-\frac{2c}{\lambda_S}-\mathcal{W}^D(Z)\right).$$

Note that $\mathcal{W}_Z^{SD}$ enters (A.6) with coefficient $-(\lambda_S-\lambda_D)$, which is what the final term of (A.14) records. Differentiating twice, $\Lambda''_{SD}(y)=-(\lambda_S-\lambda_D)\kappa/y<0$, so Λ_{SD} is strictly concave on $(0,1)$. At the upper endpoint, $\Lambda_{SD}(1)=\Lambda\left(Z;\mathcal{W}^D(Z)\right)\geq 0$ by hypothesis. At the lower endpoint, $\log(y)\,y\to 0$ and

$$\lim_{y\to 0}\Lambda_{SD}(y)=\Lambda\left(Z;\mathcal{W}^D(Z)\right)+h=\frac{\lambda_S-2\lambda_D}{\lambda_S}\,c=0, \tag{A.15}$$

where the final equality uses $\lambda_S=2\lambda_D$. Strict concavity together with $\Lambda_{SD}(1)\geq 0$ and $\lim_{y\to 0}\Lambda_{SD}(y)=0$ implies $\Lambda_{SD}(y)\geq 0$ for all $y\in(0,1)$, which is the claim. □

Equation (A.15) is where technologically innocuous staging does its work: the two endpoint values of Λ_{SD} are non-negative only because the efficiency wedge $\lambda_S-2\lambda_D$ vanishes.

Proof of Proposition 1. **(a) Long extension ($\Delta\geq\bar{\Delta}$).** Since $\mathcal{W}^D(0)=0$, (A.10) gives $\Lambda\left(0;\mathcal{W}^D(0)\right)\geq 0$. Applying Lemma A.2 with $Z=0$ yields $\Lambda\left(z;\mathcal{W}^S(z,\Delta)\right)\geq 0$ for all $z>0$, so the staged branch attains the maximum in (HJB_W) at every z. By (A.8), $\mathcal{W}^S(\cdot,\Delta)=\mathcal{W}_0^{SD}(\cdot,\Delta)$ therefore solves (HJB_W) on $\mathbb{R}_+$ with the correct boundary condition, and the staged-only policy is optimal.

(b) Short extension ($\Delta<\bar{\Delta}$). Let $\hat{T}$ be as in Lemma A.1.

If $T < \hat{T}$, then Lemma A.1(i) gives $\Lambda\left(z; \mathcal{W}^D(z)\right) < 0$ for all $z \in [0, T]$, so the direct branch attains the maximum throughout. By (A.7), $\mathcal{W}^D$ solves (HJB$_W$) on $[0, T]$ and the direct-only policy is optimal.

If $T \geq \hat{T}$, consider $\mathcal{W}^{SD}_{\hat{T}}$. For $z \leq \hat{T}$ it coincides with $\mathcal{W}^D(z)$, and Lemma A.1(i) shows the direct branch attains the maximum there. At $z = \hat{T}$, Lemma A.1(ii) delivers both value matching and smooth pasting, so $\mathcal{W}^{SD}_{\hat{T}}$ is continuously differentiable at $\hat{T}$. That same condition supplies the hypothesis of Lemma A.2 at $Z = \hat{T}$, whence $\Lambda\left(z; \mathcal{W}^{SD}_{\hat{T}}(z, \Delta)\right) \geq 0$ and the staged branch attains the maximum for all $z > \hat{T}$. Hence $\mathcal{W}^{SD}_{\hat{T}}$ is differentiable, satisfies (HJB$_W$) on $[0, T]$, and vanishes at $z = 0$; by the verification argument it is the value function, and the one-switch policy with intermediate deadline $S = T - \hat{T}$ is optimal. □

Appendix B: Proofs for Section 4

This appendix proves Proposition 2 and Theorem 1. The recursive formulation on which both rest is in Online Appendix OA.1: it computes the two parties' expected payoffs, shows that the pointwise condition (IC) suffices for incentive compatibility (Lemma OA.1), derives the Hamilton–Jacobi–Bellman equation (HJB$_V$) with its operator $\mathcal{J}$ and a verification lemma (Lemma OA.3), and establishes implementation, concavity and the absence of profitable slack (Lemmas OA.4, OA.5 and OA.6). In what follows, $V^D(u) \equiv \mathcal{W}^D(u/\varphi) - u$ and $V^{SD}(u \mid u_1) \equiv \mathcal{W}^{SD}_{u_1/\varphi}(u/\varphi, 1/\lambda_S) - u$ are the principal's payoffs from the direct-only and one-switch contracts that deliver the agent u under binding incentive conditions, and $\mathcal{J}$ is the objective maximized in (HJB$_V$).

B.1. Proof of Proposition 2

Proof of Proposition 2. We begin by showing that $\mathcal{J}$ becomes zero when the value function $\mathcal{V}$ is utilized alongside contractual terms with binding ICs.

When $u \geq \hat{u}_1$, $\mathcal{V}(u) = V^{SD}(u|\hat{u}_1)$. Then, by (OA.1.5), $\mathcal{J}(V^{SD}(u|\hat{u}_1), \cdot, u + \varphi/\lambda_S, 0) = 0$. Likewise, when $u < \hat{u}_1$, $\mathcal{V}(u) = V^D(u)$, and by (OA.1.4), $\mathcal{J}(V^D(u), u + \varphi/\lambda_D, \cdot, 1) = 0$.

Lemma OA.6 shows that $\mathcal{J}$ is nonpositive for any feasible deviations. Therefore, $\mathcal{V}$ solves (HJB_V) subject to (IC).

The concavity of $\mathcal{V}$ is shown in Lemma OA.5 (c). If $\hat{u}_1 = 0$, $\mathcal{V}(u) = V^{SD}(u|0)$ is differentiable for all $u \geq 0$. If $\hat{u}_1 > 0$, $V^D(u)$ is differentiable for all $u < \hat{u}_1$ and $V^{SD}(u|\hat{u}_1)$ is differentiable for all $u > \hat{u}_1$. By Lemma OA.5 (a), $\mathcal{V}$ is differentiable at $\hat{u}_1$ as well. Also note that $\mathcal{V}(0) = 0$. Therefore, by Lemma OA.3, for any incentive compatible contract promising the agent u units of utility, the principal's expected payoff is lower than or equal to $\mathcal{V}(u)$.

Last, by Lemma OA.4, there exists a contract implementing $(\mathcal{V}(u), u)$. Therefore, $\mathcal{V}(u)$ is the most the principal can earn from an incentive-compatible contract that delivers the agent exactly u. □

B.2. Thresholds and Proof of Theorem 1

In this section, we pin down the thresholds π_D and π_S, establish their ordering, and complete the proof of Theorem 1.

First, recall that π_S is the solution of $1/\lambda_S = \overline{\Delta}$. This gives us

$$\frac{1}{\lambda_S} = \frac{1}{\lambda_S} \log\left[\frac{2\pi - 1}{\pi - 1}\right] \qquad \Leftrightarrow \qquad \pi_S = \frac{e-1}{e-2} \approx 2.392. \tag{B.1}$$

Also recall that the threshold is relevant to whether the switching point $\hat{u}_1$ is greater than $\overline{u}$ or not. Since $\mathcal{V}$ is concave, $\hat{u}_1 \leq \overline{u}$ if and only if $\mathcal{V}'(\hat{u}_1) \geq 0$. Observe that by using the formula of V^D and (A.13), we can derive that $\mathcal{V}'(\hat{u}_1) = V^{D\prime}(\hat{u}_1) = e(\lambda_D \Pi - c)^2/[\varphi(\lambda_S \Pi - c)] - 1$. Setting the above equal to zero and discarding the smaller root, which lies below 1, it follows that $\mathcal{V}'(\hat{u}_1) \geq 0$ if and only if

$$\pi \geq \pi_D \equiv \frac{c \cdot e + \varphi + \sqrt{\varphi(c \cdot e + \varphi)}}{c \cdot e}. \tag{B.2}$$

Lemma B.1. *When $\lambda_S = 2\lambda_D$, $\pi_D \in (\pi_F, \pi_S)$.*

Proof of Lemma B.1. From $\varphi \leq c$, we have $\pi_D \leq (e + 1 + \sqrt{e+1})/e \approx 2.077$. Therefore, by (B.1), $\pi_D < \pi_S$.

Next, observe that

$$\pi_D - \pi_F = \frac{-\varphi(e-1) + \sqrt{\varphi(c \cdot e + \varphi)}}{c \cdot e} \geq \frac{\varphi(\sqrt{e+1} - (e-1))}{c \cdot e}.$$

Since $\sqrt{e+1} - (e-1) \approx .21 > 0$, $\pi_D > \pi_F$. □

Proof of Theorem 1. The ordering $\pi_F < \pi_D < \pi_S$ is Lemma B.1. By Proposition 2, $\mathcal{V}$ is concave, so (MP) admits a solution $\overline{u}$, and by Lemma OA.4 some contract implements $(\mathcal{V}(\overline{u}), \overline{u})$. It remains to locate $\overline{u}$ relative to 0 and $\hat{u}_1$.

Feasibility. Since $\mathcal{V}$ is concave with $\mathcal{V}(0) = 0$, we have $\overline{u} > 0$ if and only if $\mathcal{V}'(0) > 0$. By (6) and $\dot{\mathcal{W}}^*(0) = \lambda_D \Pi - c$, $\mathcal{V}'(0) = \dot{\mathcal{W}}^*(0)/\varphi - 1 = (\lambda_D \Pi - c - \varphi)/\varphi$, which is positive if and only if $\pi > 1 + \varphi/c = \pi_F$. For $\pi < \pi_F$ we have $\overline{u} = 0$ and no contract is offered, which is part (d).

Staging throughout. By Proposition 1 the benchmark uses the direct route only when the time remaining is below $\hat{T} = 2(\overline{\Delta} - \Delta)$, and binding incentive conditions

fix $\Delta = 1/\lambda_S$. Hence $\hat{u}_1 = \varphi\hat{T} = 2\varphi(\overline{\Delta} - 1/\lambda_S)^+$, which vanishes precisely when $\overline{\Delta} \leq 1/\lambda_S$, that is, by (B.1), when $\pi \geq \pi_S$. For such π the staged region is all of $\mathbb{R}_+$ and the implementing contract is staged-only, which is part (a).

The intermediate cases. Let $\pi_F < \pi < \pi_S$, so that $\overline{u} > 0$ and $\hat{u}_1 > 0$. Concavity of $\mathcal{V}$ gives $\overline{u} > \hat{u}_1$ if and only if $\mathcal{V}'(\hat{u}_1) > 0$, which by (B.2) holds if and only if $\pi > \pi_D$. When $\pi \in (\pi_D, \pi_S)$ the contract begins in the staged region and crosses into the direct region at $\hat{u}_1$, so it is a one-switch contract, which is part (b). When $\pi \in (\pi_F, \pi_D)$ the contract never leaves the direct region, which is part (c).

The four cases are mutually exclusive, and by the ordering of the thresholds they exhaust $\pi > 0$. □

Appendix C: Proof of Theorem 2

This appendix isolates the crossing property behind the two-switch bound and assembles the proof of Theorem 2. Online Appendix OA.3 proves Proposition 3 and constructs the value function under an efficiency loss: by Proposition OA.1, $\mathcal{V}$ has three branches, direct-only below $\hat{u}_1$, one-switch on $(\hat{u}_1, \hat{u}_2)$ and two-switch above $\hat{u}_2$, joined smoothly, where $\hat{u}_i$ is φ times the corresponding switch point of Proposition 3 at $\Delta = 1/\lambda_S$ (possibly $\hat{u}_1 = 0$).

C.1. The Crossing Property

Throughout this subsection we assume $\lambda_D < \lambda_S < 2\lambda_D$, i.e. $0 < \eta < 1$. For a candidate surplus function w it is convenient to abbreviate the gap defined in (A.6) as

$$\Lambda_z[w] \equiv \Lambda\left(z; w(z)\right) = \lambda_S\left(W^1_{z+\Delta} - w(z)\right) - \lambda_D\left(\Pi - w(z)\right), \tag{C.1}$$

the difference between the two branches of (HJB$_W$): the staged route attains the maximum in (HJB$_W$) at z if and only if $\Lambda_z[w] \geq 0$. The two substituted forms used in the proof of Proposition 1 are $\Lambda_D(x)$ of (A.11), obtained by setting $w = \mathcal{W}^D$ and $x = e^{-\lambda_D z}$, and $\Lambda_{SD}(y)$ of (A.14), obtained by setting $w = \mathcal{W}_Z^{SD}$ and $y = e^{-\lambda_S(z-Z)}$.

The proof of Proposition 1 draws its conclusions from $\lambda_S = 2\lambda_D$ in exactly two places, and they are the same place: the two substituted forms satisfy

$$\lim_{x\to 0} \Lambda_D(x) = \frac{\lambda_S - 2\lambda_D}{\lambda_D} \cdot c \qquad \text{and} \qquad \lim_{y\to 0} \Lambda_{SD}(y) = \frac{\lambda_S - 2\lambda_D}{\lambda_S} \cdot c, \tag{C.2}$$

and both vanish at $\lambda_S = 2\lambda_D$, which is what allows each of Lemmas A.1 and A.2 to conclude that a sign holds on an entire half-line.[9] When $\lambda_S < 2\lambda_D$ both limits are strictly negative, and each conclusion weakens in the same way, from a sign to a single crossing. The following lemma converts these two single crossings into a policy.

Lemma C.1 (Crossing rate). *Let w be differentiable on an interval $I \subseteq \mathbb{R}_+$ and satisfy either (A.7) or (A.8) on I. Then at every $z \in I$ with $\Lambda_z[w] = 0$,*

$$\frac{d}{dz}\Lambda_z[w] = \Phi(z) \equiv \lambda_S(\lambda_S - \lambda_D)\left(\Pi - W^1_{z+\Delta}\right) - \lambda_D c, \tag{C.3}$$

irrespective of which of the two equations w satisfies. Moreover Φ is strictly decreasing, with $\lim_{z\to\infty} \Phi(z) = (\lambda_S - 2\lambda_D)c < 0$.

Proof of Lemma C.1. From (A.5), $dW^1_{z+\Delta}/dz = \lambda_S(\Pi - c/\lambda_S - W^1_{z+\Delta})$, so

$$\frac{d}{dz}\Lambda_z[w] = \lambda_S^2\left(\Pi - \frac{c}{\lambda_S} - W^1_{z+\Delta}\right) - (\lambda_S - \lambda_D)\dot{w}(z).$$

9. The knife-edge is also what reduces Λ_D to the quadratic (A.12) and hence delivers $\hat{x}$ in closed form. Under an efficiency loss the thresholds are characterized implicitly from (A.11).

At a point with $\Lambda_z[w] = 0$ the two routes' flow gains coincide, $\lambda_S(W^1_{z+\Delta} - w(z)) = \lambda_D(\Pi - w(z))$, so the gain still outstanding after a milestone is

$$\Pi - W^1_{z+\Delta} = \Big(1 - \frac{\lambda_D}{\lambda_S}\Big)\big(\Pi - w(z)\big) = \Big(\frac{\lambda_S}{\lambda_D} - 1\Big)\big(W^1_{z+\Delta} - w(z)\big).$$

Substituting the first form into (A.7) and the second into (A.8) gives, in both cases, $(\lambda_S - \lambda_D)\dot{w}(z) = \lambda_S\lambda_D(\Pi - W^1_{z+\Delta}) - (\lambda_S - \lambda_D)c$, and (C.3) follows. Since $W^1_{z+\Delta}$ is strictly increasing in z and converges to $\Pi - c/\lambda_S$, the function Φ is strictly decreasing with limit $\lambda_S(\lambda_S - \lambda_D)\,c/\lambda_S - \lambda_D c = (\lambda_S - 2\lambda_D)c$, which is negative because $\lambda_S < 2\lambda_D$. □

Corollary C.2 (No return). *Suppose w satisfies (A.7) on $[z_0, \infty)$, with $\Lambda_{z_0}[w] = 0$ and $\Phi(z_0) < 0$. Then $\Lambda_z[w] < 0$ for all $z > z_0$.*

Proof of Corollary C.2. Since Φ is strictly decreasing, $\Phi(z) < 0$ for all $z \geq z_0$. By Lemma C.1 the derivative of $\Lambda_z[w]$ at z_0 equals $\Phi(z_0) < 0$, so $\Lambda_z[w] < 0$ for z slightly above z_0. Suppose $\Lambda_z[w] = 0$ for some $z > z_0$ and let z_1 be the smallest such point. Then $\Lambda_z[w] < 0$ on (z_0, z_1) and $\Lambda_{z_1}[w] = 0$, so the derivative at z_1 is non-negative; but Lemma C.1 gives $\Phi(z_1) < 0$, a contradiction. □

C.2. Assembling the Proof

Proof of Theorem 2. **(a)** By Proposition OA.1 the optimal contract assigns the direct route on $[0, \hat{u}_1)$, the staged route on $(\hat{u}_1, \hat{u}_2)$, and the direct route on $(\hat{u}_2, \infty)$. Under binding incentive conditions promised utility declines at the constant rate φ, so along any contract u_t is strictly decreasing and crosses each threshold at most once. The assigned route can therefore change at most twice.

The bound holds at every efficiency level because Proposition OA.1 produces exactly two thresholds for every $\eta \in (0,1)$, which in turn rests on Lemma C.1: the derivative of the staged advantage at any point where it vanishes equals Φ, irrespective of which route is in use, and Φ is strictly decreasing. A third crossing would require Φ to change sign twice.

(b) Fix an optimal contract and let $\overline{u}$ solve (MP). Promised utility declines from $\overline{u}$ to zero, so the set of promised-utility levels visited is exactly $[0, \overline{u}]$, and the routes assigned are those prevailing on that interval. Enumerating the possible positions of $\overline{u}$ relative to the two thresholds:

If $\overline{u} = 0$ the principal does not offer a contract and the project is infeasible.

If $0 < \overline{u} \leq \hat{u}_1$, then $[0, \overline{u}]$ is in the lower direct region: the contract is direct-only.

If $\hat{u}_1 = 0$ and $\overline{u} \leq \hat{u}_2$, then $[0, \overline{u}]$ is in the staged region: the contract is staged-only.

If $0 < \hat{u}_1 < \overline{u} \leq \hat{u}_2$, then $[0, \overline{u}]$ meets the lower direct and staged regions only. The contract begins on the staged route and switches to the direct route when u_t falls below $\hat{u}_1$, at calendar time $T - \hat{u}_1/\varphi$: a one-switch contract.

If $\hat{u}_1 = 0$ and $\overline{u} > \hat{u}_2$, then $[0, \overline{u}]$ meets the staged and upper direct regions only. The contract begins on the direct route and switches to the staged route for good when u_t falls below $\hat{u}_2$: a reversed one-switch contract.

If $\hat{u}_1 > 0$ and $\overline{u} > \hat{u}_2$, then $[0, \overline{u}]$ meets all three regions and the contract switches twice, at calendar times $T - \hat{u}_2/\varphi$ and $T - \hat{u}_1/\varphi$: a two-switch contract.

These are the six rows of Table 1, and they exhaust the possibilities since $\overline{u}$ must fall in one of them. In every case the reward schedules follow from binding incentive conditions exactly as in Lemma OA.4, since those conditions do not involve η. □

Appendix D: Proofs for Section 6.2

D.1. Proof of Proposition 6

Proof. Delay is free. Under binding incentive conditions every delivery is worth the promise outstanding at the date it is presented plus a premium: $u_\tau + \varphi/\lambda_S$ for a milestone, and $u_\tau + \varphi/\lambda$ for a finished project, where λ is the rate of the route whose schedule is in force. Suppose the agent holds an output from s and presents it at $\tau > s$, shirking in between. Since $\dot{u}_t = -\varphi$, he collects $\varphi(\tau - s) + u_\tau + \varphi/\lambda = u_s + \varphi/\lambda$, the value of presenting at s. An output carried into a phase in which it is inconsistent with the assigned route terminates the contract and is worth nothing. Delay therefore never gains, and it remains to compare revealing a milestone with never revealing it.

Never revealing. An unrevealed milestone is not paid for, so it is worth something only in so far as it carries the agent to completion, and completion without a milestone on record is paid only while the direct route is assigned. Let $M(z)$ be the value to the agent of holding an unrevealed milestone while the direct route is assigned with z units of time remaining, when completion pays $R(z) = \varphi(z + 1/\lambda_D)$. Then $M'(z) = \max\{\lambda_S(R(z) - M(z)), \varphi\}$ with $M(0) = 0$, and, the first branch attaining the maximum throughout,

$$M(z) = \varphi\left(z + \frac{1}{\lambda_D} - \frac{1}{\lambda_S}\right) - \varphi\left(\frac{1}{\lambda_D} - \frac{1}{\lambda_S}\right) e^{-\lambda_S z}. \tag{D.1}$$

Over the shirk-out value φz, the milestone is thus worth $M(z) - \varphi z = \varphi\eta(1 - e^{-\lambda_S z})/\lambda_S < \varphi/\lambda_S$, since $\eta \leq 1$, whereas revealing it is worth φ/λ_S. If the milestone arrives in a staged phase, holding it until a later direct phase earns shirking in the interim and then this same amount, so revealing is strictly better there too; if no

direct phase follows, an unrevealed milestone is worth nothing. On-path behavior is therefore unchanged, and so is the principal's payoff. □

D.2. Proof of Proposition 7

Proof. **(a)** By the delay step in the proof of Proposition 6, producing route k's output at s and presenting it when it is called for is worth $u_s + \varphi/\lambda_k$. The flow value of working route k at t is then $\lambda_k (u_t + \varphi/\lambda_k - u_t) = \varphi$, which is also the flow value of working route j and of shirking.

(b) Route k's output is now never paid for on its own, so it is useful only in so far as it carries the agent to completion. If k is the direct route and j the staged one, a completed project presented without a milestone on record is never paid for, so working k has flow value $0 < \varphi$. If k is the staged route and j the direct one, a milestone produced now is held as in the proof of Proposition 6, and by (D.1) the flow value of working route k with z units of time remaining is $\lambda_S(M(z) - \varphi z) = \varphi\eta(1 - e^{-\lambda_S z}) < \varphi$. □

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Online Appendix for

"To Stage or Not to Stage? Optimal Contracts and Project Management"

Yonggyun Kim Curtis Taylor

Appendix OA.1: Recursive Formulation and Value Function Verification

This appendix supplies the formal ingredients of the proof of Proposition 2 in Appendix B. Online Appendix OA.1.1 sets the problem up recursively: it computes the two parties' expected payoffs, shows that the pointwise condition (IC) suffices for incentive compatibility, and derives the Hamilton–Jacobi–Bellman equation together with a verification lemma. Online Appendix OA.1.2 constructs the candidate value function, exhibits for each promised utility a contract implementing it, and shows that no deviation to slack incentives improves upon the candidate.

OA.1.1. Recursive Formulation

OA.1.1.1. Expected Payoffs and Incentive Compatibility. We first present the two parties' expected payoffs given the agent's effort schedule.

Post-milestone payoffs. We begin with the subgame where the interim milestone is delivered at time t. Let u_M^t denote the agent's continuation utility, his expected payoff from working until completion or the deadline T^t.

Since this subgame only requires one more success, it is identical to the single-stage benchmark of Green and Taylor (2016). They show that the principal's value function V_M is characterized as follows:

$$V_M(u_M^t) = W^1_{u_M^t/\varphi} - u_M^t = \left(\Pi - \frac{c}{\lambda_S}\right)\left(1 - e^{-\frac{\lambda_S}{\varphi}u_M^t}\right) - u_M^t. \tag{OA.1.1}$$

Expected payoffs at time 0. Now consider the problem at time 0. The agent's admissible effort schedule (prior to either completion or delivery of the interim milestone) is $\mathcal{B} \equiv \{\{\beta_t\}_{0\leq t\leq T} \ : \ \beta_t \in \{0,1\}\}$. In this case, any arrival depends not only on the agent's effort (β) but also the principal's route choice ($a = \{a_t\}_{0\leq t\leq T}$). Given (a,β), the probability that neither completion nor the interim milestone has arrived by time τ is $f(a,\beta;0,\tau)$, where $f(a,\beta;t,\tau) \equiv e^{-\lambda_D \int_t^\tau a_s\beta_s ds - \lambda_S \int_t^\tau (1-a_s)\beta_s ds}$. Accordingly, the probability density of completion at time τ is $\lambda_D a_\tau \beta_\tau \cdot f(a,\beta;0,\tau)$ and that of the delivery of the interim milestone at time τ is $\lambda_S(1-a_\tau)\beta_\tau \cdot f(a,\beta;0,\tau)$. If the agent follows the recommendation, the above expressions simplify to: $\tilde{f}(a;t,\tau) \equiv e^{-\lambda_D \int_t^\tau a_s ds - \lambda_S \int_t^\tau (1-a_s) ds}$, $\lambda_D a_\tau \cdot \tilde{f}(a;t,\tau)$, and $\lambda_S(1-a_\tau)\cdot \tilde{f}(a;t,\tau)$.

Given Γ and β, and assuming the agent follows the recommendation in the updated contract, the principal's time-0 expected payoff is

$$P_0(\beta,\Gamma) \equiv \int_0^T \{(\Pi - R_\tau)\cdot \lambda_D a_\tau \beta_\tau + V_M(u_M^\tau)\cdot \lambda_S(1-a_\tau)\beta_\tau - c\}\cdot f(a,\beta;0,\tau)d\tau.$$

while the agent's expected payoff at time t is:

$$U_0(\beta,\Gamma) \equiv \int_0^T \{R_\tau \cdot \lambda_D a_\tau \beta_\tau + u_M^\tau \cdot \lambda_S(1-a_\tau)\beta_\tau + \varphi(1-\beta_\tau)\}\cdot f(a,\beta;0,\tau)d\tau.$$

Similarly, the expected payoffs of the principal and the agent when the agent follows the recommendation can be obtained by evaluating the general expressions using $\tilde{f}(a;0,\tau)$ in place of $f(a,\hat{\beta}^t;0,\tau)$. We denote these by $\tilde{P}_0(\Gamma)$ and $\tilde{U}_0(\Gamma)$, respectively.

Incentive Compatibility. Using the terms defined above, we define incentive compatibility (IC) of contracts as follows.

Definition OA.1.1. A contract $\Gamma = \left\{T, \{a_t, R_t, \hat{\Gamma}^t\}_{0\leq t\leq T}\right\}$ is *incentive compatible* if $\tilde{U}_0(\Gamma) \geq U_0(\beta,\Gamma)$ for all $\beta \in \mathcal{B}$.

The objective of the principal is to find a contract Γ that maximizes her ex ante expected payoff $P_0(\Gamma)$ subject to the incentive compatibility constraint and the limited liability constraints $R_\tau \geq 0$ for all $\tau \in [0,T]$ and $R_\tau^t \geq 0$ for all $t \in [0,T]$ and $\tau \in [t,T^t]$. Designate such a contract as an *optimal contract*.

OA.1.1.2. The Agent's Problem . Given a contract Γ, define the continuation utility that the interim milestone is delivered at t as $u_M^t = \tilde{U}^t(\hat{\Gamma}^t)$ and the continuation utility at t when neither the project is completed nor the interim milestone is delivered as

$$u_t = \tilde{U}_t(\Gamma) \equiv \int_t^T \left\{R_\tau \cdot \lambda_D a_\tau + u_M^t \cdot \lambda_S(1-a_\tau)\right\} \cdot \tilde{f}(a;t,\tau)d\tau.$$

Observe that

$$0 = \dot{u}_t + (R_t - u_t)\lambda_D a_t + (u_M^t - u_t)\lambda_S(1-a_t) \tag{HJB$_{PK}$}$$

where $\dot{u}_t \equiv du_t/dt$.

Conditions (IC$_S$) ($a_t = 0$) and (IC$_D$) ($a_t = 1$) combine into

$$(R_t - u_t)\lambda_D a_t + (u_M^t - u_t)\lambda_S(1-a_t) \geq \varphi. \tag{IC}$$

The following lemma shows that this condition serves as a sufficient condition for incentive compatibility defined in Definition OA.1.1.

LEMMA OA.1. *Given a contract Γ, suppose that there exists a continuous and differentiable process $\{u_t\}_{0\leq t\leq T}$ satisfying (HJB$_{PK}$) and $u_T = 0$, and (IC) holds for $0 \leq t \leq T$. Then, Γ is incentive compatible.*

Proof of Lemma OA.1. The proof is inspired by Proposition 3.2.1 in Bertsekas (1995).

Consider an arbitrary admissible action $\beta \in \mathcal{B}$. Using (HJB$_{PK}$), (IC$_S$) and (IC$_D$), we can derive that $0 \geq \dot{u}_t + (R_t - u_t)\lambda_D a_t \beta_t + (u_M^t - u_t)\lambda_S(1-a_t)\beta_t + \varphi(1-\beta_t)$, or equivalently,

$$-\dot{u}_t + (\lambda_D a_t + \lambda_S(1-a_t))\beta_t \cdot u_t \geq R_t \cdot \lambda_D a_t \beta_t + u_M^t \cdot \lambda_S(1-a_t)\beta_t + \varphi(1-\beta_t).$$

It is further equivalent to

$$\frac{d}{dt}\left[-u_t \cdot f(a,\beta;0,t)\right] \geq \left[\left(R_t\lambda_D a_t + u_M^t\lambda_S(1-a_t)\right)\beta_t + \varphi(1-\beta_t)\right] \cdot f(a,\beta;0,t).$$

Integrating from 0 to T and using $u_T = 0$ gives

$$u_0 \geq \int_0^T \left[\left(R_t \lambda_D a_t + u_M^t \lambda_S (1 - a_t)\right) \beta_t + \varphi(1 - \beta_t)\right] \cdot f(a, \beta; 0, t) dt = U_0(\beta, \Gamma)$$

for all $\beta \in \mathcal{B}$. Furthermore, the equality holds when $\beta_t = 1$ for all $t \in [0, T]$ from (HJB_{PK}), i.e., u_0 is equal to $\tilde{U}_0(\Gamma)$. Thus, $\tilde{U}_0(\Gamma) \geq U_0(\beta, \Gamma)$ for all $\beta \in \mathcal{B}$, which implies incentive compatibility. □

OA.1.1.3. The Principal's Problem . We now explore how the principal's value function $V(u_t)$ evolves. Note that $V(0) = 0$ since the agent will not participate in the contract when the continuation utility is zero. This will serve as a boundary condition. The value function $V(u_t)$ can be heuristically written as follows:

$$\begin{aligned} V(u_t) = \max_{R_t, u_M^t, a_t} & -cdt + (\Pi - R_t)\, \lambda_D a_t dt + V_M(u_M^t) \lambda_S (1 - a_t) dt \\ & + \{1 - \lambda_D a_t dt - \lambda_S (1 - a_t) dt\} \cdot V(u_{t+dt}) \end{aligned}$$

subject to (IC).

By using $V(u_{t+dt}) = V(u_t) + V'(u_t)\dot{u}_t dt + o(dt)$, canceling $V(u_t)$ on both sides, taking the limit as $dt \to 0$ and plugging (HJB_{PK}) in, we obtain an HJB equation:

$$0 = \max_{R, u_M, a} \mathcal{J}(V(\cdot), R, u_M, a). \tag{HJB_V}$$

where

$$\begin{aligned} \mathcal{J}(V(\cdot), R, u_M, a) \equiv & -c + (\Pi - R - V(u))\lambda_D a + (V_M(u_M) - V(u))\lambda_S (1 - a) \\ & - \{(R - u)\lambda_D a + (u_M - u)\lambda_S (1 - a)\} \cdot V'(u). \end{aligned} \tag{OA.1.2}$$

LEMMA OA.2 (Bang-bang). *Suppose the agent's effort may take any value $\beta_t \in [0, 1]$, and the principal's recommendation any value $a_t \in [0, 1]$, interpreted as the intensity with which the direct route is worked, so that arrivals occur at rate $\lambda_D\, a_t \beta_t$ on the direct route and $\lambda_S (1 - a_t)\beta_t$ on the staged route. Then:*

(a) *given any contract, the agent has an optimal effort process taking values in $\{0, 1\}$; and*

(b) *the principal's value is unchanged if recommendations are restricted to* $a_t \in \{0,1\}$, *and public randomization over recommendations does not improve upon it.*

Proof of Lemma OA.2. **(a)** Deriving the agent's Hamilton–Jacobi–Bellman equation as in the proof of Lemma OA.1, but with the supremum taken over $[0,1]$,

$$0 = \dot{U}_t + \sup_{\beta \in [0,1]} \left\{ \varphi(1-\beta) + \beta \left[(R_t - U_t)\lambda_D \, a_t + (u_M^t - U_t)\lambda_S (1 - a_t) \right] \right\}.$$

The maximand is affine in β, so the supremum is attained at $\beta = 0$ or $\beta = 1$; when the bracket equals φ every β attains it, and $\beta = 1$ is among them.
(b) Fix u and V. By (OA.1.2), $\mathcal{J}(V(\cdot), R, u_M, a)$ is affine in a for fixed (R, u_M), and the constraint (IC) is affine in a as well. Hence

$$g(a) \equiv \sup \{ \mathcal{J}(V(\cdot), R, u_M, a) \ : \ (R, u_M) \text{ satisfying (IC)} \}$$

is a supremum of affine functions of a and is therefore convex on $[0,1]$, so that $\sup_{a \in [0,1]} g(a) = \max\{g(0), g(1)\}$: the maximum in ($\text{HJB}_V$) is attained at a pure recommendation. Finally, a public randomization placing probability ρ on the recommendation a' and $1-\rho$ on a'' delivers $\rho\, g(a') + (1-\rho)\, g(a'') \leq \max\{g(0), g(1)\}$, so randomizing over recommendations cannot improve upon the best pure one. □

Then, the principal's problem is to solve (HJB_V) subject to (IC) with the boundary condition $V(0) = 0$. The following lemma shows that the solution of the problem maximizes the principal's expected payoff subject to a promise keeping constraint $U_0(\Gamma) = u$.

Lemma OA.3 (Verification Lemma). *Suppose that a differentiable and concave function* $\tilde{V}$ *solves* (HJB_V) *subject to* (IC) *with the boundary condition* $\tilde{V}(0) = 0$. *Then, for any incentive-compatible contract* Γ *with* $U_0(\Gamma) = u$, $\tilde{V}(u) \geq \tilde{P}_0(\Gamma)$.

Proof of Lemma OA.3. Fix a deterministic incentive-compatible contract Γ with promised utility u_t, and let $\tilde{f}(a; 0, t)$ be the probability that nothing has arrived by

t. Evaluating (HJB_V) at the contract's terms and using (HJB_{PK}) gives

$$\frac{d}{dt}\big(-\tilde{f}(a;0,t)\,\tilde{V}(u_t)\big) \geq \tilde{f}(a;0,t)\big((\Pi - R_t)\lambda_D a_t + V_M(u_M^t)\lambda_S(1-a_t) - c\big).$$

Integrating over $[0,T]$ with $\tilde{f}(a;0,0)=1$, $u_T=0$ and $\tilde{V}(0)=0$ yields $\tilde{V}(u_0)\geq \tilde{P}_0(\Gamma)$. This is the standard sufficiency argument for a deterministic control problem (e.g., Proposition 3.2.1 in Bertsekas 1995), applied to the state $(\tilde{f},u)$ with value $\tilde{f}\,\tilde{V}(u)$. Proposition OA.1 extends the bound to contracts that randomize. □

OA.1.2. Value Function Verification

OA.1.2.1. Value Function Candidates and Implementation . Section 4 argued that route choice under agency is the same as under observable effort, because promised utility evolves in the same way whichever route is assigned. The following lemma is the formal counterpart. Parts (a)–(c) show that a contract with deadline u/φ delivers the agent exactly u regardless of the route it follows, so that the principal's payoff from any of them is the corresponding benchmark surplus of Section 3 net of u. Part (d) records the differential equations these payoffs satisfy.

LEMMA OA.4. *The following statements hold.*

(a) *A direct-only contract with the deadline u/φ implements a pair of expected payoffs of the principal and the agent $(V^D(u),u)$ where $V^D(u)\equiv \mathcal{W}^D(u/\varphi)-u$.*

(b) *When $0<u_1<u$, a one-switch contract with the intermediate deadline $(u-u_1)/\varphi$ and the final deadline u/φ implements $(V^{SD}(u|u_1),u)$ where*

$$V^{SD}(u|u_1) \equiv \mathcal{W}^{SD}_{u_1/\varphi}\left(\frac{u}{\varphi},\frac{1}{\lambda_S}\right) - u. \tag{OA.1.3}$$

(c) *A staged-only contract with the deadline u/φ implements $(V^{SD}(u|0),u)$.*

(d) *The following differential equations hold:*

$$\varphi V^{D\prime}(u) = \lambda_D\left(\Pi - \frac{\varphi}{\lambda_D} - u - V^D(u)\right) - c, \tag{OA.1.4}$$

$$\varphi V^{SD\prime}(u|u_1) = \lambda_S\left(V_M(u+\tfrac{\varphi}{\lambda_S}) - V^{SD}(u|u_1)\right) - c. \tag{OA.1.5}$$

Proof of Lemma OA.4. **(a)** Let $\Gamma_D(T)$ denote a direct-only contract with the deadline T. The agent's expected payoff is

$$\begin{aligned} U_0(\Gamma_D(T)) &= \int_0^T R_\tau \lambda_D e^{-\lambda_D \tau} d\tau = \int_0^T \varphi\left(T - \tau + \frac{1}{\lambda_D}\right) \lambda_D e^{-\lambda_D \tau} d\tau \\ &= -\varphi\left(T-\tau\right) e^{-\lambda_D \tau}\Big|_0^T \quad = \varphi T. \end{aligned}$$

Therefore, $U_0(\Gamma_D(u/\varphi)) = u$.

Since the two payoffs sum to the direct-only surplus with deadline T, $P_0(\Gamma_D(T)) + U_0(\Gamma_D(T)) = \mathcal{W}^D(T)$, we have $P_0(\Gamma_D(u/\varphi)) = \mathcal{W}^D(u/\varphi) - u = V^D(u)$.

(b) Let $\Gamma_{SD}(T_1, T)$ denote a contract with a switch from the staged route to the direct route at T_1 and the deadline T. The subcontract at time $t \le T_1$ is denoted by $\hat{\Gamma}_{SD}(t|T_1, T)$. Then the agent's expected payoff from $\hat{\Gamma}_{SD}(t|T_1, T)$ at t is

$$\begin{aligned} U_t(\hat{\Gamma}_{SD}(t|T_1,T)) &= \int_t^{T+\frac{1}{\lambda_S}} \varphi\left(T + \frac{1}{\lambda_S} - \tau + \frac{1}{\lambda_S}\right) \lambda_S e^{-\lambda_S(\tau - t)} d\tau \\ &= -\varphi\left(T + \frac{1}{\lambda_S} - \tau\right) e^{-\lambda_S(\tau-t)}\Big|_t^{T+\frac{1}{\lambda_S}} \quad = \varphi\left(T + \frac{1}{\lambda_S} - t\right). \end{aligned}$$

Also note that

$$\begin{aligned} \int_0^{T_1} U_\tau(\hat{\Gamma}_{SD}(\tau|T_1,T)) \lambda_S e^{-\lambda_S \tau} d\tau &= \int_0^{T_1} \varphi(T + \frac{1}{\lambda_S} - \tau) \lambda_S e^{-\lambda_S \tau} d\tau \\ &= -\varphi(T-\tau) e^{-\lambda_S \tau}\Big|_0^{T_1} \quad = \varphi T - \varphi(T - T_1) e^{-\lambda_S T_1}. \end{aligned}$$

Then, the agent's expected payoff at time 0 is

$$\begin{aligned} U_0(\Gamma_{SD}(T_1,T)) =& \int_0^{T_1} U_\tau(\hat{\Gamma}_{SD}(\tau|T_1,T)) \lambda_S e^{-\lambda_S \tau} d\tau \\ &+ e^{-\lambda_S T_1} \int_{T_1}^T \varphi(T + \frac{1}{\lambda_D} - \tau) \lambda_D e^{-\lambda_D(\tau - T_1)} d\tau \\ =& \varphi T - \varphi(T-T_1) e^{-\lambda_S T_1} - e^{-\lambda_S T_1} \left[\varphi\left(T-\tau\right) e^{-\lambda_D(\tau - T_1)}\Big|_{T_1}^T\right] \quad = \varphi T. \end{aligned}$$

Thus, $U_0(\Gamma_{SD}(T_1, u/\varphi)) = u$.

As in the previous case, the sum of the expected payoffs of the principal and the agent is equal to the one-switch policy with the intermediate deadline T_1, the deadline T, and the extension $1/\lambda_S$ $P_0(\Gamma_{SD}(T_1,T)) + U_0(\Gamma_{SD}(T_1,T)) = \mathcal{W}^{SD}_{T-T_1}(T, 1/\lambda_S)$. By plugging in $T = u/\varphi$ and $T_1 = (u-u_1)/\varphi$, (OA.1.3) holds.

(c) A staged-only contract with deadline T is the case $T_1 = T$ of part (b), so it implements $(V^{SD}(u|0), u)$.

(d) By construction, $\mathcal{W}^D$ satisfies (A.7) for all $T \geq 0$, and $\mathcal{W}^{SD}_{\hat{T}}(\cdot, 1/\lambda_S)$ satisfies (A.8) with $\Delta = 1/\lambda_S$ for all $T \geq \hat{T}$ (both in remaining time). Using the definitions of V^D, V^{SD} and V_M, (OA.1.4) and (OA.1.5) follow. □

Together with Proposition 1, this lemma implies that $(\mathcal{V}(u), u)$ defined in (6) can be implemented by one of the above three contracts. Moreover, there exists $\hat{u}_1 \geq 0$ such that $\mathcal{V}$ can be rewritten as follows:

$$\mathcal{V}(u) = \begin{cases} V^D(u), & \text{if } u < \hat{u}_1, \\ V^{SD}(u|\hat{u}_1), & \text{if } u \geq \hat{u}_1. \end{cases} \tag{OA.1.6}$$

Specifically, $\hat{u}_1$ is chosen to be equal to $\varphi\hat{T}$ if $1/\lambda_S < \overline{\Delta}$, and 0 if $1/\lambda_S \geq \overline{\Delta}$. The following lemma provides useful properties of $\mathcal{V}$ and $\hat{u}_1$.

LEMMA OA.5. *Suppose that $\lambda_S = 2\lambda_D$. The following statements hold.*

(a) *if $\hat{u}_1 > 0$, $V^{SD\prime}(\hat{u}_1|\hat{u}_1) = V^{D\prime}(\hat{u}_1)$ and $V^{SD\prime}(u|u) < V^{D\prime}(u)$ for all $u < \hat{u}_1$, and if $\hat{u}_1 = 0$, $V^{SD\prime}(0|0) \geq V^{D\prime}(0)$.*

(b) *$\mathcal{V}'(u) \geq -1$ for all $u \geq 0$.*

(c) *$\mathcal{V}$ is strictly concave, and hence $\overline{u}$ is unique.*

Proof of Lemma OA.5. **(a)** Suppose that $\hat{u}_1 > 0$, which implies that $1/\lambda_S < \overline{\Delta}$. Now set $\Delta = 1/\lambda_S$. In Lemma A.1, $\hat{T}$ is chosen to satisfy $\lambda_D(\Pi - \mathcal{W}^D(\hat{T})) = \lambda_S(W^1_{\hat{T}+1/\lambda_S} - \mathcal{W}^D(\hat{T}))$ and $\lambda_D(\Pi - \mathcal{W}^D(z)) > \lambda_S(W^1_{z+1/\lambda_S} - \mathcal{W}^D(z))$ for all $z < \hat{T}$.

Using Lemma OA.4 (d) and $\mathcal{W}^D(\hat{T}) = \mathcal{W}^{SD}_{\hat{T}}(\hat{T}, 1/\lambda_S)$, we can derive that $V^{SD\prime}(\hat{u}_1|\hat{u}_1) = V^{D\prime}(\hat{u}_1)$ and $V^{SD\prime}(u|u) < V^{D\prime}(u)$ for all $u < \hat{u}_1$.

When $\hat{u}_1 = 0$, and thereby $1/\lambda_S \geq \overline{\Delta}$, it follows from (A.10) that $V^{SD\prime}(\hat{u}_1|\hat{u}_1) \geq V^{D\prime}(\hat{u}_1)$.

(b) Since a longer deadline increases the expected social surplus, $\mathcal{W}^*(T,\Delta)$ is increasing in T. Therefore, $\mathcal{V}'(u) = \dot{\mathcal{W}}^*(u/\varphi, 1/\lambda_S)/\varphi - 1 \geq -1$.

(c) If $u \leq \hat{u}_1$, $\mathcal{V}''(u) = V^{D\prime\prime}(u) = -(\Pi - c/\lambda_D)(\lambda_D^2/\varphi^2)e^{-(\lambda_D/\varphi)u} < 0$.

Now assume that $u > \hat{u}_1$. By (A.7), (A.8) and $V^{D\prime}(\hat{u}_1) \leq V^{SD\prime}(\hat{u}_1|\hat{u}_1)$,

$$\mathcal{W}^D(\hat{u}_1/\varphi) \leq \Pi - \frac{c}{\lambda_S - \lambda_D} - \frac{\lambda_S}{\lambda_S - \lambda_D}\left(\Pi - \frac{c}{\lambda_S}\right)e^{-\frac{\lambda_S}{\varphi}\hat{u}_1 - 1}. \tag{OA.1.7}$$

Using (A.9), we can derive the followings though some algebra:

$$V^{SD\prime\prime}(u|\hat{u}_1) = \left(\frac{\lambda_S}{\varphi}\right)^2 e^{\frac{\lambda_S}{\varphi}(\hat{u}_1 - u)}\left[\mathcal{W}^D(\tfrac{\hat{u}_1}{\varphi}) - \left(\Pi - \frac{2c}{\lambda_S}\right) + 2\left(\Pi - \frac{c}{\lambda_S}\right)e^{-\frac{\lambda_S}{\varphi}\hat{u}_1 - 1}\right]$$
$$- \left(\Pi - \frac{c}{\lambda_S}\right)\left(\frac{\lambda_S}{\varphi}\right)^3 (u - \hat{u}_1)e^{-\frac{\lambda_S}{\varphi}(u + \frac{1}{\lambda_S})}$$

By plugging (OA.1.7) in, we have

$$V^{SD\prime\prime}(u|\hat{u}_1) \leq \left(\frac{\lambda_S}{\varphi}\right)^2 e^{\frac{\lambda_S}{\varphi}(\hat{u}_1 - u)}\left[\frac{\lambda_S - 2\lambda_D}{\lambda_S(\lambda_S - \lambda_D)}c + \frac{\lambda_S - 2\lambda_D}{\lambda_S - \lambda_D}\left(\Pi - \frac{c}{\lambda_S}\right)e^{-\frac{\lambda_S}{\varphi}\hat{u}_1 - 1}\right]$$
$$- \left(\Pi - \frac{c}{\lambda_S}\right)\left(\frac{\lambda_S}{\varphi}\right)^3 (u - \hat{u}_1)e^{-\frac{\lambda_S}{\varphi}(u + \frac{1}{\lambda_S})}. \tag{OA.1.8}$$

Then, from $\lambda_S = 2\lambda_D$, $\mathcal{V}''(u) = V^{SD\prime\prime}(u|\hat{u}_1) \leq 0$ for all $u \geq \hat{u}_1$. □

OA.1.2.2. Proof of Proposition 2 . We show that the value function defined in the previous section solves (HJB$_V$) subject to (IC). To this end we introduce functions that specify potential deviations and establish their properties as a lemma.

First, define

$$L^D(u,R) \equiv \mathcal{J}(\mathcal{V}(\cdot), R, \cdot, 1) = \lambda_D(\Pi - R - \mathcal{V}(u)) - c - \lambda_D(R - u)\mathcal{V}'(u), \tag{OA.1.9}$$

and

$$L^S(u,w) \equiv \mathcal{J}(\mathcal{V}(\cdot), \cdot, w, 0) = \lambda_S(V_M(w) - \mathcal{V}(u)) - c - \lambda_S(w - u)\mathcal{V}'(u). \tag{OA.1.10}$$

Given u, maximizing $L^D(u,R)$ over $R \geq u + \varphi/\lambda_D$, and $L^S(u,w)$ over $w \geq u + \varphi/\lambda_S$, maximizes the right-hand side of (HJB_V) under (IC) at $a = 1$ and $a = 0$, respectively.

LEMMA OA.6. *Suppose that* $\Pi > c/\lambda_D$ *and* $\lambda_S = 2\lambda_D$. *Then, for any* $u \geq 0$, $L^D(u,R) \leq 0$ *for all* $R \geq u + \varphi/\lambda_D$, *and* $L^S(u,w) \leq 0$ *for all* $w \geq u + \varphi/\lambda_S$.

Proof of Lemma OA.6. We begin by showing $L^D(u,R) \leq 0$ for all $R \geq u + \varphi/\lambda_D$. Observe that $\partial L^D/\partial R = -\lambda_D(1 + \mathcal{V}'(u)) \leq 0$ from Lemma OA.5 (b). Also note that

$$\begin{aligned} L^D(u, u+\varphi/\lambda_D) =& \lambda_D(\Pi - u - \mathcal{V}(u)) - c - \varphi(\mathcal{V}'(u) + 1) \\ =& \lambda_D(\Pi - \mathcal{W}^*(u/\varphi, 1/\lambda_S)) - c - \dot{\mathcal{W}}^*(u/\varphi, 1/\lambda_S) \leq 0. \end{aligned}$$

The last inequality is due to (HJB_W). Therefore, $L^D(u,R) \leq 0$ for all $R \geq u + \varphi/\lambda_D$.

For L^S, observe that $\partial L^S/\partial u = -\lambda_S(w-u)\mathcal{V}''(u) \geq 0$ by the concavity of $\mathcal{V}$. Therefore, it is sufficient to check whether $L^S(u, u+\varphi/\lambda_S) \leq 0$ for all $u \geq 0$.

Note that for all $u \geq \hat{u}_1$, $L^S(u, u+\varphi/\lambda_S) = 0$ holds by (OA.1.5). Now, suppose that $u < \hat{u}_1$, thereby $\mathcal{V}(u) = V^D(u)$. Using (OA.1.5) and Lemma OA.5 (a), we have $L^S(u, u+\varphi/\lambda_S) = \varphi(V^{SD\prime}(u|u) - V^{D\prime}(u)) < 0$ for all $u < \hat{u}_1$. □

Appendix OA.2: Randomized Contracts

This appendix establishes Theorem 1 against a contract space in which the principal may randomize and need not disclose the realization to the agent. Only the upper bound requires the enlarged space: the contract exhibited in Appendix OA.1.2.1 is deterministic and delivers $\mathcal{V}(u)$, so attainment is unaffected.

Contracts. At $t = 0$ the principal offers a contract and commits fully to its terms; if the agent rejects, both parties receive zero. We restrict attention to contracts that recommend full effort at every date, which is without loss of generality: because

$c > \varphi$, any interval on which the agent is meant to shirk can be excised, advancing all subsequent events and saving $c - \varphi$ per unit of time.

To allow randomization we endow the contract with a *randomization device*: a random variable ω, drawn at $t = 0$ from a probability space $(\Omega, \mathcal{F}, \mu)$ independently of everything the agent does, on which the terms may depend. A *contract* is an object

$$\Gamma \equiv \left\{ \mu,\ \left\{ T(\omega),\ \left\{ a_t(\omega),\ R_t(\omega),\ \hat{\Gamma}^t(\omega) \right\}_{0 \le t \le T(\omega)} \right\}_{\omega \in \Omega} \right\}, \tag{OA.2.1}$$

whose components are defined at calendar time t, for each realization ω, as follows.

1. $T(\omega) \in \mathbb{R}_+$: the deadline at which the project is abandoned if no delivery has occurred.
2. $a_t(\omega) \in \{0, 1\}$: the route recommended at t, where $a_t = 1$ denotes the *direct* route and $a_t = 0$ the *staged* route.
3. $R_t(\omega) \ge 0$: the payment to the agent if he completes the project at t without having delivered an interim milestone.[10]
4. $\hat{\Gamma}^t(\omega) \equiv \left\{ T^t(\omega), \{R_s^t(\omega)\}_{t \le s \le T^t(\omega)} \right\}$: the continuation contract taking effect if the milestone arrives at t, consisting of a revised deadline $T^t(\omega) \ge t$ and payments $R_s^t(\omega) \ge 0$ for completing the project at s.

Any delivery inconsistent with the recommended route—a milestone under the direct route, or a completed project with no milestone on record under the staged route—terminates the contract without payment, so the agent follows the recommendation. A contract is *deterministic* if μ is degenerate, in which case (OA.2.1) reduces to $\Gamma = \{T, \{a_t, R_t, \hat{\Gamma}^t\}_{0 \le t \le T}\}$.

The formulation is more permissive than it may appear. A soft deadline—termination at a random date, as in Green and Taylor (2016)—is the case in which $T(\omega)$ is non-degenerate and the agent learns its realization only by being terminated; randomization over routes, payments or the length of the extension is likewise a special

10. Since both parties are risk neutral and do not discount, all payments may be backloaded to the moment of a delivery without loss of generality (see, e.g., Ray 2002).

case. Because ω is drawn once and for all at $t = 0$, devices that unfold gradually are covered too: a sequence of later randomizations is a single draw of the whole path, revealed as the contract proceeds. Nothing requires the principal to disclose ω, and this is what makes the enlarged space necessary rather than decorative: an agent who does not know which contract he faces cannot tailor a deviation to it, so a mixture of contracts that are individually not incentive compatible may be incentive compatible as a whole.[11]

The agent need not observe ω. He observes the route he is asked to work on and whether the contract is still running; beyond that, what he learns is whatever those observations reveal. Write $\mathcal{H}_t$ for the public history of deliveries, recommendations and survival up to t, and $\mu_t \equiv \mu(\cdot \mid \mathcal{H}_t)$ for his posterior over the device. The argument has three parts: Appendix OA.2.1 shows that μ_t is unaffected by the agent's own deviations, Appendix OA.2.2 uses this to show that incentive compatibility constrains only conditional means, and Appendix OA.2.3 extends the verification argument, the new content being two terms that concavity signs.

OA.2.1. Beliefs and Deviations

An *effort strategy* is an $\mathcal{H}$-adapted process $\beta = \{\beta_s\}$ together with a choice of route at each date; write $\mathbb{P}^\beta$ for the law it induces jointly with μ.

Lemma OA.1 (Beliefs are deviation-proof). *For every effort strategy β and every t, the posterior μ_t is the same under $\mathbb{P}^\beta$ as under the recommended strategy, $\mathbb{P}^1$, on every history that has positive probability under both.*

11. We are grateful to a referee for pressing this point.

Proof. Fix a public history h_t with no delivery, recommendation path $a_{[0,t]}$, and the contract still running.[12] Its likelihood under (ω, β) is

$$\mathbb{P}\left(h_t \mid \omega, \beta\right) = \underbrace{\mathbf{1}\left\{a_{[0,t]}(\omega) = a_{[0,t]},\ T(\omega) \geq t\right\}}_{\equiv\ g(h_t,\omega)} \cdot \underbrace{\exp\left(-\int_0^t \lambda(a_s)\,\beta_s\,ds\right)}_{\equiv\ f(h_t,\beta)},$$

where $\lambda(a_s) \in \{\lambda_D, \lambda_S\}$ is the arrival rate of the route worked at s. The first factor depends on ω but not on β; the second depends on β but not on ω, because the recommendation path is part of h_t and hence known to the agent. By Bayes' rule,

$$\mu_t(\omega) = \frac{g(h_t,\omega)\,f(h_t,\beta)\,\mu(d\omega)}{\int_\Omega g(h_t,\omega')\,f(h_t,\beta)\,\mu(d\omega')} = \frac{g(h_t,\omega)\,\mu(d\omega)}{\int_\Omega g(h_t,\omega')\,\mu(d\omega')},$$

and f cancels. □

The lemma does not say the agent learns nothing about ω: survival rules out realizations with $T(\omega) < t$, and each recommendation rules out those inconsistent with it. It says he learns nothing *extra* by deviating. Arrival rates depend on his effort but not on the device, contractual terms depend on the device but not on his effort, and the one channel through which the device could affect arrival rates—the recommendation—is observed. Shirking therefore slows the project without revealing which contract he is in, which is what closes the pooling channel above.

OA.2.2. Incentive Compatibility in Conditional Means

Define promised utility and the conditional means of the contractual terms: $u_t \equiv \mathbb{E}\left[\text{continuation utility} \mid \mathcal{H}_t\right]$, $\overline{R}_t \equiv \mathbb{E}\left[R_t(\omega) \mid \mathcal{H}_t\right]$, and $\overline{u}_M^t \equiv \mathbb{E}\left[u_M^t(\omega) \mid \mathcal{H}_t\right]$, all expectations under the recommended strategy.

12. Histories on which a delivery has occurred are terminal for the pre-milestone problem: completion ends the game, and delivery of the milestone starts the continuation contract $\hat{\Gamma}^t$, to which the same argument applies verbatim.

LEMMA OA.2 (Incentive compatibility). *A contract Γ is incentive compatible if and only if, at every t and every history reached with positive probability,*

$$\overline{R}_t \geq u_t + \frac{\varphi}{\lambda_D} \qquad \textit{when } a_t = 1, \qquad \textit{and} \qquad \overline{u}_M^t \geq u_t + \frac{\varphi}{\lambda_S} \qquad \textit{when } a_t = 0. \tag{OA.2.2}$$

Proof. Consider a one-shot deviation at t on the direct route. By Lemma OA.1 the agent's posterior at completion is still μ_t, since completion is driven by his own effort and carries no information about the randomization device ω, so his expected payment on completing is $\overline{R}_t$. The comparison is then that of Section 4 with $\overline{R}_t$ in place of R_t, giving the first inequality; the second is identical with λ_S and $\overline{u}_M^t$.

That one-shot deviations suffice follows from Lemma OA.1: the agent's beliefs after any deviation coincide with his beliefs on path, so the continuation problem he faces after deviating is the one he faces without deviating. The requisite integrability is automatic—payments are nonnegative and the agent's total expected payoff is $u_0 < \infty$, so each component of $U_0(\Gamma)$ is bounded in expectation by u_0, and contracts with $\mathbb{E}[T(\omega)] = \infty$ may be excluded as they deliver the principal $-\infty$. Route deviations are deterred by the off-path punishment specified above, realization by realization and hence in expectation. □

Randomization thus enters only through conditional means, and (OA.2.2) is linear in $(u_t, \overline{R}_t, \overline{u}_M^t)$, so the set of incentive-compatible continuations is convex.

OA.2.3. Verification

As u_t is a conditional expectation adapted to $\mathcal{H}$, its pre-milestone dynamics are those of the deterministic problem in conditional means plus a martingale:

$$du_t = -\left[(\overline{R}_t - u_t)\lambda_D\, a_t + (\overline{u}_M^t - u_t)\lambda_S(1 - a_t)\right] dt + dN_t, \tag{OA.2.3}$$

with drift (HJB$_{PK}$) at $\overline{R}_t$ and $\overline{u}^t_M$, where N is a mean-zero martingale collecting revisions to the agent's beliefs about ω and the jump to zero on termination. If μ is degenerate, N vanishes and (OA.2.3) reduces to $\dot{u}_t = -\varphi$ under binding incentives.

PROPOSITION OA.1 (Randomization does not help). *Every contract* Γ *in* (OA.2.1) *satisfying* (OA.2.2) *and delivering* u_0 *has* $P_0(\Gamma) \leq \mathcal{V}(u_0)$ *(Proposition 2).*

Proof. Let M_t denote the principal's accumulated payoff up to t plus $\mathcal{V}(u_t)$ if the contract is still running at t, and zero thereafter, so that $M_0 = \mathcal{V}(u_0)$ and $\mathbb{E}[M_\infty] = P_0(\Gamma)$. We show that M is a supermartingale; optional stopping then gives $\mathbb{E}[M_\infty] \leq M_0$, which is the claim. By (OA.2.3), the drift of M has three terms.

(i) The deterministic drift. Together with the flow cost and the jumps at deliveries, this is exactly the expression appearing in the proof of Lemma OA.3, evaluated at $\overline{R}_t$ and $\overline{u}^t_M$. By Lemma OA.2 those means satisfy the same constraints as in the deterministic problem, so the same argument shows the term is non-positive.

(ii) Belief revisions. Let Δ be an increment of N from a belief revision, so that $\mathbb{E}[\Delta \mid \mathcal{H}_t] = 0$. By the tangent-line characterization of concavity (Lemma OA.5),

$$\mathbb{E}\left[\mathcal{V}(u_t + \Delta) \mid \mathcal{H}_t\right] - \mathcal{V}(u_t) \leq \mathcal{V}'(u_t) \cdot \mathbb{E}\left[\Delta \mid \mathcal{H}_t\right] = 0.$$

(iii) Termination. Suppose the contract terminates at hazard $h_t \geq 0$. Termination sends both continuation values to zero, and compensating the resulting jump adds $h_t u_t$ to the drift of u. The net contribution is $h_t[u_t\,\mathcal{V}'(u_t) + \mathcal{V}(0) - \mathcal{V}(u_t)] \leq 0$, because $\mathcal{V}(0) = 0$ and concavity gives $\mathcal{V}(0) \leq \mathcal{V}(u_t) - u_t\mathcal{V}'(u_t)$.

All three terms are non-positive, so M is a supermartingale. □

COROLLARY OA.3 (Soft deadlines). *No contract that terminates the agent at a random date improves upon the optimal contract of Theorem 1.*

Proof. A soft deadline is the case of (OA.2.1) in which $T(\omega)$ is non-degenerate, and is covered by term (iii) above. □

Appendix OA.3: Proofs for Section 5

This appendix follows the structure of Section 5. Appendix OA.3.1 treats the observable-effort problem of Section 5.1: it uses the crossing property of Appendix C.1, locates the two thresholds, and proves Proposition 3. Appendix OA.3.2 constructs the principal's value function under an efficiency loss and Appendix C.2 proves Theorem 2, which together discharge the first part of Section 5.2. Appendix OA.3.3 locates the thresholds in terms of the primitives at the two extremes of the efficiency loss and proves Propositions 4 and 5.

OA.3.1. Route Choice under an Exogenous Deadline

Throughout this subsection we assume $\lambda_D < \lambda_S < 2\lambda_D$, i.e. $0 < \eta < 1$, and use the gap $\Lambda_z[w]$ of (C.1), the limits (C.2), and the crossing property of Lemma C.1 and Corollary C.2, all in Appendix C.1.

The next two lemmas are Lemmas A.1 and A.2 with $\lambda_S = 2\lambda_D$ relaxed. Neither of the two concavity properties they rely on uses the knife-edge restriction.

LEMMA OA.1 (Locating the inner switch). *Suppose that $\lambda_D < \lambda_S < 2\lambda_D$ and $\Delta \le \overline{\Delta}$. Then Λ_D is strictly concave on $(0,1]$ and attains its maximum at*

$$x^* \equiv \left[\frac{\lambda_D(\lambda_S - \lambda_D)\left(\Pi - \frac{c}{\lambda_D}\right)}{\lambda_S(\lambda_S \Pi - c)} \cdot e^{\lambda_S \Delta} \right]^{\frac{\lambda_D}{\lambda_S - \lambda_D}}. \tag{OA.3.1}$$

Moreover:

(i) *if $\Delta \le \Delta^*(\eta, \pi)$, then $\Lambda_z[\mathcal{W}^D] \le 0$ for all $z \ge 0$, with equality at most at one point;*

(ii) *if $\Delta > \Delta^*(\eta, \pi)$, there is a unique $\hat{x} \in (x^*, 1]$ with $\Lambda_D(\hat{x}) = 0$, and, setting $\hat{T} \equiv -\log(\hat{x})/\lambda_D \ge 0$, we have $\Lambda_z[\mathcal{W}^D] < 0$ for all $z < \hat{T}$, $\Lambda_{\hat{T}}[\mathcal{W}^D] = 0$, and $\Phi(\hat{T}) > 0$.*

Proof of Lemma OA.1. Differentiating (A.11) twice, $\Lambda_D''(x) = -(\lambda_S\Pi - c)e^{-\lambda_S\Delta}(\lambda_S/\lambda_D)(\lambda_S/\lambda_D - 1)x^{\lambda_S/\lambda_D - 2} < 0$, where the sign uses $\lambda_S > \lambda_D$ and $\lambda_S\Pi > \lambda_D\Pi > c$ from (1). Setting $\Lambda_D'(x) = 0$ and solving gives (OA.3.1); since $\lambda_S > \lambda_D$ we have $\Lambda_D'(x) \to C \equiv (\lambda_S - \lambda_D)(\Pi - c/\lambda_D) > 0$ as $x \to 0$, so $x^* > 0$ is the global maximizer. Evaluating (A.11) at the two ends of the domain gives $\Lambda_D(0) \equiv \lim_{x\to 0}\Lambda_D(x) = ((\lambda_S - 2\lambda_D)/\lambda_D)c < 0$ and $\Lambda_D(1) = (\lambda_S\Pi - c)(1 - e^{-\lambda_S\Delta}) - \lambda_D\Pi$; $\Lambda_D(1) \le 0$ is precisely (A.10) reversed, i.e. $\Delta \le \overline{\Delta}$ with $\overline{\Delta}$ as in (7).

The value at the maximum. Write $\Lambda_D(x) = \Lambda_D(0) + Cx - (\lambda_S\Pi - c)e^{-\lambda_S\Delta}x^{\lambda_S/\lambda_D}$. Stationarity at x^* gives $(\lambda_S\Pi - c)e^{-\lambda_S\Delta}(x^*)^{\lambda_S/\lambda_D} = (\lambda_D/\lambda_S)Cx^*$, so the power term collapses into the linear one and

$$\Lambda_D(x^*) = \Lambda_D(0) + \frac{\lambda_S - \lambda_D}{\lambda_S}\,C\,x^* = \frac{\lambda_S - 2\lambda_D}{\lambda_D}\,c + \frac{(\lambda_S - \lambda_D)^2(\lambda_D\Pi - c)}{\lambda_S\lambda_D}\,x^*. \tag{OA.3.2}$$

Consequently $\Lambda_D(x^*) \le 0$ if and only if

$$x^* \le \frac{\lambda_S(2\lambda_D - \lambda_S)c}{(\lambda_S - \lambda_D)^2(\lambda_D\Pi - c)} = \frac{1 - \eta^2}{\eta^2(\pi - 1)}. \tag{OA.3.3}$$

Case (i). Suppose first that $x^* \ge 1$. Then Λ_D is strictly increasing on $(0, 1)$, so $\Lambda_D(x) < \Lambda_D(1) \le 0$ there. Substituting η and π into (OA.3.1) and (7), $x^* \ge 1$ and $\Delta \le \overline{\Delta}$ together give $(1 + \eta)(\eta\pi - 1) \le \eta(\pi - 1)$, that is $\eta^2\pi \le 1$ (directly if $\eta\pi \le 1$, in which case $\overline{\Delta} = \infty$). By (8) we then have $\Delta^* = \overline{\Delta} \ge \Delta$, so this case lies in (i).

Suppose instead that $x^* < 1$ and $\Lambda_D(x^*) \le 0$. Then $\Lambda_D(x) \le \Lambda_D(x^*) \le 0$ for all x, with equality only possible at x^*. By (OA.3.3), raising both sides to the power η and solving for Δ, the inequality $\Lambda_D(x^*) \le 0$ is exactly $\Delta \le \Delta^*(\eta, \pi)$ with Δ^* as in (8). In either case $\Lambda_z[\mathcal{W}^D] \le 0$ for all z, which is (i).

Case (ii). If $\Delta > \Delta^*$ then, by the previous paragraph, $x^* < 1$ and $\Lambda_D(x^*) > 0 \ge \Lambda_D(1)$. Since Λ_D is strictly decreasing on $[x^*, 1]$, it has a unique zero $\hat{x}$ there, and $\hat{x} > x^*$. Strict concavity and $\Lambda_D'(x^*) = 0$ give $\Lambda_D'(\hat{x}) < 0$, and $\Lambda_D(x) < 0$ for all $x \in (\hat{x}, 1]$. Because $x = e^{-\lambda_D z}$ is strictly decreasing in z, this says $\Lambda_z[\mathcal{W}^D] < 0$ for all $z < \hat{T}$, with equality at $\hat{T}$. Finally $\mathcal{W}^D$ satisfies (A.7) on $\mathbb{R}_+$, so Lemma C.1 applies at $\hat{T}$ and the chain rule gives $\Phi(\hat{T}) = (d/dz)\Lambda_z[\mathcal{W}^D]\big|_{z=\hat{T}} = -\lambda_D\hat{x}\,\Lambda_D'(\hat{x}) > 0$. □

LEMMA OA.2 (Locating the outer switch). *Suppose that* $\lambda_D < \lambda_S < 2\lambda_D$ *and that* $Z \geq 0$ *satisfies either* $\Lambda_Z[\mathcal{W}^D] > 0$, *or* $\Lambda_Z[\mathcal{W}^D] = 0$ *and* $\Phi(Z) > 0$. *Then* Λ_{SD} *is strictly concave on* $(0,1]$ *and there is a unique* $\check{y} \in (0,1)$ *with* $\Lambda_{SD}(\check{y}) = 0$. *Setting* $\check{T} \equiv Z - \log(\check{y})/\lambda_S > Z$, *we have* $\Lambda_z[\mathcal{W}_Z^{SD}] > 0$ *for all* $z \in (Z, \check{T})$, $\Lambda_{\check{T}}[\mathcal{W}_Z^{SD}] = 0$, *and* $\Phi(\check{T}) < 0$.

Proof of Lemma OA.2. Strict concavity is $\Lambda''_{SD}(y) = -(\lambda_S - \lambda_D)\kappa/y < 0$, and the limit in (C.2) is $\Lambda_Z[\mathcal{W}^D] + h = ((\lambda_S - 2\lambda_D)/\lambda_S)c$, which is strictly negative because $\lambda_S < 2\lambda_D$. Neither computation uses the knife-edge restriction.

We first show that $\Lambda_{SD}(y) > 0$ for y slightly below 1. If $\Lambda_Z[\mathcal{W}^D] > 0$ this is immediate from $\Lambda_{SD}(1) = \Lambda_Z[\mathcal{W}^D]$ and continuity. If instead $\Lambda_Z[\mathcal{W}^D] = 0$, then $\mathcal{W}_Z^{SD}$ satisfies (A.8) on $[Z, \infty)$, so Lemma C.1 gives a derivative of $\Phi(Z) > 0$ at Z and hence $\Lambda_z[\mathcal{W}_Z^{SD}] > 0$ for z slightly above Z, which is the same statement since $y = e^{-\lambda_S(z-Z)}$ is strictly decreasing in z.

Because Λ_{SD} is concave, the superlevel set $\{y \in (0,1] : \Lambda_{SD}(y) \geq 0\}$ is an interval. It contains a left neighborhood of 1 and, by the limit above, excludes a right neighborhood of 0; hence it equals $[\check{y}, 1]$ for a unique $\check{y} \in (0,1)$, with $\Lambda_{SD} > 0$ on $(\check{y}, 1)$ and $\Lambda_{SD} < 0$ on $(0, \check{y})$. Translating back, $\Lambda_z[\mathcal{W}_Z^{SD}] > 0$ for $z \in (Z, \check{T})$ and vanishes at $\check{T}$.

Finally, Λ_{SD} is positive somewhere on $(\check{y}, 1)$, so $\check{y}$ lies strictly below the maximizer of Λ_{SD} and strict concavity gives $\Lambda'_{SD}(\check{y}) > 0$. Applying Lemma C.1 at $\check{T}$ and the chain rule, $\Phi(\check{T}) = (d/dz)\Lambda_z[\mathcal{W}_Z^{SD}]|_{z=\check{T}} = -\lambda_S \check{y}\, \Lambda'_{SD}(\check{y}) < 0$. □

It remains to extend the candidate beyond $\check{T}$. Let $\mathcal{W}_{Z_1,Z_2}^{DSD}(z, \Delta)$ denote the expected surplus of the two-switch policy that uses the direct route for $z < Z_1$, the staged route for $Z_1 \leq z \leq Z_2$, and the direct route again for $z > Z_2$. Solving (A.7)

on (Z_2, ∞) with the value $\mathcal{W}^{SD}_{Z_1}(Z_2, \Delta)$ at Z_2,

$$\mathcal{W}^{DSD}_{Z_1,Z_2}(z,\Delta) = \begin{cases} \mathcal{W}^{SD}_{Z_1}(z,\Delta), & \text{if } z \le Z_2, \\ \left(\Pi - \dfrac{c}{\lambda_D}\right) - \left[\left(\Pi - \dfrac{c}{\lambda_D}\right) - \mathcal{W}^{SD}_{Z_1}(Z_2,\Delta)\right] e^{-\lambda_D (z - Z_2)}, & \text{if } z > Z_2. \end{cases} \tag{OA.3.4}$$

Proof of Proposition 3. We construct $\hat{T}$ and $\check{T}$ and verify that $\mathcal{W}^{DSD}_{\hat{T},\check{T}}$ solves (HJB$_W$) on $\mathbb{R}_+$ with value 0 at $z = 0$. Neither threshold depends on T, because (HJB$_W$) is stated in remaining time and its boundary condition is imposed at $z = 0$.

(a) Suppose $\Delta \le \Delta^*$. Since $\Delta^* \le \overline{\Delta}$, Lemma OA.1 applies and puts us in its case (i): $\Lambda_z[\mathcal{W}^D] \le 0$ for all z, so $\mathcal{W}^D$ solves (HJB$_W$) on $\mathbb{R}_+$ by (A.7). Put $\hat{T} = \check{T} = \infty$.

(b) Suppose $\Delta^* < \Delta \le \overline{\Delta}$. Lemma OA.1 case (ii) delivers $\hat{T} \ge 0$ with $\Lambda_{\hat{T}}[\mathcal{W}^D] = 0$ and $\Phi(\hat{T}) > 0$, which is the second hypothesis of Lemma OA.2 at $Z = \hat{T}$; that lemma then delivers $\check{T} \in (\hat{T}, \infty)$.

(c) Suppose $\Delta > \overline{\Delta}$. Set $\hat{T} = 0$. Since $\mathcal{W}^D(0) = 0$, we have $\Lambda_0[\mathcal{W}^D] = \lambda_S W^1_\Delta - \lambda_D \Pi > 0$ by (A.10), so Lemma OA.2 applies at $Z = 0$ and delivers $\check{T} \in (0, \infty)$.

In each case, write $w \equiv \mathcal{W}^{DSD}_{\hat{T},\check{T}}(\cdot, \Delta)$. On $[0, \hat{T})$ and on $(\hat{T}, \check{T})$ the branch of (HJB$_W$) that w follows attains the maximum, by Lemmas OA.1 and OA.2. On $(\check{T}, \infty)$, w satisfies (A.7) from a zero of $\Lambda_z[w]$ at $\check{T}$, where $\Phi(\check{T}) < 0$ by Lemma OA.2, so Corollary C.2 gives $\Lambda_z[w] < 0$ and the direct branch attains the maximum again. At $\hat{T}$ and $\check{T}$ we have $\Lambda_z[w] = 0$, so the two branches agree and w is differentiable; since $w(0) = 0$, verification gives $w(z) = W^{0,\Delta}_z$ for all $z \ge 0$.

Fix a deadline T and translate remaining time into calendar time through $t = T - z$. The optimal policy uses the direct route when $z < \hat{T}$ or $z > \check{T}$ and the staged route when $z \in (\hat{T}, \check{T})$, which is the two-switch policy of Definition 3 with $S_1 = \max\{T - \check{T}, 0\}$ and $S_2 = \max\{T - \hat{T}, 0\}$. The three cases above give the threshold values asserted in parts (a), (b) and (c) of the proposition. □

OA.3.2. The Value Function under Efficiency Loss

Throughout this subsection $0 < \eta < 1$ and $\pi > 1$. The results of Online Appendix OA.1 that do not invoke $\lambda_S = 2\lambda_D$ remain available: in particular the incentive conditions of Lemma OA.1, the verification bound of Lemma OA.3, and the implementation results of Lemma OA.4. What must be redone is the construction of the candidate value function, since the optimal policy now has three route regions rather than two.

Two conventions carry over from Appendix OA.3.1. Under binding incentive conditions promised utility and remaining time are proportional, $u = \varphi z$, and the extension is $\Delta = 1/\lambda_S$; the staged advantage Λ_D of (A.11) therefore reappears here, in the variable $x \equiv e^{-\lambda_D u/\varphi}$ and evaluated at $\Delta = 1/\lambda_S$, as

$$\Lambda_D(x;\pi) = -\left\{(\eta+1)\pi - 1\right\} \cdot c \cdot e^{-1} x^{\eta+1} + \eta(\pi-1) \cdot c \cdot x - (1-\eta) \cdot c, \quad \text{(OA.3.5)}$$

where the dependence on π is displayed because the comparative statics of Appendix OA.3.3 vary it. The object governing the outer threshold, $\tilde{\Lambda}_{SD}$ of (OA.3.10) below, measures the advantage of the *direct* route rather than the staged one, so its sign convention is opposite to that of Λ_D; this is why it is convex where Λ_D is concave.

OA.3.2.1. A third branch. Let $V^{DSD}(u \mid u_1, u_2)$ denote the principal's payoff under a two-switch contract with switching points $u_1 < u_2$ in promised utility.

Lemma OA.3 (Implementation of the two-switch contract).

(a) *When $0 < u_1 < u_2 < u$, the two-switch contract with intermediate deadlines $(u - u_2)/\varphi$ and $(u - u_1)/\varphi$ and final deadline u/φ implements $\left(V^{DSD}(u \mid u_1, u_2), u\right)$, where*

$$V^{DSD}(u \mid u_1, u_2) \equiv \left(\Pi - \frac{c}{\lambda_D}\right)\left(1 - e^{\frac{\lambda_D}{\varphi}(u_2 - u)}\right) + \left(V^{SD}(u_2 \mid u_1) + u_2\right) e^{\frac{\lambda_D}{\varphi}(u_2 - u)} - u. \quad \text{(OA.3.6)}$$

(b) *The following differential equation holds:*

$$\varphi V^{DSD\prime}(u \mid u_1, u_2) = \lambda_D \left(\Pi - \frac{\varphi}{\lambda_D} - u - V^{DSD}(u \mid u_1, u_2)\right) - c. \quad \text{(OA.3.7)}$$

Proof of Lemma OA.3. **(a)** Let $\Gamma_{DSD}(T_1, T_2, T)$ denote a contract with two switches at $T_1 = (u - u_2)/\varphi$ and $T_2 = (u - u_1)/\varphi$ and a deadline $T = u/\varphi$. At time T_1, if the project has not been completed, the remaining contract is $\Gamma_{SD}(T_2 - T_1, T - T_1)$, and $U_0(\Gamma_{SD}(T_2 - T_1, T - T_1)) = \varphi(T - T_1)$ by Lemma OA.4(b). The agent's expected payoff at time 0 is therefore

$$\begin{aligned} U_0(\Gamma_{DSD}(T_1, T_2, T)) =& \int_0^{T_1} \varphi(T + 1/\lambda_D - \tau)\lambda_D e^{-\lambda_D \tau} d\tau + e^{-\lambda_D T_1} U_0(\Gamma_{SD}(T_2 - T_1, T - T_1)) \\ =& \varphi T - \varphi(T - T_1)e^{-\lambda_D T_1} + e^{-\lambda_D T_1}\varphi(T - T_1) \;=\; \varphi T = u. \end{aligned}$$

For the principal,

$$\begin{aligned} & P_0(\Gamma_{DSD}(T_1, T_2, T)) + U_0(\Gamma_{DSD}(T_1, T_2, T)) \\ =& \int_0^{T_1} (\Pi - c\tau)\lambda_D e^{-\lambda_D \tau} d\tau - cT_1 e^{-\lambda_D T_1} \\ & + e^{-\lambda_D T_1} \left(P_0(\Gamma_{SD}(T_2 - T_1, T - T_1)) + U_0(\Gamma_{SD}(T_2 - T_1, T - T_1))\right). \end{aligned}$$

By Lemma OA.4(b), $U_0(\Gamma_{SD}(T_2', T_1')) + P_0(\Gamma_{SD}(T_2', T_1')) = V^{SD}(\varphi T_1' \mid \varphi(T_1' - T_2')) + \varphi(T_1' - T_2')$. Substituting $T_1' = T - T_1$, $T_2' = T_2 - T_1$, $\varphi T_1 = u - u_2$ and $\varphi T_2 = u - u_1$, the right-hand side equals

$$\left(\Pi - \frac{c}{\lambda_D}\right)(1 - e^{-\lambda_D T_1}) + e^{-\lambda_D T_1} \cdot \left\{V^{SD}(u_2 \mid u_1) + u_2\right\} = V^{DSD}(u \mid u_1, u_2) + u,$$

so $P_0(\Gamma_{DSD}(T_1, T_2, T)) = V^{DSD}(u \mid u_1, u_2)$.

(b) Differentiating (OA.3.6) and multiplying by φ,

$$\begin{aligned} \varphi V^{DSD\prime}(u \mid u_1, u_2) =& \lambda_D \left(\Pi - \frac{c}{\lambda_D}\right) e^{\frac{\lambda_D}{\varphi}(u_2 - u)} - \lambda_D \left(V^{SD}(u_2 \mid u_1) + u_2\right) e^{\frac{\lambda_D}{\varphi}(u_2 - u)} - \varphi \\ =& \lambda_D \left(\Pi - \frac{\varphi}{\lambda_D} - u - V^{DSD}(u \mid u_1, u_2)\right) - c, \end{aligned}$$

which is (OA.3.7). □

The candidate value function is accordingly

$$\mathcal{V}(u) = \begin{cases} V^{D}(u), & \text{if } 0 \le u \le \hat{u}_1, \\ V^{SD}(u \mid \hat{u}_1), & \text{if } \hat{u}_1 < u \le \hat{u}_2, \\ V^{DSD}(u \mid \hat{u}_1, \hat{u}_2), & \text{if } \hat{u}_2 < u, \end{cases} \tag{OA.3.8}$$

with $0 \le \hat{u}_1 \le \hat{u}_2 \le \infty$ to be determined. Section 4 is the case $\hat{u}_2 = \infty$, in which the third branch is never reached.

OA.3.2.2. The two switching points. The thresholds are located by smooth pasting, exactly as in Appendix OA.3.1 but now in the promised-utility variable, with $\hat{u}_i = \varphi \cdot$ (the corresponding threshold in remaining time).

Proposition OA.1 (Value function under efficiency loss). *Suppose $0 < \eta < 1$ and $\pi > 1$. There exist $\hat{u}_2 \ge \hat{u}_1 \ge 0$ such that*

(i) $V^{D\prime}(u) > V^{SD\prime}(u \mid u)$ *for all* $0 \le u < \hat{u}_1$;

(ii) $V^{SD\prime}(\hat{u}_1 \mid \hat{u}_1) \ge V^{D\prime}(\hat{u}_1)$, *with* $V^{SD\prime\prime}(\hat{u}_1 \mid \hat{u}_1) > V^{D\prime\prime}(\hat{u}_1)$ *when the equality holds;*

(iii) $V^{SD\prime}(u \mid \hat{u}_1) > V^{DSD\prime}(u \mid \hat{u}_1, u)$ *for all* $\hat{u}_1 < u < \hat{u}_2$;

(iv) $V^{DSD\prime}(\hat{u}_2 \mid \hat{u}_1, \hat{u}_2) = V^{SD\prime}(\hat{u}_2 \mid \hat{u}_1)$ *and* $V^{DSD\prime\prime}(\hat{u}_2 \mid \hat{u}_1, \hat{u}_2) > V^{SD\prime\prime}(\hat{u}_2 \mid \hat{u}_1)$;

(v) $V^{DSD\prime}(u \mid \hat{u}_1, \hat{u}_2) > (1/\varphi)\left[\lambda_S\left(V_M(u + \varphi/\lambda_S) - V^{DSD}(u \mid \hat{u}_1, \hat{u}_2)\right) - c\right]$ *for all* $u > \hat{u}_2$.

With these thresholds, $\mathcal{V}$ of (OA.3.8) is the principal's value function.

Conditions (i)–(ii) locate $\hat{u}_1$ and are the promised-utility counterparts of Lemma OA.1; conditions (iii)–(iv) locate $\hat{u}_2$ and correspond to Lemma OA.2; condition (v) says that the direct route remains optimal above $\hat{u}_2$, corresponding to Corollary C.2. Lemmas OA.4 and OA.5 deliver (i) and (ii) and construct the thresholds $\tilde{\pi}_S(\eta)$ and $\tilde{\pi}_M(\eta)$ used in Appendix OA.3.3; Lemma OA.6 delivers (iii) and (iv); Lemma OA.7 delivers (v). The final claim follows from Lemmas OA.8 and OA.3.

LEMMA OA.4 (Staging at the deadline). *If $\eta \le 1/(e-1)$, the inequality $V^{D\prime}(0) > V^{SD\prime}(0|0)$ always holds. If $\eta > 1/(e-1)$, $V^{D\prime}(0) > V^{SD\prime}(0|0)$ is equivalent to $\pi < \tilde{\pi}_S(\eta) \equiv (e-1)/(\eta(e-1)-1)$. Moreover, if $\pi = \tilde{\pi}_S(\eta)$, then $V^{D\prime}(0) = V^{SD\prime}(0|0)$ and $V^{D\prime\prime}(0) < V^{SD\prime\prime}(0|0)$.*

Proof of Lemma OA.4. By $V^D(0) = V^{SD}(0|0) = 0$ and Lemma OA.4(d),

$$\varphi V^{D\prime}(0) = \lambda_D \Pi - \varphi - c,$$

$$\varphi V^{SD\prime}(0|0) = \lambda_S V_M(\tfrac{\varphi}{\lambda_S}) - c = \lambda_S \left(\Pi - \frac{c}{\lambda_S}\right)(1 - e^{-1}) - \varphi - c.$$

Therefore $V^{D\prime}(0) > V^{SD\prime}(0|0)$ is equivalent to $(\eta(e-1)-1)\,\lambda_D \Pi < c(e-1)$. When $\eta \le 1/(e-1)$ the left-hand side is non-positive and the inequality always holds; when $\eta > 1/(e-1)$ it is equivalent to $\pi < \tilde{\pi}_S(\eta)$.

Next assume $\eta > 1/(e-1)$ and $\pi = \tilde{\pi}_S(\eta)$. With some algebra, $\varphi^2 V^{SD\prime\prime}(0|0) - \varphi^2 V^{D\prime\prime}(0) = \lambda_D c\left[(e-1)\eta^2/((e-1)\eta - 1)\right]$, which is positive since $\eta > 1/(e-1)$, so $V^{SD\prime\prime}(0|0) > V^{D\prime\prime}(0)$. □

LEMMA OA.5 (Staging at some deadline). *There exists $\tilde{\pi}_M(\eta) \ge 2\lambda_D/\lambda_S = 2/(\eta+1)$ with $\tilde{\pi}_M(1) = 1$ such that the following statements hold.*

(a) *If $1 \le \pi < \tilde{\pi}_M(\eta)$, then $V^{D\prime}(u) > V^{SD\prime}(u|u)$ for all $u \ge 0$.*

(b) *Suppose that either (i) $\eta \le 1/(e-1)$ and $\pi > \tilde{\pi}_M(\eta)$, or (ii) $\eta > 1/(e-1)$ and $\tilde{\pi}_S(\eta) \ge \pi > \tilde{\pi}_M(\eta)$. Then there exists $\hat{u}_1 > 0$ such that $V^{D\prime}(\hat{u}_1) = V^{SD\prime}(\hat{u}_1|\hat{u}_1)$, $V^{D\prime\prime}(\hat{u}_1) < V^{SD\prime\prime}(\hat{u}_1|\hat{u}_1)$, and $V^{D\prime}(u) > V^{SD\prime}(u|u)$ for all $u \in [0, \hat{u}_1)$.*

Proof of Lemma OA.5. By (OA.1.4) and (OA.1.5), $\varphi V^{SD\prime}(u|u) - \varphi V^{D\prime}(u)$ equals $\Lambda_D(x;\pi)$ of (OA.3.5) with $x = e^{-\lambda_D u/\varphi}$. Since $(\partial^2 \Lambda_D/\partial x^2)(x;\pi) = -(\eta+1)\eta\{(\eta+1)\pi - 1\} \cdot c \cdot e^{-1} x^{\eta-1} < 0$ for $\pi \ge 1$, the function Λ_D is strictly concave in x. Let $x^*(\pi)$ solve $\max_{0 \le x \le 1} \Lambda_D(x;\pi)$. The first-order condition gives

$$x^*(\pi) = \left[\frac{\eta(\pi-1)}{(\eta+1)\{(\eta+1)\pi - 1\}e^{-1}}\right]^{\frac{1}{\eta}}. \tag{OA.3.9}$$

[13] Define

$$h(\pi) \equiv \Lambda_D(x^*(\pi);\pi) = K\left(\frac{\pi-1}{(\eta+1)\pi-1}\right)^{\frac{1}{\eta}}(\pi-1)\cdot c-(1-\eta)\cdot c,$$

where $K=(\eta^2/(\eta+1))\,(\eta e/(\eta+1))^{1/\eta}$. Then

$$h\left(\frac{2}{\eta+1}\right)=(1-\eta)c\left[\frac{\eta^2}{(\eta+1)^2}\left(\frac{\eta(1-\eta)e}{(\eta+1)^2}\right)^{\frac{1}{\eta}}-1\right]<0$$

because $\eta<1$ and $\eta(1-\eta)e\le e/4<1\le(\eta+1)^2$. Moreover $\lim_{\pi\to\infty}h(\pi)=\infty$ and

$$h'(\pi)=\frac{K\pi(1+\eta)c}{\pi-1}\cdot\left(\frac{\pi-1}{(\eta+1)\pi-1}\right)^{\frac{1}{\eta}+1}>0,$$

so there is a unique $\pi\ge 2/(\eta+1)$ with $h(\pi)=0$; denote it $\tilde{\pi}_M(\eta)$. At $\eta=1$ we have $h(2/(\eta+1))=h(1)=0$, so $\tilde{\pi}_M(1)=1$.

(a) If $1\le\pi<\tilde{\pi}_M(\eta)$ then $0>h(\pi)=\Lambda_D(x^*(\pi);\pi)\ge\Lambda_D(x;\pi)$ for all $0\le x\le 1$, which is $V^{D\prime}(u)>V^{SD\prime}(u|u)$ for all $u\ge 0$.

(b) Suppose first $\eta\le 1/(e-1)$ and $\pi>\tilde{\pi}_M(\eta)$, so $h(\pi)=\Lambda_D(x^*(\pi);\pi)>0$. By Lemma OA.4, $\Lambda_D(1;\pi)=\varphi(V^{SD\prime}(0|0)-V^{D\prime}(0))<0$. Concavity of Λ_D and $(\partial\Lambda_D/\partial x)(x^*(\pi);\pi)=0$ then give $x_1\in(x^*(\pi),1]$ with $\Lambda_D(x_1;\pi)=0$, $(\partial\Lambda_D/\partial x)(x_1;\pi)<0$, and $\Lambda_D(x;\pi)<0$ for all $x\in(x_1,1]$. Setting $\hat{u}_1\equiv-(\varphi/\lambda_D)\log x_1$ translates these into the three stated conditions.

Suppose instead $\eta>1/(e-1)$. By the definition of $\tilde{\pi}_S(\eta)$, if $\pi\ge\tilde{\pi}_S(\eta)$ then $\Lambda_D(1;\pi)\ge 0$, whence $h(\pi)\ge\Lambda_D(1;\pi)\ge 0$ and $\pi\ge\tilde{\pi}_M(\eta)$; therefore $\tilde{\pi}_S(\eta)\ge\tilde{\pi}_M(\eta)$. If $\tilde{\pi}_S(\eta)\ge\pi>\tilde{\pi}_M(\eta)$ then $\Lambda_D(x^*(\pi);\pi)>0>\Lambda_D(1;\pi)$, and the same argument applies. □

LEMMA OA.6 (Locating the outer threshold). *Suppose that $\eta<1$, $\Pi>c/\lambda_D$, and either (i) $V^{D\prime}(\hat{u}_1)<V^{SD\prime}(\hat{u}_1|\hat{u}_1)$, or (ii) $V^{D\prime}(\hat{u}_1)=V^{SD\prime}(\hat{u}_1|\hat{u}_1)$ and $V^{D\prime\prime}(\hat{u}_1)<$*

13. When $\pi\le((\eta+1)e^{-1}-\eta)/((\eta+1)^2e^{-1}-\eta)$ the maximizer is $x^*(\pi)=1$. Since $((\eta+1)e^{-1}-\eta)/((\eta+1)^2e^{-1}-\eta)<1$ for every $\eta>0$, the interior solution applies whenever $\pi\ge 1$.

$V^{SD''}(\hat{u}_1|\hat{u}_1)$. *Then there exists a unique* $\hat{u}_2 > \hat{u}_1$ *such that* $V^{SD'}(\hat{u}_2|\hat{u}_1) = V^{DSD'}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$ *and* $V^{SD''}(\hat{u}_2|\hat{u}_1) < V^{DSD''}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$. *Moreover* $V^{SD'}(u|\hat{u}_1) > V^{DSD'}(u|\hat{u}_1, u)$ *for all* $u \in (\hat{u}_1, \hat{u}_2)$.

Proof of Lemma OA.6. Using (OA.3.7), (OA.1.5) and $V^{DSD}(u|\hat{u}_1, u) = V^{SD}(u|\hat{u}_1)$, the difference $\varphi V^{DSD'}(u|\hat{u}_1, u) - \varphi V^{SD'}(u|\hat{u}_1)$ equals

$$\lambda_D \Pi - \lambda_S \left\{V_M(u + \varphi/\lambda_S) + u + \varphi/\lambda_S\right\} + (\lambda_S - \lambda_D)(V^{SD}(u|\hat{u}_1) + u).$$

Proceeding as in (A.14) and writing $y \equiv e^{-(\lambda_S/\varphi)(u-\hat{u}_1)}$, this becomes

$$\begin{aligned} \tilde{\Lambda}_{SD}(y) \;\equiv\; & \frac{1-\eta}{1+\eta} c + (\lambda_S \Pi - c) e^{-1-\frac{\lambda_S}{\varphi}\hat{u}_1} \left[1 + \frac{\eta}{1+\eta} \log y\right] y \\ & + \eta \left[\frac{1-\eta}{1+\eta} c - (\lambda_D \Pi - c) e^{-\frac{\lambda_D}{\varphi}\hat{u}_1}\right] y. \end{aligned} \tag{OA.3.10}$$

By hypothesis $\tilde{\Lambda}_{SD}(1) = \varphi V^{D'}(\hat{u}_1) - \varphi V^{SD'}(\hat{u}_1|\hat{u}_1) \leq 0$. Differentiating twice, $\tilde{\Lambda}''_{SD}(y) = (\eta/(1+\eta))(\lambda_S \Pi - c) e^{-1-(\lambda_S/\varphi)\hat{u}_1} 1/y > 0$ since $\Pi > c/\lambda_D > c/\lambda_S$, so $\tilde{\Lambda}_{SD}$ is strictly convex; and $\lim_{y\to 0} \tilde{\Lambda}_{SD}(y) = ((1-\eta)/(1+\eta))c > 0$. Convexity then yields a unique $y_2 \in (0,1)$ with $\tilde{\Lambda}_{SD}(y) < 0$ on $(y_2, 1)$, $\tilde{\Lambda}_{SD}(y_2) = 0$, and $\tilde{\Lambda}'_{SD}(y_2) < 0$. Setting $\hat{u}_2 = \hat{u}_1 - (\varphi/\lambda_S) \log y_2$ gives $V^{SD'}(u|\hat{u}_1) > V^{DSD'}(u|\hat{u}_1, u)$ for all $u \in (\hat{u}_1, \hat{u}_2)$ and equality at $\hat{u}_2$; and since y is decreasing in u, $\tilde{\Lambda}'_{SD}(y_2) < 0$ gives $V^{SD''}(\hat{u}_2|\hat{u}_1) < V^{DSD''}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$. □

Lemma OA.7 (No return to staging). *Suppose that* $\pi > 1$, $V^{SD'}(\hat{u}_2|\hat{u}_1) = V^{DSD'}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$ *and* $V^{SD''}(\hat{u}_2|\hat{u}_1) < V^{DSD''}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$. *Then, for all* $u > \hat{u}_2$,

$$\varphi V^{DSD'}(u|\hat{u}_1, \hat{u}_2) > \lambda_S \left(V_M(u + \varphi/\lambda_S) - V^{DSD}(u|\hat{u}_1, \hat{u}_2)\right) - c.$$

Proof of Lemma OA.7. Differentiating (OA.1.4) and (OA.1.5),

$$\varphi V^{SD''}(u|\hat{u}_1) = \lambda_S \left(V'_M\left(u + \varphi/\lambda_S\right) + 1\right) - \lambda_S \left(V^{SD'}(u|\hat{u}_1) + 1\right),$$

$$\varphi V^{DSD''}(u|\hat{u}_1, \hat{u}_2) = -\lambda_D \left(V^{DSD'}(u|\hat{u}_1, \hat{u}_2) + 1\right),$$

so the hypotheses imply

$$(\lambda_S - \lambda_D)(1 + V^{SD\prime}(\hat{u}_2|\hat{u}_1)) > \lambda_S(V_M'(\hat{u}_2 + \varphi/\lambda_S) + 1)$$

$$\iff \qquad \eta(1 + V^{SD\prime}(\hat{u}_2|\hat{u}_1)) > (\eta + 1)\left(\frac{\lambda_S \Pi - c}{\varphi}\right) e^{-\frac{\lambda_S}{\varphi}\hat{u}_2 - 1}. \quad \text{(OA.3.11)}$$

Define $\Lambda_{DSD} : [\hat{u}_2, \infty) \to \mathbb{R}$ by

$$\Lambda_{DSD}(u) \equiv \lambda_S \left[V_M(u + \varphi/\lambda_S) - V^{DSD}(u|\hat{u}_1, \hat{u}_2)\right] - \varphi V^{DSD\prime}(u|\hat{u}_1, \hat{u}_2) - c,$$

so that

$$\Lambda_{DSD}(u) = (\eta - 1)\, c - (\lambda_S \Pi - c) e^{-\frac{\lambda_S}{\varphi}\hat{u}_2 - 1} \cdot e^{\frac{\lambda_S}{\varphi}(\hat{u}_2 - u)} + \eta\varphi \left(V^{SD\prime}(\hat{u}_2|\hat{u}_1) + 1\right) e^{\frac{\lambda_D}{\varphi}(\hat{u}_2 - u)},$$

and

$$\begin{aligned}\Lambda_{DSD}(\hat{u}_2) &= \lambda_S \left[V_M(\hat{u}_2 + \varphi/\lambda_S) - V^{SD}(\hat{u}_2|\hat{u}_1)\right] - c - \varphi V^{DSD\prime}(\hat{u}_2|\hat{u}_1, \hat{u}_2) \\ &= \varphi V^{SD\prime}(\hat{u}_2|\hat{u}_1) - \varphi V^{DSD\prime}(\hat{u}_2|\hat{u}_1, \hat{u}_2) = 0.\end{aligned}$$

Writing $x \equiv e^{(\lambda_D/\varphi)(\hat{u}_2 - u)}$,

$$\tilde{\Lambda}_{DSD}(x) = (\eta - 1)c - (\lambda_S \Pi - c) e^{-\frac{\lambda_S}{\varphi}\hat{u}_2 - 1} x^{\eta + 1} + \eta\varphi \left(V^{SD\prime}(\hat{u}_2|\hat{u}_1) + 1\right) x,$$

with $\tilde{\Lambda}_{DSD}(1) = \Lambda_{DSD}(\hat{u}_2) = 0$. By (OA.3.11),

$$\tilde{\Lambda}_{DSD}'(1) = -(\eta + 1)(\lambda_S \Pi - c) e^{-\frac{\lambda_S}{\varphi}\hat{u}_2 - 1} + \eta\varphi \left(V^{SD\prime}(\hat{u}_2|\hat{u}_1) + 1\right) > 0,$$

and $\tilde{\Lambda}_{DSD}''(x) = -(\eta + 1)\eta(\lambda_S \Pi - c) e^{-(\lambda_S/\varphi)\hat{u}_2 - 1} x^{\eta - 1} < 0$. Hence $\tilde{\Lambda}_{DSD}' > 0$ on $(0, 1)$, and since $\tilde{\Lambda}_{DSD}(1) = 0$ we get $\tilde{\Lambda}_{DSD}(x) < 0$ for all $x \in (0, 1)$, which is the claim. □

Proof of Proposition OA.1. Conditions (i) and (ii) are Lemmas OA.4 and OA.5: the former settles whether $\hat{u}_1 = 0$, and the latter constructs $\hat{u}_1 > 0$ when it is positive, together with the inequality $V^{D\prime} > V^{SD\prime}$ below it. Condition (iii) and the first half

of (iv) are Lemma OA.6, whose hypotheses are exactly (ii); the second half of (iv) is the same lemma's second-order conclusion. Condition (v) is Lemma OA.7, whose hypotheses are exactly (iv).

For the final claim, let $\mathcal{V}$ be given by (OA.3.8) with these thresholds. It satisfies $\mathcal{V}(0) = 0$. It is differentiable: each branch is differentiable on its interior, and where two branches meet the one-sided derivatives agree—at $\hat{u}_2$ by (iv), and at $\hat{u}_1$ by (ii), which holds with equality whenever $\hat{u}_1 > 0$ and is vacuous otherwise, since the first branch is then empty. It is concave by Lemma OA.8. By (OA.1.4), (OA.1.5) and (OA.3.7), $\mathcal{J}$ vanishes on each branch when evaluated at the binding-incentive terms, and by Lemma OA.9 it is non-positive at every feasible deviation; hence $\mathcal{V}$ solves (HJB_V) subject to (IC). Lemma OA.3 then bounds the payoff of every incentive-compatible contract delivering u by $\mathcal{V}(u)$, and Lemmas OA.4 and OA.3 exhibit a contract attaining it. □

OA.3.2.3. Concavity and verification. Two facts complete the argument, and neither uses the knife edge.

Lemma OA.8 (Concavity). *Suppose $\pi > 1$ and $0 < \eta < 1$. Then $\mathcal{V}$ defined in (OA.3.8) is concave.*

Proof of Lemma OA.8. When $u < \hat{u}_1$, $\mathcal{V}''(u) = V^{D\,\prime\prime}(u) < 0$ by Lemma OA.5(c).

When $\hat{u}_1 < u < \hat{u}_2$, the bound (OA.1.8) still applies, and $\lambda_S < 2\lambda_D$ gives $\mathcal{V}''(u) = V^{SD\,\prime\prime}(u|\hat{u}_1) \leq 0$.

When $u > \hat{u}_2$, differentiating (OA.3.6) twice,

$$V^{DSD\,\prime\prime}(u|\hat{u}_1, \hat{u}_2) = -\left(\frac{\lambda_D}{\varphi}\right)^2 \cdot \left(\Pi - \frac{c}{\lambda_D} - (V^{SD}(\hat{u}_2 \mid \hat{u}_1) + \hat{u}_2)\right) e^{-\frac{\lambda_D}{\varphi}(u-\hat{u}_2)},$$

which is negative because the total surplus $V^{SD}(\hat{u}_2 \mid \hat{u}_1) + \hat{u}_2$ cannot exceed the first-best surplus $\Pi - c/\lambda_D$. Since the three branches are smoothly pasted at $\hat{u}_1$ and $\hat{u}_2$, $\mathcal{V}$ is concave throughout. □

Part of this is worth a remark, because it is the one place where an efficiency loss makes the argument *easier* rather than harder. On the middle branch the bracketed term of (OA.1.8) carries the factor $\lambda_S - 2\lambda_D$. At the knife edge that term vanishes and concavity follows from the remaining term alone; when $\lambda_S < 2\lambda_D$ the factor is strictly negative, so the bracket reinforces rather than offsets.

LEMMA OA.9 (No profitable slack). *Suppose $\pi > 1$ and $0 < \eta < 1$. Then for every $u \geq 0$, $L^D(u, R) \leq 0$ for all $R \geq u + \varphi/\lambda_D$, and $L^S(u, w) \leq 0$ for all $w \geq u + \varphi/\lambda_S$.*

Proof of Lemma OA.9. As in the case without efficiency loss, $\mathcal{V}(u) + u$ is increasing in u, so $\mathcal{V}'(u) \geq -1$ and $\partial L^D/\partial R \leq 0$. It therefore suffices to show $L^D(u, u + \varphi/\lambda_D) \leq 0$ for all $u \geq 0$. From (OA.1.4), (OA.1.5) and (OA.3.7),

$$L^D(u, u + \varphi/\lambda_D) = \begin{cases} 0, & \text{if } u \leq \hat{u}_1 \text{ or } u \geq \hat{u}_2, \\ \varphi V^{DSD'}(u|\hat{u}_1, u) - \varphi V^{SD'}(u|\hat{u}_1), & \text{if } u \in (\hat{u}_1, \hat{u}_2), \end{cases}$$

which is non-positive because $\hat{u}_1$ and $\hat{u}_2$ satisfy $V^{DSD'}(u|\hat{u}_1, u) < V^{SD'}(u|\hat{u}_1)$ on $(\hat{u}_1, \hat{u}_2)$ by Lemma OA.6.

Likewise, concavity of $\mathcal{V}$ gives $\partial L^S/\partial u \geq 0$, so it suffices to show $L^S(u, u + \varphi/\lambda_S) \leq 0$ for all $u \geq 0$. Here

$$L^S(u, u + \varphi/\lambda_S) = \begin{cases} \varphi V^{SD'}(u|u) - \varphi V^{D'}(u), & \text{if } u \leq \hat{u}_1, \\ 0, & \text{if } u \in (\hat{u}_1, \hat{u}_2), \\ \lambda_S(V_M(u + \varphi/\lambda_S) - V^{DSD}(u|\hat{u}_1, \hat{u}_2)) & \\ \qquad -c - \varphi V^{DSD'}(u|\hat{u}_1, \hat{u}_2), & \text{if } u \geq \hat{u}_2, \end{cases}$$

which is non-positive by Lemma OA.5 on the first branch and Lemma OA.7 on the third. □

REMARK OA.3.1. The reversed one-switch contract arises only at intermediate efficiency losses. It requires $\hat{u}_1 = 0$ together with $\overline{u} > \hat{u}_2$. When staging is

technologically innocuous the second condition fails, since $\hat{u}_2 = \infty$; under a small efficiency loss it fails as well, since Lemma OA.11 gives $\hat{u}_2 \geq \overline{u}$. Under a large efficiency loss the first condition fails instead: $\hat{u}_1 = 0$ requires $V^{SD\prime}(0|0) \geq V^{D\prime}(0)$, which by Lemma OA.4 is impossible when $\eta \leq 1/(e-1)$, and $\underline{\eta} \leq 1/(e-1)$ by definition.

OA.3.3. Extreme Efficiency Losses

Theorem 2 classifies the optimal contract by the position of $\overline{u}$ relative to $\hat{u}_1$ and $\hat{u}_2$, but leaves all three quantities implicit. This subsection locates them in terms of the primitives at the two extremes of the efficiency loss. Three lemmas do the work: two compare $\overline{u}$ with $\hat{u}_2$, in opposite directions and under opposite hypotheses, and the third compares $\overline{u}$ with $\hat{u}_1$. Since $\mathcal{V}$ is concave, each comparison reduces to a sign condition on $\mathcal{V}'$ evaluated at a threshold, which is what the three lemmas exploit.

Throughout, $\tilde{\pi}_S(\eta)$ of Lemma OA.4 is the return above which the staged route survives to the deadline ($\hat{u}_1 = 0$), and $\tilde{\pi}_M(\eta)$ of Lemma OA.5 the return above which the staged route is used at all ($\hat{u}_1 < \infty$).

LEMMA OA.10 (Reaching the outer region). *$\hat{u}_2 < \overline{u}$ if $\pi > \tilde{\pi}_M(\eta)$ and $\eta \leq c/(c+\varphi)$.*

Proof of Lemma OA.10. Since $\mathcal{V}$ is strictly concave, $\hat{u}_2 < \overline{u}$ is equivalent to $0 < \mathcal{V}'(\hat{u}_2) = V^{SD\prime}(\hat{u}_2|\hat{u}_1) = V^{DSD\prime}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$, and by (OA.3.7) the last inequality is

$$\lambda_D \left(\hat{u}_2 + V^{SD}(\hat{u}_2|\hat{u}_1)\right) < \lambda_D \Pi - c - \varphi. \tag{OA.3.12}$$

Since $V^{SD\prime}(\hat{u}_2|\hat{u}_1) = V^{DSD\prime}(\hat{u}_2|\hat{u}_1, \hat{u}_2)$ and $V^{DSD}(\hat{u}_2|\hat{u}_1, \hat{u}_2) = V^{SD}(\hat{u}_2|\hat{u}_1)$, equations (OA.1.4) and (OA.1.5) give

$$\lambda_D \left(\Pi - \hat{u}_2 - V^{SD}(\hat{u}_2|\hat{u}_1)\right) = \lambda_S \left(V_M\left(\hat{u}_2 + \varphi/\lambda_S\right) + \hat{u}_2 + \varphi/\lambda_S\right) - \lambda_S \left(V^{SD}(\hat{u}_2|\hat{u}_1) + \hat{u}_2\right).$$

Substituting (OA.1.1),

$$(\lambda_S - \lambda_D)\left(V^{SD}(\hat{u}_2|\hat{u}_1) + \hat{u}_2\right) = \lambda_S\left(\Pi - \frac{c}{\lambda_S}\right)\left(1 - e^{-\frac{\lambda_S}{\varphi}\hat{u}_2 - 1}\right) - \lambda_D\Pi$$

$$\iff \qquad \eta\lambda_D\left(V^{SD}(\hat{u}_2|\hat{u}_1) + \hat{u}_2\right) = \eta\lambda_D\Pi - c - (\lambda_S\Pi - c)e^{-\frac{\lambda_S}{\varphi}\hat{u}_2 - 1},$$

so that (OA.3.12) is equivalent to $\eta(c+\varphi) - c < (\lambda_S\Pi - c)\, e^{-\lambda_S\hat{u}_2/\varphi - 1}$. The right-hand side is strictly positive because $\Pi > c/\lambda_D > c/\lambda_S$, and the left-hand side is non-positive because $\eta \leq c/(c+\varphi)$. The inequality therefore holds, and $\hat{u}_2 < \overline{u}$. □

LEMMA OA.11 (Not reaching the outer region). *If $\eta > \sqrt{c/(c+\varphi)}$, then $\hat{u}_2 \geq \overline{u}$.*

Proof of Lemma OA.11. Following the computation in the proof of Lemma OA.10, $\hat{u}_2 \geq \overline{u}$ is equivalent to

$$\overline{y} \equiv \frac{(\eta-1)c + \eta\varphi}{(\lambda_S\Pi - c)e^{-\frac{\lambda_S}{\varphi}\hat{u}_1 - 1}} \geq e^{\frac{\lambda_S}{\varphi}(\hat{u}_1 - \hat{u}_2)} = \hat{y}_2, \tag{OA.3.13}$$

where, by the proof of Lemma OA.6, $\hat{y}_2$ is the root of $\tilde{\Lambda}_{SD}(y) = 0$ other than 1. Since $\hat{u}_2 \geq \hat{u}_1$, the inequality holds trivially if $\overline{y} \geq 1$, so assume $\overline{y} < 1$. Because $\tilde{\Lambda}_{SD}$ is strictly convex with $\tilde{\Lambda}_{SD}(1) \leq 0$, the inequality $\tilde{\Lambda}_{SD}(\overline{y}) \leq 0$ implies (OA.3.13).

Writing $\hat{x}_1 \equiv e^{-(\lambda_D/\varphi)\hat{u}_1}$ and using (OA.3.5), rewrite $\tilde{\Lambda}_{SD}$ of (OA.3.10) as

$$\tilde{\Lambda}_{SD}(y) = \frac{1-\eta}{1+\eta}c - \Lambda_D(\hat{x}_1)\, y + \left[-\frac{1-\eta}{1+\eta}c + \frac{\eta}{1+\eta}(\lambda_S\Pi - c)e^{-\frac{\lambda_S}{\varphi}\hat{u}_1 - 1}\log y\right] y,$$

where $\hat{x}_1$ is chosen so that $\Lambda_D(\hat{x}_1) \geq 0$. Substituting the definition of $\overline{y}$,

$$\tilde{\Lambda}_{SD}(\overline{y}) = \frac{1-\eta}{1+\eta}c(1-\overline{y}) - \Lambda_D(\hat{x}_1)\overline{y} + \frac{\eta}{1+\eta}\left((\eta-1)c + \eta\varphi\right)\log\overline{y}.$$

It therefore suffices to show that

$$G(y) \equiv \frac{1-\eta}{1+\eta}c(1-y) - \Lambda_D(\hat{x}_1)y + \frac{\eta}{1+\eta}\left((\eta-1)c + \eta\varphi\right)\log y$$

satisfies $G(y) \leq 0$ for all $y < 1$. Since $\eta \geq \sqrt{c/(c+\varphi)} > c/(c+\varphi)$, $G''(y) = -\eta\left((\eta-1)c + \eta\varphi\right)/\left((1+\eta)y^2\right) < 0$, and, using $\eta \geq \sqrt{c/(c+\varphi)}$ together with

$\Lambda_D(\hat{x}_1) \geq 0$, $G'(1) = -\Lambda_D(\hat{x}_1) + (1/(1+\eta))\left((\eta^2-1)c + \eta^2\varphi\right) < 0$. Finally $G(1) = -\Lambda_D(\hat{x}_1) \leq 0$, so concavity gives $G(y) \leq G(1) + G'(1)(1-y) \leq 0$ for all $y < 1$. Hence $\tilde{\Lambda}_{SD}(\overline{y}) \leq 0$ and $\hat{u}_2 \geq \overline{u}$. □

LEMMA OA.12 (Reaching the inner threshold). *Suppose $\eta > \overline{\eta}$. There exists $\tilde{\pi}_D(\eta) \in (\tilde{\pi}_M(\eta), \tilde{\pi}_S(\eta))$ such that $\hat{u}_1 < \overline{u}$ if and only if $\pi > \tilde{\pi}_D(\eta)$.*

Proof of Lemma OA.12. Since $\eta > \overline{\eta} \geq 1/(e-1)$, the threshold $\tilde{\pi}_S(\eta)$ exists. If $\pi \geq \tilde{\pi}_S(\eta)$ then $\hat{u}_1 = 0$ by Lemma OA.4, and $\tilde{\pi}_S(\eta) = (e-1)/((e-1)\eta - 1) \geq (e-1)/(e-2) > 2 \geq (c+\varphi)/c = \pi_F$, so the project is feasible, $\overline{u} > 0$, and $\overline{u} > \hat{u}_1$.

Now suppose $\tilde{\pi}_M(\eta) < \pi < \tilde{\pi}_S(\eta)$. Since $\mathcal{V}$ is strictly concave, $\hat{u}_1 < \overline{u}$ is equivalent to $0 < \mathcal{V}'(\hat{u}_1) = V^{D\prime}(\hat{u}_1)$, which in turn is equivalent to

$$\frac{\varphi}{\lambda_D \Pi - c} < e^{-\frac{\lambda_D}{\varphi}\hat{u}_1} = \hat{x}_1, \tag{OA.3.14}$$

where $\hat{x}_1$ is the root of $\Lambda_D(x) = 0$ identified in Lemma OA.5. The hypothesis $\tilde{\pi}_M(\eta) < \pi < \tilde{\pi}_S(\eta)$ gives $\Lambda_D(1) < 0 < \Lambda_D(x^*)$ with x^* as in (OA.3.9).

Two cases satisfy (OA.3.14): (i) $x^* \geq \varphi/(\lambda_D\Pi - c)$; and (ii) $\varphi/(\lambda_D\Pi - c) > x^*$ together with $\Lambda_D(\varphi/(\lambda_D\Pi - c)) < 0$. Case (i) is equivalent to $\Lambda_D'(\varphi/(\lambda_D\Pi - c)) < 0$, which by algebra is

$$\frac{(\eta+1)\pi - 1}{(\pi-1)^{\eta+1}} < \frac{\eta e}{\eta+1}\varphi^{-\eta}c^{\eta}, \tag{OA.3.15}$$

and case (ii) is equivalent to $\Lambda_D'(\varphi/(\lambda_D\Pi - c)) \geq 0$ and $\Lambda_D(\varphi/(\lambda_D\Pi - c)) < 0$, that is

$$\frac{\eta e}{\eta+1}\varphi^{-\eta}c^{\eta} \leq \frac{(\eta+1)\pi - 1}{(\pi-1)^{\eta+1}} < (\eta(c+\varphi) - c)\, e\varphi^{-\eta-1}c^{\eta}. \tag{OA.3.16}$$

By the proof of Lemma OA.5, the condition $\pi > \tilde{\pi}_M(\eta)$ is equivalent to

$$\frac{(\eta+1)\pi - 1}{(\pi-1)^{\eta+1}} < \left(\frac{\eta^2}{1-\eta^2}\right)^{\eta}\frac{\eta e}{1+\eta}. \tag{OA.3.17}$$

Using $\eta > \sqrt{c/(c+\varphi)}$, simple algebra gives

$$\frac{\eta e}{\eta+1}\varphi^{-\eta}c^{\eta} < (\eta(c+\varphi) - c)e\varphi^{-\eta-1}c^{\eta} < \left(\frac{\eta^2}{1-\eta^2}\right)^{\eta}\frac{\eta e}{1+\eta},$$

so the single inequality $((\eta+1)\pi-1)/(\pi-1)^{\eta+1} < (\eta(c+\varphi)-c)\, e\varphi^{-\eta-1}c^{\eta}$ implies each of (OA.3.15), (OA.3.16) and (OA.3.17). Let $\tilde{\pi}_D(\eta)$ be the value of π at which its two sides are equal. Then $\tilde{\pi}_D(\eta) > \tilde{\pi}_M(\eta)$, since $\pi < \tilde{\pi}_M(\eta)$ implies $\pi < \tilde{\pi}_D(\eta)$, and $\hat{u}_1 < \overline{u}$ if and only if $\pi > \tilde{\pi}_D(\eta)$. □

The two bounds on η are what separate the regimes: Lemma OA.11 requires η above $\sqrt{c/(c+\varphi)}$ and Lemma OA.10 requires η at or below $c/(c+\varphi)$. Since $c/(c+\varphi) < 1$, the square root is the larger of the two, and the interval between them is exactly the band that neither proposition covers.

Proof of Proposition 4. Suppose $\eta \in (\overline{\eta}, 1)$. Since $\overline{\eta} \geq \sqrt{c/(c+\varphi)}$, Lemma OA.11 gives $\hat{u}_2 \geq \overline{u}$, so the contract never enters the upper direct region and the last two rows of Table 1 are unavailable. The third row is then the only possibility when $\hat{u}_1 = 0$, and the second and fourth the only possibilities when $\hat{u}_1 > 0$; in particular at most one switch occurs.

As $\overline{\eta} \geq 1/(e-1)$, $\tilde{\pi}_S(\eta)$ exists. When $\pi > \tilde{\pi}_S(\eta)$ we have $\hat{u}_1 = 0$, so by Theorem 2(b) the contract is staged-only. When $\pi \leq \tilde{\pi}_S(\eta)$ we have $\hat{u}_1 > 0$, and Lemma OA.12 supplies $\tilde{\pi}_D(\eta) < \tilde{\pi}_S(\eta)$ with $\hat{u}_1 < \overline{u}$ precisely when $\pi > \tilde{\pi}_D(\eta)$. The contract is therefore one-switch for $\pi \in (\tilde{\pi}_D(\eta), \tilde{\pi}_S(\eta))$ and direct-only for $\pi \in (\pi_F, \tilde{\pi}_D(\eta))$.

Finally $\tilde{\pi}_D(\eta) \geq \pi_F$: when $\pi < \pi_F$ the project is infeasible, so $\overline{u} = 0 \leq \hat{u}_1$, and Lemma OA.12 then gives $\pi \leq \tilde{\pi}_D(\eta)$. □

Proof of Proposition 5. Let $\eta < \underline{\eta}$. As $\underline{\eta} \leq c/(c+\varphi)$, Lemma OA.10 applies whenever $\pi > \tilde{\pi}_M(\eta)$ and gives $\hat{u}_2 < \overline{u}$: the contract reaches the upper direct region.

Since $\underline{\eta} \leq 1/(e-1)$, Lemma OA.4 shows that the staged route is never preferred at the deadline, so $\hat{u}_1 > 0$ whenever the staged route is used at all. Two cases remain.

If $\pi \leq \tilde{\pi}_M(\eta)$, the staged route is never used and $\hat{u}_1 = \hat{u}_2 = \infty$; the second row of Table 1 applies and the contract is direct-only.

If $\pi > \tilde{\pi}_M(\eta)$, then $0 < \hat{u}_1 < \hat{u}_2 < \overline{u}$, which is the sixth row of Table 1: a two-switch contract with intermediate deadlines $T - \hat{u}_2/\varphi$ and $T - \hat{u}_1/\varphi$. □

REMARK OA.3.2 (*The intermediate band*). Propositions 4 and 5 leave $\eta \in [\underline{\eta}, \overline{\eta}]$ uncharacterized, because neither Lemma OA.11 nor Lemma OA.10 applies there and the comparison between $\overline{u}$ and $\hat{u}_2$ can go either way. Theorem 2 continues to hold, so the optimal contract still takes one of the forms in Table 1; what is not available is a closed-form map from (η, π) to forms. The band is not merely a technical residue: by Remark OA.3.1 it is the only region in which the reversed one-switch contract arises.

Appendix OA.4: Discounting

This appendix supports the discussion of discounting in Section 6.3. Both parties discount at a common rate $r > 0$. Incentive compatibility is unaffected, but the reduction that organizes the main analysis fails, so we characterize the optimal contract numerically.

OA.4.1. What Changes

Working route j over $[t, t+dt)$ still delivers $\lambda_j(R_t - u_t)\,dt$ against $\varphi\,dt$ from shirking, so (IC_D) and (IC_S) hold as stated and the premium for a delivery remains φ/λ_j. What changes is the law of motion for promised utility. Discounting adds a return on the outstanding promise, so binding incentives now give $\dot{u}_t = r\,u_t - \varphi$, and the deadline at which promised utility reaches zero is $T(u) = -(1/r)\log(1 - ru/\varphi)$, which is finite only for $u < \varphi/r$ and converges to u/φ as $r \to 0$. Promised utility remains a clock, but one that runs at a rate depending on how much is owed.

The extension is therefore no longer a constant. Delivering the milestone still raises promised utility by φ/λ_S, but the time this buys is

$$\Delta(u) = T\left(u + \tfrac{\varphi}{\lambda_S}\right) - T(u) = -\frac{1}{r}\log\left[1 - \frac{r/\lambda_S}{1 - ru/\varphi}\right], \tag{OA.4.1}$$

which is increasing in u and tends to $1/\lambda_S$ as $r \to 0$. Because (6) prices the extension at $1/\lambda_S$ independently of the state, it no longer holds, and the principal's problem cannot

be read off the observable-effort benchmark. We therefore solve the HJB equations for $\mathcal{V}$ and V_M directly, with the route chosen pointwise as in (HJB_V).

OA.4.2. Why Discounting Favors the Direct Route

Consider the two routes with no deadline. If completion occurs at a random date τ, the expected discounted surplus is $(\Pi + c/r)\,\mathbb{E}[e^{-r\tau}] - c/r$, so the route with the larger expected discount factor yields the larger surplus. That factor is $\lambda_D/(\lambda_D + r)$ on the direct route and $(\lambda_S/(\lambda_S + r))^2$ on the staged route. At $\lambda_S = 2\lambda_D$ the two routes have the same expected completion time, yet the direct route has the larger factor for every $r > 0$, since $(2\lambda_D + r)^2 > 4\lambda_D(\lambda_D + r)$. The reason is convexity: $e^{-r\tau}$ is convex in τ, so it rewards the more dispersed completion time. Far from the deadline, where the deadline effect has washed out, discounting thus plays the role that an efficiency loss plays in Section 5, and it generates the same outer direct region.

OA.4.3. Numerical Results

Fix $\lambda_S = 2\lambda_D$, so that staging is technologically innocuous and Theorem 1 applies at $r = 0$. There the outer threshold is absent: $\hat{u}_2 = \infty$, and the contract takes one of three forms. Any positive discount rate makes $\hat{u}_2$ finite.

Figure OA.4.1 plots $\hat{u}_2(r)$ against the optimal promised utility $\overline{u}(r)$. Both decline in r—a more impatient principal buys less time—but the outer threshold falls much faster, and the two cross at $r^* \approx 0.335$. Below r^* the outer direct region exists but is never reached: the principal does not buy enough time for it to matter, and the contract looks exactly as it does under patience. Above r^* the contract begins on the direct route and turns to staging as the deadline approaches. This is the reversed one-switch contract of Table 1, a form that Theorem 1 excludes.

Table OA.4.1 records the form of the optimal contract across project returns and discount rates. Every cell is one of the six forms of Table 1: discounting relabels

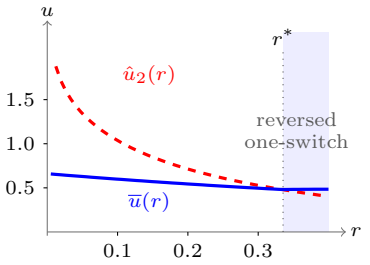


FIGURE OA.4.1. The outer threshold and the optimal promised utility as functions of the discount rate ($\lambda_D = 1$, $\lambda_S = 2$, $c = 1$, $\varphi = 0.5$, $\pi = 3$)

TABLE OA.4.1. Form of the optimal contract under discounting ($\lambda_S = 2\lambda_D$, $c = 1$, $\varphi = 0.5$)

	Discount rate r			
π	0	0.10	0.20	0.40
1.8	one-switch	one-switch	one-switch	direct-only
2.0	one-switch	one-switch	one-switch	two-switch
2.4	staged-only	one-switch	one-switch	two-switch
3.0	staged-only	staged-only	staged-only	reversed one-switch
4.0	staged-only	staged-only	staged-only	reversed one-switch

the parameter regions but generates nothing new. The two-switch and reversed one-switch contracts, which require $\eta < 1$ under patience, arise here at $\eta = 1$ once the principal is impatient enough.

Two comparative statics are worth noting. Staging survives impatience best when the project return is high: at $\pi = 2.4$ a discount rate of 0.10 is already enough to reintroduce the direct route near the deadline, while at $\pi = 3$ the contract is staged-only up to r^*. And the direction of the switch reverses as r grows: at moderate returns the contract stages first and turns direct, as under patience, whereas at high returns and high r it does the opposite. The reason is that a high return buys a long deadline, and a long deadline is where impatience penalizes staging most.

OA.4.4. Scope

These are numerical results at $\lambda_S = 2\lambda_D$, not a characterization. Two features of the discounted problem stand in the way of one. The extension (OA.4.1) depends on the state, so the thresholds are no longer determined by a benchmark problem in remaining time; and the domain of $\mathcal{V}$ is bounded by φ/r, so the comparative statics

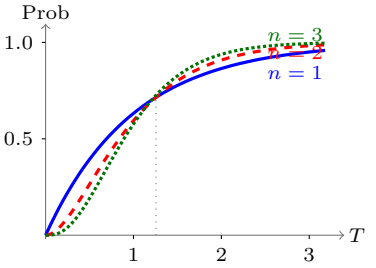


FIGURE OA.5.1. Completion probability under stage decompositions, holding the expected completion time fixed at $1/\lambda_D$ ($\lambda_D = 1$)

in r cannot be taken far without also raising φ. What the exercise establishes is the qualitative claim: impatience is another route to the outer direct region, alongside efficiency loss, and it produces no contract that Theorem 2 has not already classified.

Appendix OA.5: More Than Two Stages

The project admits exactly two decompositions in the main text: one stage or two. In practice the number of interim deliverables is itself a design variable. Figure OA.5.1 plots completion probabilities for $n = 1, 2, 3$ stages with the expected completion time held fixed, so that only the shape of the distribution varies. The deadline effect strengthens with n: at a short deadline, more stages means a lower completion probability, and the gap widens as n grows. In this example the crossing point moves slightly in and the curves are close for $n = 2$ and $n = 3$, so that most of the difference from the direct route is already present at $n = 2$. Characterizing the optimal n-stage contract requires n promised utilities and is left to future work.